

Imbalance and Reallocation of Funds under Chinese Quantity-Based Monetary Policy*

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Abstract

This paper studies the reallocation of funds through wholesale funding markets and its effects on credit supply. The negotiable certificate of deposit (NCD) market plays a primary role in the reallocation of funds following monetary policy interventions. State banks are conservative in their lending, which prevents full reallocation of funds when they lend on the NCD market, but not when they borrow. Following a shift in state banks' positions on the NCD market from lending to borrowing, nonstate banks' lending relative to state banks' lending increased, and firms borrowed more loans from nonstate banks.

Keywords: Quantity-based monetary policy, capital allocation, bank heterogeneity, wholesale funding market, bank loan

JEL Codes: E02, E5, G21

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1 Introduction

Price-based monetary instruments prevail in Western economies such as the United States, where interest rates on reserve balances (IORB), overnight reverse repos, the discount window, and the Standing Repo Facility (SRF) determine the shadow costs of funds for most banks. In contrast, monetary policy in many emerging economies largely relies on quantity-based instruments, such as required reserves ([Bank for International Settlements, 2019](#)). Directly injecting predetermined amounts of funds to banks, quantity-based instruments do not naturally equalize banks' funding costs or eliminate systematic funding imbalances within the banking system. Therefore, in addition to the usual monetary transmission channels, the reallocation of funds among banks may constitute a crucial part of resource allocation. Wholesale funding markets are potentially of first-order importance in resource allocation rather than merely for providing short-term liquidity.

This paper studies how the reallocation of funds operates within the banking system and its effects on lending to the real economy, using China as a laboratory. Specifically, we shed light on the roles of different types of banks and institutions in the reallocation process and the implications for the implementation of quantity-based monetary policy instruments. We begin our analysis with several stylized facts about China's monetary policy and banking system. First, the two major instruments for base money supply, required reserve ratio (RRR) cuts and Medium-term Lending Facilities (MLFs), do not supply funds to banks according to their funding needs, and a systematic reallocation of funds among banks typically follows. MLFs provide a subset of banks, mostly primary dealer (PD) banks, with medium-term loans. RRR cuts, instead, effectively supply funds to all banks according to their deposit holdings rather than their funding needs.¹ Second, the Chinese banking system features considerable heterogeneity across banks. State banks, which are majority-owned by the central government, are more conservative in their lending practices than nonstate banks, perhaps driven by their non-market objectives. Third, among the various wholesale funding markets, the newly developed Negotiable Certificate of Deposit (NCD) market is particularly active, competitive, and potentially important in facilitating the reallocation of funds. Additionally, money market funds (MMFs) have rapidly expanded over the past decade, significantly influencing resource allocation among banks.

Building upon these observations, we develop a simple model of the Chinese banking system. The model consists of one state bank, many small nonstate banks, and MMFs. Banks borrow and lend in the competitive wholesale funding market and extend loans to firms, while MMFs generate profits by lending to banks. State banks exhibit conservatism in the sense that compared to nonstate banks, they have additional aversion to wholesale lending and loans, but not to wholesale borrowing. Through the model, we characterize the reallocation of funds following quantity-based

¹We provide more details in Section 2.1.

monetary policy interventions. The reallocation operates through the substitution of funds injected by the monetary authority for banks' wholesale funding positions, which we refer to as the substitution effect. State banks' conservatism prevents their "excessive" financial resources from being fully reallocated to other banks. As a result, when state banks are endowed with ample financial resources and lend in the wholesale funding market, they originate more loans than nonstate banks. However, when state banks are endowed with limited financial resources and borrow in the wholesale funding market, they behave similarly to nonstate banks. Hence, the distribution of banks' endowed financial resources—comprising central bank funds and deposits—crucially shapes the eventual allocation of resources across banks. The model illustrates that the overall impact of quantity-based monetary policy hinges not only on the volume of injected funds but also on the funding condition of state banks.

Guided by the model, we conduct a series of empirical analyses. We first characterize the substitution effect of monetary policy by examining how banks adjust various wholesale funding positions in response to variations in central bank borrowing. The analysis faces the challenges that 1) central bank borrowing may be affected by a bank's idiosyncratic demand for funds, and 2) central bank borrowing is affected by monetary policy intervention, which itself is endogenous to the state of the economy. To mitigate potential endogeneity concerns, our identification exploits the cross-sectional variation of the change in banks' endowed funds caused by MLF operations. Specifically, we instrument a bank's central bank borrowing using the interaction of the aggregate liquidity supplied via MLF and a dummy variable indicating whether a bank is a PD bank. This instrument is motivated by the observation that MLF operations predominantly supply funds to PD banks, so only their central bank borrowing is substantially affected by MLFs. This empirical design is similar in spirit to a difference-in-differences (DiD) framework or a Bartik instrument (Autor et al., 2013; Cong et al., 2019; Goldsmith-Pinkham et al., 2020; Breuer, 2021; Borusyak et al., 2022), where 1) MLF causes a shift in the aggregate liquidity supply, and 2) PD banks are directly exposed to this shift but non-PD banks are not.² Additionally, our analysis controls for other sources of endowed funds, including deposits and required reserves.

The two-stage least squares (2SLS) estimation results corroborate the model's predictions regarding both direction and magnitude of substitution effects on the NCD market. First, we estimate the elasticity of substitution between central bank borrowing and borrowing through NCDs for all PD banks at -1.339 , close to the theoretical prediction of -1 . Moreover, no significant difference exists between state and nonstate banks in this respect. Second, the estimated elasticity of substitution between central bank borrowing and lending through NCDs is significantly higher for nonstate banks compared to state banks, by more than one-third. These findings lend support to the central model prediction that state banks' conservatism prevents full reallocation of funds when lending

²We discuss the validity of the instrument in Section 5.1.

but not when borrowing in wholesale markets. However, we do not find robust evidence for similar substitution effects in traditional wholesale funding markets, such as interbank deposits, interbank placements, repos, and reverse repos.

We also investigate the reallocation of funds following RRR cuts. Our model predicts that, following RRR cuts, the correlation between banks' wholesale funding and deposits becomes more negative due to an increased substitution of deposits for wholesale funding. We test this prediction by examining the cross-sectional relationship between banks' wholesale funding and deposits around the onset of the wave of RRR cuts in 2018Q2. The results confirm a significantly more negative correlation between borrowing through NCDs and deposits. Consistent with our findings from MLF operations, banks do not significantly adjust their traditional wholesale transactions in response to the wave of RRR cuts.

In summary, our analyses of substitution effects following MLF operations and RRR cuts underscore three key insights. First, the substitution of central bank funds for banks' wholesale funding positions serves as an essential transmission channel for monetary policy within the banking system. Second, state banks' conservatism prevents full reallocation of funds when they lend in wholesale markets but does not when they borrow. Third, the NCD market plays a primary role in the reallocation of funds following monetary policy interventions, whereas traditional wholesale markets play a relatively minor role.

Next, we examine the effect of the reallocation mechanism on bank lending to the real economy. According to the model, when state banks shift from the lending side to the borrowing side in the wholesale funding market, it indicates an improvement in the reallocation of funds. This improvement arises because state banks' conservatism prevents full substitution for wholesale lending, but not for wholesale borrowing. Consequently, the gap between banks' marginal costs of funds narrows: nonstate banks finance more investment compared to state banks. To test this implication empirically, we focus on episodes around 2018. A conventional view of the Chinese banking system is that state banks hold ample financial resources and provide funds to other banks. However, state banks began to systematically issue NCDs in 2018 and have been net issuers since 2019 onward, indicating their shift from the lending side in the NCD market to the borrowing side.³ Comparing the periods before and since 2018, the second part of our empirical analysis assesses the impacts of the reallocation mechanism on bank lending.

We first check whether the model prediction is consistent with the macro pattern of bank lending using banks' monthly *Sources and Uses of Credit Funds* published by the People's Bank of China (PBC). After detrending state and nonstate banks' loans and endowed funds, we find that

³A potential force driving this shift is the deposits channel established by [Drechsler et al. \(2017\)](#). In 2017 and the first half of 2018, high interest rates caused massive deposits outflows to non-bank investment vehicles such as MMFs, which eventually reduced banks' endowed funds. See [2.3](#) for more details.

since 2018, nonstate banks' loans have persistently been above their trend more than state banks', but their endowed funds have not. This finding suggests that nonstate banks' higher levels of loans were financed by sources other than endowed funds, consistent with our argument that the reallocation of funds improved and led to higher lending by nonstate banks. We then examine the impact of the reallocation mechanism using granular bank-level data in a DiD framework. We find that the difference in loans between state and nonstate banks was stable before 2018 but increased by 9.1% afterward. Across different types of banks, joint-stock banks and urban commercial banks, which rely more on wholesale funding markets to obtain funds, exhibited significantly higher levels than state banks, but rural commercial banks did not.

Furthermore, we provide firm-level evidence of the impact of the reallocation mechanism. In China, firms listed in the National Equities Exchange and Quotations (NEEQ) are required to fully disclose their borrowing activities, which allows for a comparison between different sources of loans within a firm. To measure loan composition, we calculate the fraction of the average daily balance of loans from nonstate banks to that from all banks. This fraction naturally captures the relative difference between borrowing from nonstate banks and that from state banks and controls for differences in firms' characteristics and demands for loans. For firms that borrow substantially from both state and nonstate banks, the fraction has increased by 2.7% since 2018. Our back-of-the-envelope calculation suggests that this estimate corresponds to a relative growth in lending of nonstate banks to state banks by 10.81%. To extend the firm-level analysis to a larger sample, we study firms listed in A share markets. A similar within-firm borrowing analysis is not feasible, since we do not observe these firms' loan composition. Alternatively, we conduct a between-firm borrowing analysis based on the well-documented tendency of state banks to disproportionately lend to state-owned enterprises (SOEs) (Cong et al., 2019). A further implication of the model is that the difference between the loans made to non-SOEs and SOEs is greater when state banks shift to the borrowing side in the wholesale funding market. We find that non-SOEs' borrowings increased more than SOEs' by 16.3% since 2018.

Taken together, all our bank-level and firm-level analyses support the model prediction that the reallocation of funds improves as state banks move to the borrowing side on the wholesale funding market. Moreover, they suggest a similarly large magnitude of the impact of the reallocation mechanism.

One concern might be that other trends or policy changes in China drive our lending results regarding the difference between state and nonstate banks. The main concerns are related to the real estate sector and shifts in firms' borrowings from other sources to bank loans. In the past two decades, real estate became the most important sector in the Chinese economy. A large fraction of bank loans is related to the real estate sector, so shocks to the credit demand of the sector and regulation related to the credit supply to the sector may cause substantial changes in banks' loan

portfolios. Since NEEQ has few firms related to this sector, the within-firm borrowing analysis is not substantially affected by this alternative mechanism. For the bank-level lending analysis and the between-firm borrowing analysis, we redo the analysis excluding loans to the real estate sector and excluding firms in this sector. We find similar results, suggesting that the change in bank loans since 2018 was not solely driven by the real estate sector.

Firms obtain external financing not only through bank loans but also through nonbank lending, for example shadow banking, and credit markets such as the bond market. Changes in the supply of these alternative sources may affect firms' demand for bank loans. In 2018, an important regulatory change in the financial sector, a new regulation on asset management (NRAM), may have caused such changes on a large scale. NRAM imposed clear restrictions on the investment of wealth management companies in nonstandardized debt and may have diverted some firms to bank loans. We provide evidence that this alternative explanation does not drive our firm-level results. Regarding the within-firm borrowing analysis, we split NEEQ firms into two subsamples: one consisting of firms with non-bank borrowing and one consisting of firms without. Our lending results are similarly present in both of the two subsamples, suggesting the shift from non-bank borrowing is unlikely to be the main driving force. Our between-firm analysis examines firms' borrowings including both bank loans and non-bank borrowings, so it cannot be driven by the shift from non-bank borrowing either. To exclude the impact of the shift from credit markets to bank loans, we redo the two analyses on the subsamples of firms without bond issuance, and still find results aligning closely with the full sample analysis. We also discuss the impact of the regulation on NCDs in 2018 and policies that facilitate lending to small and micro business (SMB), and illustrate why they are unlikely the driving force of our findings.

Conceptually, the fact that state banks' positions in wholesale funding markets have implications for lending to the real economy hinges on both uneven monetary policy instruments and frictions in wholesale funding markets. If monetary policy injects funds according to banks' needs and equalizes their funding costs, wholesale funding markets should be irrelevant for lending to the real economy. Conversely, if monetary policy induces funding imbalances but wholesale funding markets are frictionless, reallocation will be complete and the initial endowment of funds will not matter. Our empirical analysis jointly tests both uneven monetary policy instruments and frictions in wholesale funding markets.

Our findings have several policy implications. First, our results highlight that the incompleteness of reallocation originates primarily from state banks' conservatism on the lending side of wholesale funding markets. The pronounced improvement in allocative efficiency after 2018, when state banks shifted from being net holders of NCDs (lenders) to net issuers (borrowers), reveals that the root of market segmentation lies in state banks' reluctance to supply funds when they are endowed with ample liquidity. Policymakers could encourage state banks to participate more

actively as NCD investors by improving disclosure and collateral practices, deepening secondary-market liquidity, and developing credit-risk mitigation tools. This would help channel excess funds through wholesale markets even when state banks are not liquidity constrained.

Second, to reduce the initial funding imbalances created by quantity-based instruments, policymakers could broaden the set of eligible counterparties for MLF beyond PD banks, or design targeted liquidity tools that reach banks with higher marginal lending productivity, particularly nonstate banks facing binding funding constraints. The People’s Bank of China has already made some attempts in this direction. The Targeted Medium-Term Lending Facility (TMLF, 2018–2021) periodically extended liquidity to banks with strong lending to private enterprises and SMEs, and the 2024 reform of MLF operations adopted a more market-based framework, further expanding market-based access to central bank liquidity. These developments are consistent with our implication that broadening direct access to central bank funds can improve the initial allocation of liquidity.

Taken together, these considerations underscore that the effectiveness of quantity-based monetary policy hinges not only on the volume of injected liquidity, but also on whether the design of policy instruments and the structure of wholesale markets jointly ensure that funds flow to banks with the strongest lending incentives.

The paper is organized as follows. The remainder of this section reviews the related literature. Section 2 provides necessary institutional background and highlights the stylized facts related to our analysis. Section 3 builds a simple model of the Chinese banking system. Section 4 provides a detailed description of the data we use. Section 5 characterizes the reallocation of funds following the two major quantity-based instruments. Section 6 investigates the effects of the reallocation mechanism on banks’ lending to the real economy. Section 7 examines potential alternative explanations for our empirical findings. Section 8 concludes.

Related Literature Our paper is related to the large literature on the different transmission channels of monetary policy, such as the bank lending channel (Bernanke, 1983; Bernanke and Blinder, 1988, 1992; Kashyap and Stein, 1994) and the balance sheet channel (Bernanke and Gertler, 1989; Kiyotaki and Moore, 1997; Gertler and Kiyotaki, 2010; He and Krishnamurthy, 2013; Brunnermeier and Sannikov, 2014). Recently, researchers have uncovered new channels through which monetary policy transmits and generates real effects, including wholesale funding markets (Vari, 2020; Bianchi and Bigio, 2022; Eisenschmidt et al., 2022; Altavilla et al., 2022b) and banks’ market power (Scharfstein and Sunderam, 2016; Drechsler et al., 2017; Xiao, 2020; Wang et al., 2022). Bianchi and Bigio (2022) consider the wholesale funding market and its impact on banks’ asset allocations. However, the funding imbalance and frictions in our paper originate from the conduct of monetary policy and the characteristics of the Chinese banking system and are different from

the ones they consider. [Drechsler et al. \(2017\)](#) establish the deposits channel of monetary policy, that is, deposits flow out of the banking system in the period of high interest rates since banks do not sufficiently increase deposit rates. Our paper contributes to this literature by highlighting the conduct of monetary policy and the characteristics of the Chinese banking system as a source of the funding imbalance and frictions in the monetary transmission mechanism. Particularly, we argue that high interest rates changed banks' funding conditions through the deposit channel in 2017 and 2018, which further affected the reallocation of funds within the banking system and lending to the real economy.

We also contribute to a growing literature on the Chinese banking system, especially on capital allocation and bank heterogeneity. The heterogeneous nature of financial intermediaries can generate endogenous macroeconomic risks through risk-shifting behavior ([Coimbra and Rey, 2024](#)). In the Chinese context, bank heterogeneity along the dimension of local versus nationwide banks also shapes credit allocation, as local government debt issuance can crowd out private investment through geographically segmented banking markets ([Huang et al., 2020](#)). [Cong et al. \(2019\)](#) study credit allocation across firms through the banking system during China's economic stimulus plan of 2009–2010, and find that the stimulus-driven credit expansion disproportionately favored SOEs over non-SOEs. [Hachem and Song \(2021\)](#) develop a model with interbank market power and liquidity regulation to study China's shadow banking activities. [Geng and Pan \(2024\)](#) document that the implicit government guarantee leads to an SOE premium in bond markets, and the premium increased when liquidity conditions deteriorate. [Fang et al. \(2020\)](#) document the collateral channel of monetary policy in China.⁴ [Chen et al. \(2025\)](#) document that nonstate banks' reliance on wholesale funding enhances the transmission of interest-rate-based monetary policy to productive firms, yet increases systemic fragility during downturns. To the best of our knowledge, we are the first to consider the nature of China's major quantity-based monetary policy instruments and analyze the associated reallocation of funds. We highlight that quantity-based instruments do not eliminate or even induce funding imbalances within the banking system, so the subsequent reallocation of funds among banks constitutes a crucial part of resource allocation.

Finally, our paper is also related to the literature studying the uneven distribution of excess reserves across banks. After the Global Financial Crisis, central banks around the world have substantially expanded their balance sheets and provided excess reserves to banks with suitable assets. Most of the recent literature has focused on the possible effects of this asymmetric distribution of excess reserves on financial stability and money markets ([Copeland et al., 2021](#); [Acharya et al., 2023](#); [D'Avernas and Vandeweyer, 2023](#); [Du et al., 2024](#)). [Altavilla et al. \(2022a\)](#) instead focus on whether this asymmetric distribution of excess reserves hampers the monetary policy transmission mechanism because banks might have reached a satiation point. Exploiting the introduction of the

⁴See also [Chen et al. \(2018, 2023\)](#); [Sun et al. \(2021\)](#); [Wang and Jin \(2021\)](#); [Li et al. \(2022\)](#).

tiering system by the European Central Bank, the authors find that a reallocation of funds to banks with higher liquidity needs enhanced the transmission mechanism.

2 Institutional Background and Stylized Facts

In this section, we detail the institutional background and document several stylized facts, which inform our modeling choices. We refer to [Amstad and He \(2020\)](#); [Huang et al. \(2020\)](#); [Sun \(2020\)](#); [Wang \(2020\)](#) for a more detailed description.

2.1 Monetary policy

During the period between 2002 and 2013, China saw a large influx of foreign exchange (FX) from current account surpluses and, in some years, even financial account surpluses. Because the People’s Bank of China (PBC) accumulated FX reserves and at the same time kept the Renminbi (RMB) exchange rate stable, a large degree of RMB liquidity was injected into the economy. As a form of passive base money supply, funds outstanding for FX were so strong that the PBC’s monetary policy instruments were mainly used to sterilize FX inflows during this period.⁵ Since 2013, the growth of funds outstanding for FX has been much slower, even negative at times. As a consequence, Chinese monetary policy became more dependent on the PBC’s active money supply, which is mainly implemented through quantity-based instruments. Table 2 presents an overview of the quantity-based instruments used extensively from 2013 to 2019, and Panel A of Figure 1 plots the amount of base money supplied through these instruments. Among them, RRR cuts, MLF, and Pledged Supplementary Lending (PSL) dominate the base money supply.

In September 2014, the PBC created MLF to supply medium-term base money to the economy. At a monthly frequency, MLF provides banks with loans, whose maturity is typically one year but can also be three or six months. Banks need to provide collateral in the form of high-quality bonds such as government bonds, central bank bills, policy financial bonds, and high-grade credit bonds. MLF targets commercial banks and policy banks that meet the requirements of macro-prudential management. PD banks are de facto the main counterparty of MLF. Although MLF supplies funds to a subset of banks, it is used as a flexible instrument to “maintain the overall stability and moderate growth of bank system liquidity and support reasonable growth of monetary credit,” according to the PBC’s description.⁶ PSL, instead, is designed to provide collateralized loans to only three

⁵For example, the RRR was raised from 7.5% in 2006 to 21% in 2011 despite the global financial crisis; central bank bills were issued to absorb commercial banks’ excessive liquidity.

⁶On its official website, the PBC introduces MLF as follows: “To maintain the overall stability and moderate growth of bank system liquidity and support reasonable growth of monetary credit, the central bank needs to continuously enrich and improve the tool combination based on the term, subject, and purpose of liquid-

policy banks and is basically more a fiscal than a pure monetary policy instrument.⁷ The left panel of Figure 1 shows that RRR cuts and MLF have been the two main monetary policy instruments in China since 2014, which is the period we focus on in this study.

The importance of the two instruments has been widely recognized. RRR cuts are considered the most effective policy instrument in supplying liquidity and signaling the PBC's commitment to expansionary monetary policy. The interest rate on excess reserves is very low in China: during the period from 2008 to 2020, it was constantly 0.72%. Banks have little incentive to hold excess reserves, and the requirement for reserves is a binding constraint. Hence, RRR cuts have large impacts on banks' funding conditions. To prevent liquidity crunches, the PBC launched two waves of RRR cuts in 2015Q1 and in 2018Q2. The RRR was adjusted from 20% to 17% and from 17% to 13.5%, respectively, which successfully reduced money market rates. On the other hand, MLF is considered a more moderate and flexible way to control the liquidity in the banking system and is implemented on a monthly basis. Although the PBC operates these instruments at its discretion, their use follows a systematic rule. In Online Appendix C.1, we estimate a Taylor-type policy rule for the quantity-based instruments, in which the instrument responds to lagged inflation and to the output gap and the output-gap response switches endogenously according to whether growth is above or below the government's target. The estimates show that the quantity-based instruments we study follow essentially the same rule as the M2 growth rate in Chen et al. (2018) and the composite price- and quantity-based policy indicator in Miranda-Agrippino et al. (2026): they expand when growth falls short of the target and do not respond significantly when growth is at or above it.

A common feature of the two instruments is that they do not inject funds to banks according to their needs. By design, RRR cuts inject funds proportional to banks' deposits, so banks with a higher share of deposits in liabilities instead of those with higher demand for funds are injected with relatively more funds. MLF injects funds to a subset of banks, mostly PD banks, which we discuss in the next subsection. Panel B of Figure 1 shows the injection of MLF and the one-year MLF rate over time. The MLF rate was stable despite substantial variation in the money market reference rate with the same maturity, the one-year Shibor. As a result, the MLF rate was substantially lower than the one-year Shibor in 2017 and 2018 and higher in 2019 and the first half of 2020. The substantial difference between them suggests that the PBC actively manages the quantity of MLF instead of the rate. Further evidence of quantity management is that the MLF injection amounts to a multiple of 0.1 trillion RMB in many months. Controlling the quantity of MLF, the PBC may ration banks to a fraction of what they demand or ask them to borrow certain amounts, so banks are

ity demand, in order to further improve the flexibility, specificity, and effectiveness of regulation." Please see <http://www.pbc.gov.cn/zhengcehuobisi/125207/125213/125437/125446/125870/2809123/index.html>.

⁷In practice, PSL has been mainly used to fund policy banks' special loans for shanty-town renovation.

probably not able to freely borrow from MLF at the MLF rate.⁸ Taken together, it seems unlikely that MLF is designed to completely satisfy banks' needs for funds at the MLF rate and the PBC uses the rate to directly control the market interest rate.⁹ Consequently, banks may have different shadow costs of funds after policy interventions. In this sense, we describe Chinese monetary policy as uneven.¹⁰ Uneven monetary policy results in a systematic funding imbalance within the banking system, so the subsequent reallocation of funds among banks constitutes a crucial part of monetary policy transmission.

2.2 The banking system

Banks dominate the Chinese financial system. As of 2017, banking institutions' total assets amounted to 252 trillion RMB and accounted for 95% of all financial institutions' total assets. At that time, China's banking industry consisted of over 4,000 commercial banks, including 6 state banks, 12 joint-stock commercial banks, 134 urban commercial banks, and many rural commercial banks, private banks, and foreign subsidiary banks.

The six state banks, also referred to as the "Big 6" banks, are majority-owned by the central government and have similar business models. They include the "Big 4" banks—the Industrial & Commercial Bank of China (ICBC), the China Construction Bank (CCB), the Bank of China (BoC), and the Agricultural Bank of China (ABC), and two minor ones—the Bank of Communications and the Postal Savings Bank of China. Due to their large coverage of bank branches, state banks have considerable advantage in taking deposits at low deposit rates and focus on traditional financial intermediation between depositors and borrowers. Joint-stock banks are held mainly by nonstate entities. Although joint-stock banks are not completely privately owned, the government holds significantly smaller stakes in them than in the state banks. Urban commercial banks and large rural commercial banks were formerly owned by local governments. During the 2000s, they were transformed into joint-stock companies, in which the local governments became the main but not necessarily dominant shareholders.

It is widely acknowledged that state and nonstate banks differ substantially in their objectives and operations. Many previous studies treat the distinction between them as a crucial feature of Chinese banking system (Chen et al., 2018; Hachem and Song, 2021; Wang and Jin, 2021; Sun et al., 2021). Figure 2 shows that state banks have lower loan rates, lower nonperforming loan

⁸For example, in 2019 and the first half of 2020, the Shibor remained persistently lower than the MLF rate, but the PBC still lent positive amounts to banks and the amounts were usually multiples of 0.1 trillion RMB.

⁹This does not mean that the MLF rate has no impact on the market at all. In practice, it is the benchmark to determine the Loan Prime Rate and is considered to signal the PBC's goal and future monetary policy.

¹⁰We show that these instruments respond systematically to macroeconomic conditions by estimating an endogenous-switching policy rule (Chen et al., 2018; Miranda-Agrippino et al., 2026). The tools expand counter-cyclically when growth falls below target (see Online Appendix C.1).

ratios, and lower returns on bond investment. These observations are consistent with state banks being more conservative in their investment than nonstate banks.

Similar to their counterparts in the U.S., PDs in China are trading counterparties of the PBC in its implementation of monetary policy. PDs are selected by the PBC every year. As of 2019, there are 49 PDs consisting of 2 policy banks, 6 state banks, 38 nonstate banks, and 3 non-bank financial institutions. During our sample period (from 2013 to 2019), the biggest 31 banks were consistently designated as PD banks. As mentioned before, only PDs can receive the injection of MLF by the PBC and therefore play important roles in the current monetary policy framework. We refer to the remaining banks as non-PD banks. Given the conservatism of state banks, we divide PD banks into two groups: state banks and nonstate PD banks.

2.3 Wholesale funding markets

Since the implementation of Chinese monetary policy frequently induces systematic funding imbalance within the banking system, wholesale funding markets that reallocate funds among banks and mitigate the imbalances are crucial to the allocation of financial resources. Traditionally, banks borrow and lend in three types of wholesale transactions: deposits from/to financial institutions, placement from/to financial institutions, and repos/reverse repos. Throughout the paper, we refer to the three jointly as traditional wholesale transactions. Placement from/to financial institutions, along with repos/reverse repos, are predominantly short-term instruments. Deposits from/to financial institutions, while they can extend to longer maturities, serve not only the purpose to generate interest income but also to facilitate payment and settlement flows. The non-competitive nature of traditional wholesale funding markets renders them rather unresponsive to monetary policy interventions.

In addition to traditional wholesale transactions, NCDs were introduced at the end of 2013. An NCD is a nonsecured, fixed-term certificate of deposit issued by depository institutions to financial institutions including banks and non-bank investment vehicles. The maturities of NCDs typically range from one to twelve months, with twelve months being the most popular. The NCD market is considered to be competitive because NCDs are tradable and have excellent secondary market liquidity ([Amstad and He, 2020](#)). As [Figure 3](#) shows, since its inception in December 2013, the NCD market grew rapidly, reaching 10.7 trillion RMB by the end of 2019, and has become comparable to the three types of traditional wholesale transactions combined. Given its sheer size and high liquidity, the NCD market is widely acknowledged to be one of the most important wholesale funding markets in China, and the interest rates on the NCD market serve as a good barometer of banks' funding costs ([Adrian et al., 2019](#)). When NCDs were first introduced in 2013, state banks were the biggest holders of NCDs, but were later overtaken by rural commercial

banks and mutual funds broadly defined, including money market funds and wealth management products, which we will discuss in more detail below. On the issuance side, joint-stock and urban commercial banks have always been the main issuers. Notably, state banks began to systematically issue NCDs in 2018 and have become net issuers in 2019. This latter fact seems at odds with the typical view that state banks have ample resources and basically provide funds to other banks.

What triggered the shift in 2018? Panel A of Figure 4 shows that money market interest rates remained high from 2017m1 to 2018m7. Specifically, the three-month Shibor was almost always above 4%. While investment vehicles such as money market funds offered a return of at least 4% to their investors, banks offered much lower deposit rates, typically around 1.5% during that time period. As documented by Drechsler et al. (2017), banks have market power in providing various services to depositors, so they can keep a large fraction of deposits with low deposit rates. This pattern is even more true in China, as banks are subject to deposit rate ceilings and cannot sufficiently compete with respect to deposit rates.

Given the return gap between money markets and deposits, non-bank investment vehicles attracted a large amount of funds from depositors, who traditionally deposit most funds at banks (Drechsler et al., 2017). Panel B of Figure 4 shows that in the 18 months from 2017m1 to 2018m7, money market funds' NAV ballooned from 3.6 trillion RMB to 8.6 trillion RMB. The migration of funds put pressure on banks to attract deposits and reduced deposits for all banks. Specifically, Panel C of Figure 4 plots the year-on-year growth rates of bank deposits. Here, bank deposits include only domestic individual deposits and domestic corporate deposits, but not deposits of financial institutions, since they are essentially wholesale borrowing. The figure shows that state banks' deposit growth started to decrease in early 2017, and the decrease was reversed shortly after 2018m7 when interest rates decreased drastically.

The migration of funds due to high interest rates significantly increased state banks' marginal costs of funds, which explains why state banks likely resorted to the NCD market in 2018. One piece of evidence supporting this conjecture is the dramatic increase in banks' structural deposits, which banks used to increase the effective returns of deposits and to circumvent the deposit ceiling.¹¹ Panel B of Figure 4 shows that both state banks' and nonstate banks' structural deposits increased rapidly from 2017m1 to 2018m7. More broadly, in the five years from 2016 to 2020, money market funds' NAV and banks' structural deposits exhibited a strong positive correlation. The dramatic increase in banks' structural deposits hints at banks facing deteriorating funding

¹¹ Structural deposits combine traditional deposits and financial options. The deposit part determines the minimum return (covers the principle), and the option part determines the upside potential. In many cases, banks set very low thresholds to trigger executions of options, which made the products pseudo-structured and almost equivalent to high-return deposit products. In 2017 and 2018, the effective returns of structural deposits were usually between 3.5% and 4.5%. In 2019, the China Banking and Insurance Regulatory Commission imposed new regulations that discourages banks from issuing pseudo-structured deposits (https://www.gov.cn/xinwen/2019-10/18/content_5442092.htm).

conditions. Otherwise, banks would not resort to such costly deposits. Complementary to our study, using proprietary data from a leading Chinese FinTech company, [Buchak et al. \(2021\)](#) provide causal evidence that MMFs attracted deposits from households by offering more competitive interest rates.

2.4 Fund flows in Chinese banking system and wholesale funding markets

Figure 5 sketches the fund flows in the Chinese banking system and wholesale funding markets. The PBC injects funds to banks mainly through quantity-based instruments such as RRR cuts and MLF, which may leave a substantial funding imbalance within the banking system. Therefore, the reallocation of funds in the wholesale funding market is crucial to monetary policy transmission and resource allocation. Traditionally, depositors mainly deposit money at banks. Recently, attracted by higher returns, depositors also deposit money at money-market mutual funds, which further allocate funds to banks. Directing funds in the wholesale funding market, money-market mutual funds play a profound role in the resource allocation across banks.

State banks exhibit a broadly conservative investment style that shapes all of their lending decisions and has implications for wholesale funding markets. This conservative orientation likely reflects three related institutional features. First, state banks are systemically important institutions explicitly tasked with safeguarding financial system stability ([Chen et al., 2018](#); [Hachem and Song, 2021](#)). Because their failure or distress would have outsized contagion effects, they internalize a high shadow cost for any exposure that is volatile or difficult to monitor, such as uncollateralized wholesale lending to smaller nonstate banks. Second, their business model is built around vast branch networks and lending to large corporate clients, giving them a comparative advantage in traditional deposit-taking and corporate credit rather than in active risk-taking in financial markets ([Ma and He, 2020](#); [Drechsler et al., 2017](#)). Third, their governance structure emphasizes asset safety and nonperforming loan control over short-term profit maximization ([Gao et al., 2020](#)). Unlike executives of nonstate banks who are rewarded primarily according to profitability, senior managers of state banks are essentially government officials whose career advancement depends on avoiding major financial accidents ([Deng et al., 2011](#); [Huang et al., 2020](#)).¹²

3 A Simple Model of Chinese Banking System

Guided by these stylized facts, we now build a simple model of the Chinese banking system with a wholesale funding market. We use the model to characterize the reallocation of funds following

¹²The welfare implications of this conservatism are ambiguous. It may reflect desirable macroprudential restraint by systemically important institutions and thus improve social welfare, or it may result from poor corporate governance within state banks and thus reduce social welfare.

Table 1: A bank’s balance sheet at date 0.

Asset		Liability	
Required reserves	ρd_j	Regular Deposits	d_j
Excess reserves	x_j	High-cost Deposits	h_j
Loans	k_j	CB borrowing	m_j
Wholesale lending	l_j	Wholesale borrowing	b_j
		Equity	0

quantity-based monetary policy, to explain the dynamics of banks and wholesale funding markets in 2017 and 2018, and to derive implications regarding wholesale funding and bank loans that we subsequently bring to the data.

3.1 Setup

The model consists of one state bank, many small nonstate banks, and money market funds (MMFs).¹³ Nonstate banks have a total mass of one and are indexed by $i \in [0, 1]$. We normalize the discount factor to 1. The subscript “s” denotes the state bank, while “i” denotes one of the N nonstate banks. When we generally talk about all banks, we use the subscript “j”.

Table 1 gives an overview of a bank j ’s balance sheet. We assume 0 equity for simplicity since the model does not feature insolvency. Banks have four sources of funding: regular deposits d_j , high-cost deposits h_j , central bank borrowing m_j , and wholesale borrowing from other banks and MMFs b_j . On the asset side, banks must hold at least a fraction ρ of their regular deposits as reserves, where ρ is the required reserve ratio set by the central bank. Hence, bank j ’s reserves consist of required reserves ρd_j and excess reserves x_j . The rest of its funds are allocated between wholesale lending to other banks l_j and loans to firms k_j . Here, we use the terms wholesale borrowing/lending and loans so that model implications can be easily mapped to our empirical results. More broadly, banks can also obtain funds through or allocate funds to tradable debt securities. Taken together, bank j faces the following budget constraint

$$\rho d_j + x_j + l_j + k_j = d_j + h_j + m_j + b_j.$$

We assume that regular deposits d_j are exogenous and the deposit rate is fixed at R_D . As documented by Drechsler et al. (2017), banks have market power in providing various services to depositors, so they can keep a large fraction of deposits with low deposit rates. Furthermore, a bank’s ability to provide these services is crucial to the size of deposits. This is even more

¹³In practice, big Chinese state banks are considered to have similar business models and funding conditions. Our empirical analysis also treats state banks as one group and does not distinguish between them. Hence, we assume one state bank for simplicity.

true in China, as banks are subject to deposit rate ceilings and cannot sufficiently compete with respect to deposit rates. Regular deposits can be interpreted as the deposits that are largely driven by services and insensitive to returns. On the other hand, depositors who are sensitive to returns decide between banks and MMFs. To compete for them with MMFs, banks issue special deposit products with high rates. We call these high-cost deposits. To obtain h_j high-cost deposits, a bank needs to promise a return of $R_D + \frac{1}{2}\kappa h_j$. More broadly, high-cost deposits can be interpreted as any funding source that is not fully competitive, whether it is deposit-based or nondeposit-based. With all banks taking a total of $h_s + \int_0^1 h_i di$ high-cost deposits, MMFs hold $W - h_s - \int_0^1 h_i di$ funds.

The interest rates on required reserves, excess reserves, and central bank borrowing are 0, $R_X - 1$, and $R_M - 1$, respectively.¹⁴ They are determined by the central bank. We detail the wholesale funding market and the loan market structure below. For the time being, denote bank j 's expected profits from wholesale funding by Σ_j and those from loans by Π_j . Nonstate banks are profit-maximizing, so a nonstate bank i 's utility can be represented by its expected profits as follows:

$$[\rho d_i + R_X \cdot x_i] - R_D \cdot d_i - \left(R_D + \frac{1}{2}\kappa h_i\right) \cdot h_i - R_M \cdot m_i + \Sigma_i + \Pi_i. \quad (1)$$

The state bank cares not only about its expected profits but also the safety of its lending. Specifically, the state bank has additional aversion to risky wholesale lending l_{risky} and risky loans k_{NSOE} , and its utility is represented as follows:

$$[\rho d_s + R_X \cdot x_s] - R_D \cdot d_s - \left(R_D + \frac{1}{2}\kappa h_s\right) \cdot h_s - R_M \cdot m_s + \left[\Sigma_s - \frac{1}{2}\delta l_{risky}^2\right] + \left[\Pi_s - \frac{1}{2}\eta k_{risky}\right]. \quad (2)$$

This additional aversion captures state banks' conservatism discussed in Section 2.2.

On the wholesale funding market, banks and MMFs borrow and lend funds, achieving reallocation of funds. As shown by our empirical analysis in Section 5, the competitive uncollateralized NCD market plays a dominant role in the reallocation of funds. Therefore, we consider the wholesale funding market to be competitive and uncollateralized, where all players take the wholesale funding rate R_{WF} as given and R_{WF} is endogenously determined to clear the market. We focus on the cases in which R_{WF} is always greater than the deposit rate R_D . A nonstate bank i 's utility from wholesale funding is

$$\Sigma_i = R_{WF} \cdot (l_i - b_i). \quad (3)$$

For the state bank, wholesale lending is not completely safe, as it is essentially uncollateralized lending to nonstate banks.¹⁵ Therefore, all wholesale lending is risky for the state bank and given

¹⁴In practice, the interest rate on required reserves may be positive. Since it does not affect banks' decisions, we assume it is zero.

¹⁵Although defaults on NCDs are rare, they have happened recently. Baoshang bank, an urban commercial bank, failed in 2019 and caused turmoils in the NCD market.

its risk aversion, its utility from wholesale funding is

$$\Sigma_s - \frac{1}{2} \delta l_{risky}^2 = R_{WF} \cdot (l_s - b_s) - \frac{1}{2} \delta l_s^2. \quad (4)$$

MMFs lend all their funds $W - h_s - \int_0^1 h_i di$ to banks, since they have no other investment opportunities with positive returns.

Each bank faces two types of firms, SOEs and non-SOEs. In China, it is widely accepted that SOEs enjoy implicit government guarantees and are associated with low financial risks from a lender's perspective (Geng and Pan, 2024), while lending to non-SOEs exposes banks to significantly higher financial risks. Therefore, we assume that loans made to non-SOEs incur the state bank's aversion to originate risky loans, and loans made to SOEs do not. Making k_{SOE} loans to SOEs and k_{NSOE} loans to non-SOEs, a nonstate bank's utility from loans is

$$[R_K \cdot k_{SOE} - \gamma k_{SOE}] + [R_K \cdot k_{NSOE} - \gamma k_{NSOE}],$$

and the state bank's is

$$[R_K \cdot k_{SOE} - \gamma k_{SOE}] + [R_K \cdot k_{NSOE} - \gamma k_{NSOE}] - \frac{1}{2} \eta k_{NSOE}.$$

This functional form of expected profits implies a decreasing return on loans. This property could originate from the fact that on a bank's local lending market, the marginal quality of the borrower deteriorates as lending increases. R_K represents the maximum return on loans expected by banks, which potentially depends on banks' expectations about the fundamental, and γ determines the rate at which the return decreases as lending increases. The two parameters could be potentially different for SOEs and non-SOEs, but here we assume they are identical for simplicity.

Solving for banks optimal strategies of making loans, we readily obtain the following results.

Proposition 1. *Under banks' optimal strategies of making loans, the following properties hold.*

- *The state bank originates a higher fraction of loans to SOEs than non-SOEs.*
- *A nonstate bank's utility follows*

$$\begin{aligned} & [\rho d_i + R_X \cdot x_i] - R_D \cdot d_i - \left(R_D + \frac{1}{2} \kappa h_i \right) \cdot h_i - R_M \cdot m_i \\ & + R_{WF} \cdot (l_i - b_i) + \left[R_K k_i - \frac{1}{2} \gamma k_i^2 \right], \end{aligned} \quad (5)$$

whereas the state bank's utility follows

$$\begin{aligned}
& [\rho d_s + R_X \cdot x_s] - R_D \cdot d_s - \left(R_D + \frac{1}{2} \kappa h_s \right) \cdot h_s - R_M \cdot m_s \\
& + \left[R_{WF} \cdot (l_s - b_s) - \frac{1}{2} \delta l_s^2 \right] + \left[R_K k_s - \frac{2\gamma + \eta}{4\gamma + \eta} \gamma k_s^2 \right].
\end{aligned} \tag{6}$$

The difference in loan composition between banks stem from the conservatism in loans. Proposition 1 is supported by various pieces of empirical evidence. Figure 2 shows that state banks have lower loan rates and lower non-performing loan ratios than nonstate banks. Based on loan portfolios of the 19 largest Chinese banks, Cong et al. (2019) show a positive correlation between banks' state ownership and their borrowers' state ownership, implying that state banks tend to originate loans to SOEs more than nonstate banks.

Remark. Here, we model the implications of the state bank's conservatism on both its wholesale lending and loans. It is worth noting that the implications regarding reallocation within the banking system and the empirical analysis center around the state bank's conservatism on wholesale lending and does not rely on the conservatism on loans. Specifically, all the implications below hold as long as $\delta > 0$, even if $\eta = 0$.

3.2 Price-based monetary policy

As a benchmark, we first consider the case when the central bank uses price-based monetary policy instruments. The central bank sets the interest rate on excess reserves, R_X , and that on central bank borrowing, R_M , to R_{CB} . Banks are allowed to freely deposit reserves to or borrow from the central bank. Then R_{CB} pins down the wholesale funding rate, the marginal cost of high-cost deposits, and the marginal return of loans for all banks as follows:

$$\begin{aligned}
R_{WF} &= R_{CB}, \\
R_D + \kappa h_s &= R_D + \kappa h_i = R_{CB}, \\
R_K - \frac{4\gamma + 2\eta}{4\gamma + \eta} \gamma k_s &= R_K - \gamma k_i = R_{CB}.
\end{aligned}$$

Since banks can always attain their desired funding amounts through transactions with the central bank, the state bank's conservatism on wholesale lending will not have any effect on banks' decisions.

3.3 Quantity-based monetary policy

In the rest of the section, we focus on the case when the central bank uses quantity-based monetary policy instruments. To capture the policy observed in China, we make two further assumptions. First, banks earn zero interest on excess reserves: $R_X = 0$. In the sample period we consider, the interest rate on excess reserves in China was only 0.72%, which was substantially lower than the returns of other investment opportunities. Hence, Chinese banks have little incentive to hold excess reserves besides using them as necessary liquidity buffers. Second, central bank borrowing and the required reserve ratio (RRR) are controlled by the central bank. As a way to implement its monetary policy, the central bank injects funds to or withdraws funds from banks by changing central bank borrowing or the RRR. We then rewrite a bank j 's budget constraint as

$$x_j + k_j - h_j + l_j - b_j = (1 - \rho) d_j + m_j. \quad (7)$$

We refer to the right-hand side of the equation, $e_j \triangleq (1 - \rho) d_j + m_j$, as a bank's endowed funds, which are exogenous to it.

Banks choose $(x_j, k_j, h_j, l_j, b_j)$ to maximize expression (5) or (6) subject to equation (7), taking endowed funds as given. Nonstate banks' equilibrium choices are characterized by the following first-order conditions:

$$\begin{cases} x_i = 0, \\ R_D + \kappa h_i = R_{WF}, \\ R_K - \gamma k_i = R_{WF}. \end{cases} \quad (8)$$

Since nonstate banks borrow and lend at a positive rate of R_{WF} on the wholesale funding market, holding a positive amount of excess reserves is dominated. The marginal cost of high-cost deposits and the marginal return of loans must be equal to R_{WF} .

The state bank's equilibrium choices are characterized by the following first-order conditions:

$$\begin{cases} x_s = 0, \\ R_D + \kappa h_s = R_{WF} - \delta l_s \cdot \mathbf{1}\{l_s > 0\}, \\ R_K - \frac{4\gamma+2\eta}{4\gamma+\eta} \gamma k_s = R_{WF} - \delta l_s \cdot \mathbf{1}\{l_s > 0\}. \end{cases} \quad (9)$$

If the state bank lends on the wholesale funding market, its marginal return of loans must be equal to the wholesale funding rate minus its marginal aversion to wholesale lending. However, if the state bank borrows on the wholesale funding market, its aversion is immaterial to its choices. Still, holding a positive amount of excess reserves is dominated.

We characterize banks' equilibrium choices for any positive wholesale funding rate R_{WF} as follows:

Proposition 2. *Given a positive wholesale funding rate R_{WF} , banks' equilibrium choices are given as follows:*

- *for nonstate banks,*

$$\begin{aligned} h_i &= \frac{R_{WF} - R_D}{\kappa}, \\ k_i &= \frac{R_K - R_{WF}}{\gamma}, \\ b_i - l_i &= k_i - h_i - e_i; \end{aligned}$$

- *for state banks,*

$$\begin{aligned} h_s &= \frac{R_{WF} - R_D - \delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\kappa}, \\ k_s &= \frac{R_K - R_{WF} + \delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma}, \\ b_s - l_s &= k_s - h_s - e_s. \end{aligned}$$

For the wholesale funding market to clear, the following equation must hold:

$$b_s - l_s + \int_0^1 (b_i - l_i) di = W - h_s - \int_0^1 h_i di.$$

Combining it with Proposition 2, we obtain

$$k_s + \int_0^1 k_i di = e_s + e_{ns} + W, \quad (10)$$

where $e_{ns} = \int_0^1 e_i di$ represents the total endowed funds of nonstate banks. Equation (10) reflects the resource constraint of the banking system: the total loans made by all banks must be equal to all the funds minus required reserves.

Based on equation (10), Proposition 3 solves for the wholesale funding rate, R_{WF} .

Proposition 3. *Let R_{WF}^1 be the solution to:*

$$\begin{aligned} &\left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma + \frac{4\gamma+2\eta}{4\gamma+\eta} \gamma \frac{\delta}{\kappa} + \delta}\right) \frac{R_K - R_{WF}^1}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma} + \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma + \frac{4\gamma+2\eta}{4\gamma+\eta} \gamma \frac{\delta}{\kappa} + \delta} \frac{R_{WF}^1 - R_D}{\kappa} + \frac{R_K - R_{WF}^1}{\gamma} \\ &= \left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma + \frac{4\gamma+2\eta}{4\gamma+\eta} \gamma \frac{\delta}{\kappa} + \delta}\right) e_s + e_{ns} + W, \end{aligned}$$

and R_{WF}^2 be the solution to

$$\frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} + \frac{R_K - R_{WF}^2}{\gamma} = e_s + e_{ns} + W.$$

- If $e_s > \frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF}^2 - R_D}{\kappa}$, the equilibrium wholesale funding rate is $R_{WF}^* = R_{WF}^1$.
- If $e_s \leq \frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF}^2 - R_D}{\kappa}$, the equilibrium wholesale funding rate is $R_{WF}^* = R_{WF}^2$.

To pin down the wholesale funding rate, we need to figure out whether the state bank lends or borrows on the wholesale funding market. If the state bank has ample endowed funds e_s (that is greater than $\frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF}^2 - R_D}{\kappa}$), it will lend on the wholesale funding market, and its conservatism kicks in. We refer to this case as Scenario 1. Otherwise, it will borrow on the wholesale funding market, and its conservatism does not play a role. We refer to this case as Scenario 2. As suggested by Proposition 3, due to the state bank's conservatism, the reallocation of resources among banks is not complete, and thus the state bank's funding condition affects how the equilibrium plays out.

Proposition 4. *Consider Scenario 1 and Scenario 2 defined above and hold the total available funds $e_s + e_{ns} + W$ fixed.*

- In Scenario 1, the state bank's marginal cost of high-cost deposits is lower than the wholesale funding rate, and it does not borrow on the wholesale funding market. In Scenario 2, its marginal cost of high-cost deposits is equal to the wholesale funding rate, and it borrows on the market.
- In both scenarios, nonstate banks' marginal cost of high-cost deposits is equal to the wholesale funding rate, and they may borrow or lend.

In Scenario 1, the state bank has ample endowed funds, but due to its conservatism on wholesale lending, it does not sufficiently reallocate the funds to nonstate banks. As a result, the state bank holds “excessive” funds in the sense that its marginal cost of high-cost deposits equals $R_{WF} - \delta l_s$ and is thus lower than the wholesale funding rate. Since wholesale funding is more expensive than high-cost deposits at the margin, the state bank does not borrow on the wholesale funding market. In Scenario 2, the state bank has limited endowed funds and borrows on the wholesale funding market. Since its conservatism does not kick in, it will borrow up to the point that its marginal cost of funds is equal to the wholesale funding rate.

Proposition 4 suggests that the state bank with ample endowed funds does not immediately tap the wholesale funding market for funds when its funding condition deteriorates, because high-cost deposits are still cheaper than wholesale funding at the margin. This mechanism helps explain

why we observe that state banks did not systematically issue NCDs in 2017, although their funding conditions have deteriorated. At that time, high-cost deposits such as structural deposits discussed in Section 2.3 were still less expensive than NCDs, so state banks chose to obtain more funds through high-cost deposits instead of NCDs. As state banks issued more high-cost deposits, their marginal costs approached NCDs'. In early 2018, state banks found NCDs attractive as well and began to issue NCDs systematically. Another way to see what happened is that the migration of deposits out of state banks reduced their endowed funds in 2017 and eventually induced a shift from Scenario 1 to Scenario 2 in 2018.

Proposition 4 also suggests that the state bank borrowing on the wholesale funding market can serve as an indicator of Scenario 2. Since reallocation of funds is impeded only in Scenario 1 but not in Scenario 2, a comparison between the two scenarios can inform us about the impact of reallocation. In Section 6, we conduct such an analysis, using the point when state banks begin to systematically issue NCDs to empirically identify the periods that the two scenarios correspond to.

3.4 Testable implications about quantity-based monetary policy

In general, quantity-based monetary policy has three effects on banks' equilibrium choices. The first is the expectation effect. Monetary policy changes banks' expectations about the fundamental. The second is the general equilibrium effect. Monetary policy changes the wholesale funding rate and thus affects banks' marginal cost of loans. A loosening (tightening) policy pushes the wholesale funding rate on the wholesale funding market down (up), and as a response, banks increase (decrease) loans. The third effect is unique to quantity-based monetary policy. Since it directly changes banks' endowed funds, banks will adjust their wholesale funding in the opposite direction so that they have the funds to make optimal amounts of loans. This effect originates from the substitution between endowed funds and wholesale funding, and we refer to it as the substitution effect.

Next, we derive implications about the reallocation of funds following quantity-based monetary policy and its impact on loans. These results not only illustrate the role of conservatism on wholesale lending, but also guide our empirical design. Assume the central bank implements a new monetary policy. For any term z , let Δz represent the change in z following the new monetary policy stance. The change in wholesale funding is given by

$$\begin{aligned}\Delta(b_i - l_i) &= \Delta k_i - \Delta h_i - \Delta e_i \\ &= \frac{R_K - \Delta R_{WF}}{\gamma} - \frac{\Delta R_{WF} - \Delta R_D}{\kappa} - \Delta e_i\end{aligned}\tag{11}$$

for nonstate banks, and

$$\begin{aligned}\Delta(b_s - l_s) &= \Delta k_s - \Delta h_s - \Delta e_s \\ &= \frac{R_K - \Delta R_{WF} + \delta \Delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{\Delta R_{WF} - \Delta R_D - \delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\kappa} - \Delta e_s.\end{aligned}\quad (12)$$

for the state bank. Implication 1 characterizes the substitution effect by considering how the change in wholesale funding varies with the change in endowed funds.

Implication 1. *Consider banks that borrow on the wholesale funding market. The substitution of endowed funds for wholesale borrowing is the same for the state bank and nonstate banks:*

$$\frac{\partial \Delta b_s}{\partial \Delta e_s} = \frac{\partial \Delta b_i}{\partial \Delta e_i} = -1.$$

Consider banks that lend on the wholesale funding market. The substitution of endowed funds for wholesale lending is smaller for the state bank than for nonstate banks:

$$\frac{\partial \Delta l_s}{\partial \Delta e_s} < \frac{\partial \Delta l_i}{\partial \Delta e_i} = 1.$$

Given the fundamental and the wholesale funding rate, an increase in a nonstate bank's endowed funds induces a one-to-one decrease in its wholesale borrowing or a one-to-one increase in its wholesale lending. However, for the state bank, this substitution pattern only holds for its wholesale borrowing but not for its wholesale lending. The state bank's marginal aversion to wholesale lending is increasing in the amount, so an increase in endowed funds induces an increase in wholesale lending that is less than one-to-one. Hence, the state bank and nonstate banks have the same substitution of endowed funds for wholesale borrowing, but the state bank has weaker substitution of endowed funds for wholesale lending than nonstate banks. This point is crucial because it implies that the state bank's conservatism prevents full reallocation of funds when the state bank lends on the wholesale funding market, but not when it borrows.

Specifically, RRR cuts increase banks' endowed funds proportional to their regular deposits: $-\partial e_j / \partial \rho = d_j$. Hence, RRR cuts result in a higher substitution of regular deposits for wholesale funding.

Implication 2. *After RRR cuts, wholesale borrowing is more negatively correlated with regular deposits across banks, whereas wholesale lending is more positively correlated.*

The last implication concerns banks' loans. In Scenario 1, the state bank holds ample funds, and also originates more loans relative to nonstate banks, with the state bank's marginal return of loans being lower than nonstate banks'. In Scenario 2, funds are fully reallocated to the point that

all banks' marginal returns of loans are equal. As a result, when the banking system shifts from Scenario 1 to Scenario 2, nonstate banks originate more loans compared with the state bank.

Implication 3. *When the banking system shifts from Scenario 1 to Scenario 2, the change in nonstate banks' loans, $k_i^2 - k_i^1$, is greater than the change in state banks' loans, $k_s^2 - k_s^1$.*

On the other hand, according to Proposition 1, the state bank originates a higher fraction of loans to SOEs than nonstate banks. Therefore, nonstate banks making more loans relative to the state bank implies that non-SOEs receive more loans than SOEs.

Implication 4. *When the banking system shifts from Scenario 1 to Scenario 2, the change in the loans made to non-SOEs is greater than the change in the loans made to SOEs.*

4 Data and Summary Statistics

We now detail our data sources and provide summary statistics.

4.1 Data sources

We construct a bank-level dataset at the quarterly frequency for a sample period from 2013Q4 to 2019Q4, which covers the period from the emergence of NCDs until the outbreak of COVID-19. Quarterly bank balance sheet and income statement data are from the Wind database. Due to regulation, publicly listed banks and banks that intend to issue bonds on the interbank markets are required to disclose audited financial statements at a regular frequency. These disclosures can be found on banks' official website or on the National Interbank Funding Center. Disclosed bank information includes total assets, total liabilities, central bank borrowing, various wholesale borrowing and lending, reserves, loans, financial investments, deposits, and return on assets (ROA). Besides constructing this quarterly bank-level dataset, we also obtain banks' monthly balance sheet information from *Sources and Uses of Credit Funds* published by the PBC. The PBC publishes the aggregate data for three categories of banks: nonstate banks, the Big 4 banks, and the Big 7 banks. The Big 7 banks include the 6 state banks and the biggest policy bank, China Development Bank.¹⁶

Data on NCD issuance are also from the Wind database, which collects the information from the National Interbank Funding Center. For each NCD issued, we have information on issue volume, issuer bank, issuer credit rating, issue rate, issue date, and term. These data allow us to calculate NCD outstanding balances at the bank level. We also obtain monthly NCD holdings and NCD outstanding balances aggregated at the bank-type level from the Shanghai Clearing House. However, the data on NCD holdings at the bank level are not available.

¹⁶See Section 2.2 for more details.

In terms of economy-wide data, money market interest rates such as Shibor and R007 are from the National Interbank Funding Center. The total assets under management of money market funds are collected from the Asset Management Association of China (AMAC). Data on firms listed in National Equities Exchange and Quotations (NEEQ) are from their annual reports. Data on firms listed in A share markets are from China Stock Market & Accounting Research (CSMAR).

4.2 Summary Statistics

Table 3 reports the summary statistics of banks' quarterly financial information. Banks are divided into three groups: state banks, nonstate PD banks and non-PD banks. The sizes of banks across the three groups differ significantly. State banks have an average asset holdings of RMB 17,346.7 billion, which is about 6 (132) times larger than that for nonstate PD banks (non-PD banks). We scale other financial variables by banks' total assets in the previous quarter. While Figure 2 shows that state banks have lower loan rates, they dominate the deposit market and offer lower deposit rates than other banks (Ma and He, 2020). Overall, state banks' average return on assets (ROA) is higher than that of nonstate PD banks and non-PD banks.

On the liability side, deposits are the most important funding source for all banks. State banks have the highest deposit-to-asset ratios (75.65%) on average, which is largely ascribed to their broad networks of branches and special political status. Non-PD banks have slightly lower deposits-to-asset ratios (74.95%), because many of them are rural and small urban banks that take deposits but have limited investment opportunities. In contrast, nonstate PD banks have substantially lower deposits-to-asset ratios (61.93%) on average, because they have ample investment opportunities that cannot be satisfied by their deposits. Central bank (CB) borrowing and wholesale borrowing are the other two important funding sources. Wholesale borrowing mainly consists of the three types of traditional wholesale borrowing (deposits from financial institutions, placements from financial institutions, and repos) and debt securities issued. Consistent with their lower deposits-to-asset ratios, nonstate PD banks rely much more heavily on wholesale borrowing than state banks and non-PD banks. NCD outstanding is part of debt securities issued and not reported separately in banks' financial reports. However, since NCD issuance is public, we can directly calculate each bank's amount of NCD outstanding. Across the three groups, nonstate PD banks have the highest relative level of NCD outstanding, which on average accounts for 8.54% of their total assets. This pattern suggests that nonstate PD banks engage most actively in borrowing through NCDs. State banks' NCD outstanding accounts for only 0.38% of their total assets on average, since state banks did not systematically issue NCDs until 2018 as Figure 3 shows. Among non-PD banks, urban commercial banks rely heavily on borrowing through NCDs, but rural commercial banks do not. On average, non-PDs' NCD outstanding accounts for about 5.31% of their total

assets.

On the asset side, bank loans and financial investments are the two dominant assets and together account for roughly 80% of total assets for all banks, whereas for state banks bank loans and for nonstate PD banks financial investment are more important. Within the category of loans, real estate loans constitute a relatively minor proportion, with the average loan to asset ratio ranging from 2.61% to 3.98% across the three types of banks. Conversely, the real estate sector plays a larger role within the shadow banking loans, which we will discuss in more detail in Section 7. Financial investments include investment in bonds, mutual funds, and account receivable investments, including NCDs. Since banks do not systematically report their NCD holdings, we cannot separate them from financial investments.¹⁷ In terms of the three types of traditional wholesale lending, they together account for around 8% of total assets for all three groups. Due to low interest rates on excess reserves, banks hold relatively small amounts of them, which on average account for less than 1% of their total assets. We calculate excess reserves as total reserves minus the product of deposits and the required reserve ratios.

To study the impact of the reallocation mechanism on bank lending, we conduct two types of analysis: within-firm and between-firm. For the within-firm analysis, we employ a sample of firms from the National Equities Exchange and Quotations (NEEQ) system, which is the third national securities trading venue in China, following the Shanghai Stock Exchange and Shenzhen Stock Exchange, and a platform specifically for small and medium-sized enterprises (SMEs). The advantage of using NEEQ firms is that they are required to fully disclose their borrowing activities, which allows for a comparison between different lending sources within a firm.¹⁸ Following the sample selection criteria discussed in Section 6.3, we construct a sample of 671 firms with 11,626 firm quarter observations. Total assets average 1.368 billion RMB. At the end of each quarter, firms are required to disclose all active loans with their respective start and end dates. This facilitates the precise calculation of the daily outstanding loan balance. Panel A of Table 4 reports the corresponding summary statistics. The average daily balance of loans is 0.5 billion RMB, whereas the average of loans from state and nonstate banks are 0.228 and 0.272 billion RMB, respectively. The average fraction of loans from nonstate banks is about 53%.

For the between-firm analysis, we employ firms listed in the entire A share markets. The sample comprises 3,394 firms, of which 1,013 are SOEs and the remainder are Non-SOEs. Panel B of Table 4 reports the summary statistics. SOEs have an average loan volume of 9.884 billion RMB, whereas non-SOEs have a lower average loan size of 1.949 billion RMB. This indicates that SOEs typically have greater financing needs or easier access to larger loans, which may be

¹⁷Only a few publicly listed banks report NCD holdings in their annual reports.

¹⁸The main board of the Chinese stock market does not require full disclosure of borrowing activities, so for firms listed there, we do not have access to a complete profile of their borrowing activities.

related to their larger scale. The average revenue for SOEs is 16.689 billion RMB, substantially higher than the 3.254 billion RMB for Non-SOEs. This difference likely reflects the dominant position of SOEs in key industries and sectors, as well as their larger market share. The ROA for SOEs averages at 2.354%, compared to 3.201% for Non-SOEs. This gap suggests that Non-SOEs may be more efficient in using their assets to generate profits. The quick ratio for SOEs averages at 117.964%, while for Non-SOEs it is 177.885%, suggesting that Non-SOEs may hold a larger proportion of liquid assets relative to their short-term liabilities, which may be related to the market risks they face. The operating cash flow ratio for SOEs averages at 8.351%, compared to 7.087% for Non-SOEs. The book-to-market ratio for SOEs averages at 70.252%, while for Non-SOEs it is 58.047%. This pattern suggests that the market valuation of SOEs is closer to their book values. The price-to-earnings ratio for SOEs averages at 93.718, while for Non-SOEs it is substantially higher at 157.525. This spread indicates that the stocks of Non-SOEs may be traded at a higher premium, likely due to market expectations for their future growth.

5 Reallocation Following Monetary Policy Interventions

As discussed above, China’s uneven monetary policy implementation may result in or exacerbate the funding imbalance within the banking system, and the transmission of monetary policy depends heavily on the reallocation of wholesale funding among banks. This section characterizes how the reallocation operates by testing the first two implications of Section 3.4. In Section 5.1 and Section 5.2, we exploit MLF to examine the substitution of endowed funds for wholesale funding, which is central in the transmission of quantity-based monetary policy. In Section 5.3, we show the reallocation of wholesale funding following RRR cuts.

5.1 MLF: Substitution for wholesale funding

This subsection estimates banks’ substitution of endowed funds for various wholesale funding terms using quarterly bank-level data. Through this estimation, we can gauge the importance of different markets in the reallocation of wholesale funding. Implication 1 suggests that an increase in endowed funds leads to a decrease in wholesale borrowing or an increase in wholesale lending. To bring this prediction to the data, we examine the impact of the change in central bank borrowing. Since deposits are the most important part of banks’ endowed funds, we directly control the changes in deposits and the required reserve ratio. Specifically, we estimate the following model for all banks:

$$\Delta Y_{i,t} = \beta_1 \Delta CBBorrow_{i,t} + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \alpha_i + \lambda_t + u_{i,t}. \quad (13)$$

The dependent variable $\Delta Y_{i,t}$ is the quarterly change in a wholesale funding term at time t , and $\Delta CBBorrow_{i,t}$ is the quarterly change in central bank borrowing. We include the following variables as controls:

- lagged central bank borrowing $CBBorrow_{i,t-1}$;
- bank controls $D_{i,t} = \{Deposit_{i,t-1}, \Delta Deposit_{i,t}, \Delta RRR_t \times Deposit_{i,t}, XReserve_{i,t-1}\}$, where $Deposit_{i,t-1}$ is lagged deposits, $\Delta Deposit_{i,t}$ is the quarterly change in deposits, and ΔRRR_t is the quarterly change in the required reserve ratio, $XReserve_{i,t-1}$ is lagged excess reserves;
- fundamental controls $X_{i,t} = \{PD_{i,t}, \Delta r_t \times SB_i, \Delta r_t \times PD_{i,t}\}$, where SB_i equals one if bank i is a state bank, $PD_{i,t}$ equals one if bank i is a PD bank at time t , Δr_t is the change in the interbank rate measured by the 3-month Shibor rate; and
- bank fixed effects and quarter fixed effects.

Since NCD was introduced recently and banks adopted it gradually over time, for each bank, we include the quarters since the first time that its NCD outstanding is greater than both 1 billion RMB and 0.1% of its total assets. We cluster standard errors by bank. All balance sheet terms, whether they are lagged or quarterly changes, are scaled by lagged total assets.

Two challenges identifying the substitution effect arise in the above regression. First, a bank's central bank borrowing may be affected by its idiosyncratic demand for funds. Second, a bank's central bank borrowing is affected by monetary policy interventions, which are endogenous to economic condition. To address the endogeneity issue, we include time fixed effect and rely on the cross-sectional variation of the change in wholesale funding terms and the change in central bank borrowing. Specifically, we instrument $\Delta CBBorrow_{i,t}$ by the interaction of ΔMLF_t and $PD_{i,t}$, $\Delta MLF_t \times PD_{i,t}$, as well as the interaction of ΔMLF_t and SB_i , $\Delta MLF_t \times SB_i$, where ΔMLF_t is the quarterly change in the aggregate liquidity supply by MLF.¹⁹ The instruments are motivated by two observations. First, the central bank actively controls the quantity of MLF, as discussed in Section 2.1. Second, MLF supplies funds mostly to PD banks, so only their central bank borrowing is substantially affected by MLF. Potentially, MLF supplies funds to state banks and nonstate PD banks in different ways, which can be accommodated by $\Delta MLF_t \times SB_i$. Therefore, our identification exploits the cross-sectional variation of the change in banks' endowed funds caused by MLF and investigates how the cross-sectional variation of the changes in banks' wholesale funding follows.

The specification is similar in spirit to a difference-in-differences framework or a Bartik instrument (Autor et al., 2013; Cong et al., 2019; Goldsmith-Pinkham et al., 2020; Breuer, 2021;

¹⁹State banks are all PDs.

Borusyak et al., 2022), where (1) MLF causes a shift in the aggregate liquidity supply, and (2) PD banks are directly exposed to this shift but non-PD banks are not. As emphasized by Goldsmith-Pinkham et al. (2020), the Bartik instrument is equivalent to using the exposure shares as instruments, with the aggregate shock contributing only to instrument relevance. In the ideal case, identification therefore relies on the exogeneity of the exposure shares—that is, PD_i must be uncorrelated with the error term after conditioning on controls and fixed effects.

However, PD status is likely correlated with unobserved bank characteristics. When the shares are potentially endogenous in this way, Borusyak et al. (2022) show that identification can instead rest on the quasi-random assignment of the aggregate shocks. In our setting, this requires that the PBC’s quarterly MLF injections are uncorrelated with the idiosyncratic wholesale-funding demand shocks of PD banks relative to non-PD banks, after conditioning on controls.

We attempt to address this requirement through our fixed-effect structure. First, banks’ PD statuses are mostly constant and pre-determined, and do not vary with their idiosyncratic demand for funds. This stability ensures that bank fixed effects absorb any persistent, time-invariant heterogeneity between PD and non-PD banks, including differences in liquidity-demand elasticities, macro-prudential regulatory constraints, and wholesale-market access. Second, the shift is common and not dependent on the idiosyncrasies of particular banks. According to the PBC’s description, MLF is used to supply medium-term base money to “maintain the overall stability and moderate growth of bank system liquidity and support reasonable growth of monetary credit.” The aggregate nature of the shock implies that quarter fixed effects absorb any aggregate monetary-policy stance or signaling that might coincide with MLF operations and affect all banks equally. What remains identified is therefore the within-bank response to changes in MLF availability, conditional on the bank’s predetermined PD status and the common quarterly macroeconomic environment. Furthermore, to address the concern that time-varying, PD-specific non-funding channels might confound our estimates, we include leverage, capital adequacy ratio (CAR), loans-to-deposits ratio (LDR), and log assets as controls.

Columns (1) and (2) of Table 5 report the results of model (13) when the dependent variable is NCD outstanding. The OLS estimate of β_1 is not significantly different from 0, whereas the 2SLS estimate is close to -1 and significantly negative at the 5% level, consistent with Implication 1. The F-statistic in the first stage of 2SLS estimation is 20.82, suggesting that weak instruments are not a concern. As mentioned earlier, the OLS estimate could potentially be biased due to unobservable idiosyncrasies of banks. For example, banks’ idiosyncratic demand for funds moves NCD outstanding and central bank borrowing in the same direction and thus attenuates the negative correlation induced by the substitution effect. The Bartik instrument ameliorates this concern, allowing us to uncover the substitution effect.

In Columns (9) and (10), the OLS estimate regarding repo is significantly negative at the 10%

level, and its 2SLS estimate is similar but becomes insignificant, suggesting that weak substitution of central bank borrowing for repo may exist.²⁰ For each of the other seven wholesale funding terms, neither of their OLS and 2SLS estimates are significant, consistent with the uncompetitive nature of traditional wholesale funding markets discussed in Section 2.3. Taken together, the results in Tables 5 suggest that banks respond to the liquidity supply by MLF mainly through NCDs, whereas other wholesale funding markets play minor roles. The estimated magnitude of the substitution of central bank borrowing for NCD outstanding is close to the model prediction of -1 .

Remark 5. As discussed in Section 2.2, whether a bank is a PD is determined by the PBC, and this status may change over time. A potential concern is that the endogeneity of PD designation could bias the estimation. However, it is worth noting that the 31 largest banks—which constitute the majority of PD banks in any given year—maintained their PD status consistently throughout our sample period. Thus, we do not expect fluctuations in PD status to significantly influence our results. In fact, when we fix banks’ PD statuses to their 2013 classifications, the estimates remain similar in magnitudes and statistically significant.

A natural concern with these results is that the PD and state-bank dummies may stand in for bank characteristics such as leverage, capital adequacy, size, or the loan-to-deposit ratio. We address this concern while keeping the structure of the baseline specifications, first at the bank level and then at the bank-type level. At the bank level, we re-estimate the substitution regressions with lagged leverage, capital adequacy ratio (CAR), loans-to-deposits ratio (LDR), and log assets added as controls. In Table 6, which augments Table 5, the 2SLS estimate on $\Delta CBBorrow_{i,t}$ moves from -1.339 to -1.375 when the four controls are added and remains significant at the 5% level.

5.2 MLF: Difference in substitution for wholesale funding

This subsection tests the implication regarding the difference in substitution for wholesale funding between state and nonstate banks. According to Implication 1, state banks and nonstate banks have the same substitution of endowed funds for wholesale borrowing, but state banks have weaker substitution of endowed funds for wholesale lending than nonstate banks. Since the previous section has shown that banks respond to the liquidity supply by MLF mainly through NCDs, we focus on the substitution for NCD outstanding and holdings.

We first analyze the quarterly bank-level data, which allow us to test the difference in substitution for NCD outstanding. To capture the difference in substitution, we add the interaction of the change in central bank borrowing and the state dummy variable, $\Delta CBBorrow_{i,t} \times SB_i$, to model (13) as follows:

²⁰Although the repo market is competitive, its maturities are predominantly short-term and therefore misaligned with the duration of MLF operations, which helps explain the weak substitution effects.

$$\begin{aligned}\Delta Y_{i,t} = & \beta_0 \Delta CBBorrow_{i,t} \times SB_i + \beta_1 \Delta CBBorrow_{i,t} \\ & + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \alpha_i + \lambda_t + u_{i,t}.\end{aligned}\quad (14)$$

We consider two specifications. First, similar to the way we estimate model (13), we use the sample of all banks and instrument $\Delta CBBorrow_{i,t} \times SB_i$ and $\Delta CBBorrow_{i,t}$ by $\Delta MLF_t \times PD_{i,t}$ and $\Delta MLF_t \times SB_i$ in the 2SLS estimation. Second, we use the sample of all PD banks and instrument $\Delta CBBorrow_{i,t} \times SB_i$ by $\Delta MLF_t \times SB_i$ because $\Delta MLF_t \times PD_{i,t}$ is a quarterly time series and absorbed by quarter fixed effects. Note in the first specification, the 2SLS estimates of the difference in substitution and of the level of substitution are both identified, but for the second specification, only the difference in substitution is. The reason is that the first specification can exploit the cross-sectional variation between PD banks and non-PD banks to estimate the level of substitution, and that variation is absent in the second specification. We again cluster standard errors at the bank level. Table 7 shows that when the dependent variable is NCD outstanding, neither of the OLS and 2SLS estimates of β_0 are significantly different from 0, regardless of the specification. Panels A and B of Table 7 report the two specifications, with Columns (1) and (2) reproducing the baseline estimates and Columns (3) and (4) adding the lagged bank characteristics as level controls. The interaction results are likewise unchanged by the controls. The results support the model prediction that the substitution of central bank borrowing for NCD outstanding does not differ significantly between state and nonstate banks.

To test the difference in substitution for NCD holdings, due to the fact that NCD holding is not disclosed in banks' statements, we use the monthly NCD holdings published by the Shanghai Clearing House. The data are available at the monthly frequency starting in October 2014. Shanghai Clearing House categorizes commercial banks into the Big 6 banks, joint-stock banks, urban commercial banks, rural commercial banks, and other banks, and reports the aggregate NCD holdings of each category. We obtain banks' monthly balance sheet information from *Sources and Uses of Credit Funds* published by the PBC. The PBC publishes the aggregate data for three categories of banks: nonstate banks, the Big 4 banks, and the Big 7 banks. The Big 7 banks include the 6 state banks and the biggest policy bank, China Development Bank.²¹

Merging the two sets at the monthly level, we construct panel data with two bank types: state banks and nonstate banks. For nonstate banks, NCD holdings and balance sheet information are exactly matched. For state banks, we match the Big 6 banks' NCD holdings with either the Big 4 banks' balance sheet information or the Big 7 banks'. Since the only cross-sectional variation we can employ is between the two bank types, we estimate model (14) with the dependent vari-

²¹See Section 2.2 for more details.

able being NCD holdings using the second specification. That means, only the coefficient of the difference in substitution (β_0) is properly identified in the 2SLS estimation, but that of the level of substitution (β_1) is not. We scale all balance sheet items by lagged total funds of the corresponding bank type, and scale ΔMLF_t by lagged overall total funds of both types because the change in MLF is shared by all banks. Since there are only two bank types, we use Huber-White robust standard errors.

We think the Big 4 banks are a better approximation of the Big 6 banks than the Big 7 banks. China Development Bank is the biggest policy bank and considered to be China's second Ministry of Finance. Its objectives and operations are quite different from commercial banks. Particularly, its central bank borrowing contains a large amount of funds granted through facilities that are not available to commercial banks, such as Pledged Supplementary Lending (PSL). As discussed in Section 2.1, PSL is designed to fund policy banks for shanty-town renovations, and its size is comparable to all funds supplied by MLF. Given its special objective and large size, PSL makes policy banks' central bank borrowing fluctuate in a different way than the Big 6 banks'. On the other hand, we can confirm that the Big 4 banks are indeed a good approximation using the quarterly bank-level data. For the sample period from 2014Q4 to 2019Q4, we calculate the Big 6 banks' quarterly changes in central bank borrowing scaled by their lagged total assets and those of the Big 4 banks, and then regress the former on the latter. Figure 6 shows the resulting scatter plot and the fitted OLS regressions line. The OLS estimates of the slope and the intercept are almost 0.99 and 0, respectively, and the R-squared is 0.97, suggesting that the two time series are almost identical. With this observation, we expect that using the Big 4 banks' information does not cause a substantial bias when estimating the difference in substitution.

Table 8 reports the estimates using the Big 4 banks' information. Again, weak instruments are not a concern. Panel A shows the estimates of the substitution for NCD holdings. In Columns (1) and (2), the 2SLS estimate is significantly different from 0 at the 1% level and suggests that state banks' substitution for NCD holdings is weaker than nonstate banks' by 38%. As a comparison, we use the Big 4 banks' information and the same specification to estimate the difference in the substitution for NCD outstanding. Panel B shows the estimates for the Big 6 banks' NCD outstanding, which is affected by the same mismatching in data, and Panel C shows the estimates for the Big 4 banks' NCD outstanding, which is not. In Columns (1) and (2) of both panels, all the estimates are not significantly different from 0, which is consistent with the estimation using the quarterly bank-level data. Moreover, the point estimates are similar, which is consistent with our expectation that the imperfect overlap does not cause a substantial bias.

At the bank-type level, we re-estimate the three panels with the lagged bank-type leverage, log aggregate assets, and LDR added as level controls. CAR is not included because it cannot be constructed using original data at the bank-type level in the way we construct other bank-type

controls. In Table 8, Panel A, the 2SLS coefficient on $\Delta CBBorrow_{i,t} \times SB_i$ for the Big 6's NCD holdings goes from -0.382 to -0.299 and stays significant at the 5% level. The issuance-side results in Panels B and C remain insignificant. Taken together, we find that state banks' substitution for NCD holdings is weaker than nonstate banks', but their substitution for NCD outstanding is not significantly different. This finding is consistent with the model assumption that state banks are conservative in lending, but not in borrowing.

5.3 RRR cuts

In this subsection, we characterize the reallocation of funds following RRR cuts. In our sample period, two waves of RRR cuts occurred, one starting in 2015Q1 and the other starting in 2018Q2. We focus on the second wave, since the NCD market was newly established and not yet fully developed around 2015. As Figure 3 shows, by the end of 2015Q3, the size of the NCD market was about 2.5 trillion RMB, which was quite small compared to its subsequent levels. In addition, many banks were granted the permission to issue NCDs only during 2014 and 2015, and learning how to use NCDs efficiently might also take time. According to Implication 2, after RRR cuts, wholesale borrowing (wholesale lending) will be more negatively (positively) correlated with deposits because of a stronger substitution of deposits for wholesale funding. The simplest way to capture the cross-sectional relationship between wholesale funding and deposits is to regress the former on the latter period by period. However, the estimation may be biased since deposits are not fully exogenous. To mitigate the concern, we use the deposits in a predetermined period instead of the deposits in the current period. Note that deposits are the dominant part of Chinese banks' liabilities. Even a small percentage change in deposits can cause substantial changes in wholesale funding terms. Hence, we also add the difference between the deposits in the current period and the deposits in the predetermined period as a separate regressor to control for the impact of the change in deposits over time. We fix the period to 2018Q1, the quarter right before the second wave of RRR cuts. When we use an alternative period, results effectively remain unchanged.²²

Specifically, we estimate the following model for each quarter around the start of the second wave of RRR cuts:

$$Y_{i,t} = \beta_1 Deposit_{i,2018Q1} + \beta_2 (Deposit_{i,t} - Deposit_{i,2018Q1}) + \beta_3 CBBorrow_{i,t} + \lambda_t + \gamma X_{i,t} + u_{i,t}. \quad (15)$$

Here, the dependent variable $Y_{i,t}$ is a wholesale funding term and $CBBorrow_{i,t}$ controls for the impact of central bank borrowing. Fundamental controls are the same as in equation (13). Since

²²Fixing to earlier periods causes two problems. First, the deposits used in the estimation deviate more from the actual deposits that determine the impact of the RRR cuts on a bank. Second, the sample size will be smaller, as fewer banks report deposits in earlier periods.

we use the quarter before the start of the wave, 2018Q1, as the benchmark to assess the subsequent change, we include only banks with observations in that quarter. We cluster standard errors by banks.

Labeling 2018Q2 as quarter 0, Figure 7 plots the estimates of β_1 for NCD outstanding and their 95% confidence intervals from the eighth quarter before the start of the wave to the end of 2019. The estimates of β_1 are close to 0 before the start of the wave. With the onset of the wave of RRR cuts, the estimated coefficients become consistently more negative, and the difference is significant or nearly significant at the 5% level. This pattern confirms Implication 2. Online Appendix C.3 shows the figures of β_1 for other wholesale funding terms. Repo also displays a similar pattern in that the estimates of β_1 are stable before the start of the wave and become more negative afterwards but the change is much weaker economically and its significance is sparse. The other seven wholesale funding terms do not follow a clear pattern, again consistent with Implication 2. Similar to the earlier findings regarding the reallocation following MLF in Section (5.1), the results here suggest that banks respond to the liquidity supply by RRR cuts mainly through NCDs, whereas other wholesale funding markets play minor roles.

6 Banks' Lending to the Real Economy

In the previous section, we provide empirical evidence for the reallocation mechanism proposed by the model. According to the model, in Scenario 1, in which state banks are endowed with ample funds and lend on the wholesale funding market, their conservatism impedes the reallocation of funds to nonstate banks, whereas in Scenario 2, in which state banks are endowed with limited funds and borrow on the wholesale funding market, their conservatism plays no role, and the efficiency of the reallocation mechanism improves. Next, we assess the effects of the reallocation mechanism on banks' lending to the real economy.

Figure 3 shows that state banks began to systematically issue NCDs in 2018 and have become net issuers since 2019, suggesting that they were moving from the lending side of the wholesale market to the borrowing side. Figure 4 indicates that the high interest rate and the resulting boom in MMFs in 2017 and 2018 dampened state banks' deposit growth and could be a driving force behind this change.²³ Mapping this shift to our model, we interpret the periods before and since 2018 as Scenarios 1 and 2, respectively. Our empirical strategy is then to test Implication 3 by comparing the two subsamples before and since 2018. Other factors might possibly drive the difference in the two subsamples, which we will address in Section 7.

²³Buchak et al. (2021) provide causal evidence that MMFs in China attract households' deposits from banks by offering more competitive interest rates.

6.1 The macro pattern of bank lending

Implication 3 states that moving from Scenario 1 to Scenario 2, nonstate banks witness more growth in lending than state banks. We first check whether it is consistent with the macro pattern of bank lending using banks' monthly *Sources and Uses of Credit Funds* published by the PBC. We focus on the period from 2015 to 2019, because the relevant data are available from 2015 and the Covid-19 pandemic broke out at the beginning of 2020.

Panel A of Figure 8 plots the cumulative arithmetic growth of loans using the level of loans in 2017m12 as the benchmark. Specifically, the time series of the level of loans are scaled by the level in 2017m12, and normalized to 0 in 2017m12. The plot shows that nonstate banks exhibited more growth than state banks throughout the whole period. To control for the pretrends, we use OLS models to estimate the average monthly growth rates before 2018. We then use the fitted slope to predict the trends since 2018, which are the dashed lines in Panel A of Figure 8, and plot the detrended time series in Panel B of Figure 8. We can see that nonstate banks' cumulative growth of loans was persistently more above its prior trend than state banks' since 2018.

On the other hand, to see whether the higher growth of loans is due to the improved reallocation mechanism, we also want to control for the variation in banks' endowed funds. We calculate endowed funds, which are held fixed in our model, as the sum of central bank borrowing and deposits in excess of required reserves, where deposits include only individual deposits and corporate deposits, but not deposits of financial institutions, since they are essentially wholesale borrowing. Endowed funds are relatively more exogenous than other sources of funds that banks can obtain from other financial institutions and the broad financial market. Similar to loans, we plot the cumulative growth of endowed funds and its detrended time series in Panels C and D of Figure 8. Unlike loans, nonstate banks' cumulative growth of endowed funds was relatively below its trend compared to state banks' in the two years since 2018. This finding suggests that nonstate banks' higher cumulative growth of loans was probably not backed by endowed funds but backed by other sources. This pattern is consistent with our arguments that the reallocation of funds improved since 2018 and led to higher lending by nonstate banks.

6.2 Bank-level lending

Next, we formally test Implication 3 using granular bank-level data in a DiD framework. We use yearly data for this test for two reasons. First, a large fraction of banks are not publicly listed and have complete observations only at an annual frequency.²⁴ Second, the sector-level breakdown of bank loans is available only in annual reports.²⁵ For consistency, we still focus on the period from

²⁴Only a few big banks are public firms and consistently release quarterly reports.

²⁵Section 7 analyzes sector-level bank loans to rule out alternative explanations.

2015 to 2019.²⁶ Specifically, we estimate the following model:

$$\log(Lending_{i,t}) = \beta_0 After_t \times NSB_i + \beta_1 \log(Endowed_{i,t}) + \alpha_i + \lambda_t + u_{i,t}. \quad (16)$$

Here, the dependent variable $\log(Lending_{i,t})$ is the log of bank i 's loans in year t , $After_t$ is a dummy variable that equals one since the year 2018, NSB_i is a dummy variable that equals one if a bank is a nonstate bank, and $\log(Endowed_{i,t})$ is the log of bank i 's endowed funds in year t . We also control bank fixed effects and year fixed effects. We cluster standard errors by banks. Implication 3 predicts the coefficient β_0 is positive.

To test the parallel trend assumption which we require for identification, we replace $After_t$ in equation (16) with year dummies and use the year 2017 as benchmark. Panel A of Figure 9 reports the estimates of β_0 when $\log(Endowed_{i,t})$ is not included as a regressor, and Panel B reports those when it is included. In both panels, we see that in the three years leading up to 2018, the difference in lending between state and nonstate banks is stable, but since 2018, nonstate banks have experienced significantly higher growth. With the variation of endowed funds controlled, the estimated increase in growth since 2018 is smaller but still significantly positive. Panel A of Table 9 reports estimated coefficients of equation (16). Column (2) suggests that since 2018, nonstate banks' lending growth increased by 9.1% more than state banks', and the increase is significant at the 1% level.

Substantial heterogeneity across banks is present. Joint-stock banks and urban commercial banks rely more on wholesale funding markets to obtain funds than rural commercial banks. Figure 3 shows that they are dominant borrowers on the NCD market, whereas rural commercial banks as a whole are more likely to be on the lending side of the market. We expect that joint-stock banks and urban commercial banks are affected more by the improved reallocation mechanism and experience higher growth than rural commercial banks. Columns (3) to (8) show the difference in lending between state banks and different types of nonstate banks. The estimates of β_0 are significantly positive at the 1% level for joint-stock banks and urban commercial banks but not significantly different from 0 for rural commercial banks.

6.3 Within-firm borrowing analysis

Thirdly, we provide firm-level evidence suggesting that the reallocation mechanism affects lending to the real economy. A corollary of Implication 3 is that firms that borrow both from state and nonstate banks should receive higher fractions of lending from nonstate banks relative to state banks since 2018. We test this prediction using the sample of firms listed in the NEEQ system. NEEQ firms are required to fully disclose their borrowing activities, which allows for a comparison

²⁶Our results remain unchanged if we extend the sample to earlier periods.

between different borrowing sources within a firm.²⁷ By comparing loans from state and nonstate banks to the same firm, we are conducting a within-firm borrowing analysis. The advantage of the within-firm analysis is that it naturally controls for firms' demand for loans and focuses on loan supply.

We use a firm's average daily balance (ADB) of bank loans to measure its overall borrowing in a period. To measure loan composition, we calculate the fraction of the average daily balance of loans from nonstate banks to that from nonstate and state banks as follows:

$$NSBFraction_{i,t} \triangleq \frac{\text{firm } i\text{'s ADB of loans from NSBs in period } t}{\text{firm } i\text{'s ADB of loans from NSBs and SBs in period } t}. \quad (17)$$

We consider this fraction of borrowing from nonstate banks a desirable measure for three reasons. First, it captures the relative difference between borrowing from nonstate banks and that from state banks. A higher fraction actually indicates more borrowing from nonstate banks relative to state banks but not more borrowing in the absolute sense. Second, it naturally controls for differences across firms such as their respective size. Hence, we do not need to scale borrowing by variables like total assets or sales, which may not adequately control for the size heterogeneity of firms across different industries. Third, it is a measure of loan source distribution within a firm other than the level of the firm's total loans. Therefore, it is immune to the shocks related to demand for loans such as business conditions and investment opportunities.

Similar to Section 6.2, we need to control for variation in the overall endowed funds of state and nonstate banks. To control for this effect, we use the aggregate data. We include $NSBEndowedFraction_t$ as a control, which we define as follows:

$$NSBEndowedFraction_t \triangleq \frac{\text{the average endowed funds of all NSBs in period } t}{\text{the average endowed funds of all NSBs and the Big 4 banks in period } t}. \quad (18)$$

Here $NSBEndowedFraction_t$ is the fraction of the average endowed funds of all nonstate banks to those of all nonstate banks and the Big 4 banks, calculated using banks' monthly *Sources and Uses of Credit Funds* published by the PBC.

For consistency, we still focus on the period from 2015 to 2019. Since all NEEQ firms release quarterly reports, we use quarterly data. We restrict the sample to firms whose ADBs of loans from state banks and nonstate banks both exceed 25 million RMB in at least one quarter by the end of 2016. We impose this restriction for two reasons. First, the loan composition of firms that actively borrow from both state and nonstate banks is a more sensitive indicator for the change in loan supply from state and nonstate banks than that of other firms. Second, including only

²⁷ The main board of Chinese stock market does not require full disclosure of borrowing activities, so for firms listed there, we do not have access to a complete profile of their borrowing activities.

firms with large amounts of loans helps avoid the influence of policies targeting small and micro business (SMB). In our sample period, the PBC had various policies to facilitate lending to SMB.²⁸ Meanwhile, news reports suggest that to control their risk exposure to SMBs, banks reduced loans to small and medium-sized enterprises that did not qualify as SMB loans. The selected firms' total ADB accounts for 50% of all firms' ADB. Two concerns led us to pick the year 2016 as part of the sample selection criterion. First, we want to pick a year before 2017. The selection criterion may affect the lending dynamics of the selected firms we observe. We want to have several quarters of observations by the end of 2017 that are not affected by the selection criterion to characterize the trend before the shift. Second, picking an earlier year will lead to a smaller number of selected firms. Picking the year 2016 achieves a balance between the two considerations.

We estimate the following model:

$$NSBFraction_{i,t} = \beta_0 After_t + \beta_1 NSBEndowedFraction_t + \alpha_i + u_{i,t}. \quad (19)$$

We include firm fixed effects and cluster standard errors at the firm level. To test for parallel trends, we control for $NSBEndowedFraction_t$ and show the time trend of β_0 . However, since $NSBEndowedFraction_t$ is a time series, it will be absorbed when we add quarter fixed effects. Hence, we replace $After_t$ in model (19) with a year dummy for quarters in 2015 and quarter dummies for later quarters, and use the quarter 2017Q4 as the omitted category. Panel A of Figure 10 reports the estimates of β_0 when $NSBEndowedFraction_t$ is not included as a regressor, and Panel B reports those when it is included. Controlling for endowed funds, we see that in the three years leading up to 2018, $NSBFraction_{i,t}$ was stable, but that it has increased significantly since 2018.

Columns (1) and (2) of Table 10 shows the estimation results of equation (19). Compared with Column (1), the estimates of β_0 in Column (2) is smaller because we control for endowed funds, but it is still positive and significant at the 1% level. The point estimate suggests that the fraction of loans from nonstate banks has increased by 2.7% since 2018, which is consistent with Implication 3. To facilitate comparison with estimates from other analyses in this section, we do a back-of-the-envelope calculation of the difference between nonstate banks' and state banks' loan growth around 2018 based on this estimate as follows:²⁹

$$\frac{\Delta NSBFraction}{NSBFraction \cdot (1 - NSBFraction)}.$$

Given that the mean value of $NSBFraction$ before 2018 is 51.39% and the point estimate of $\Delta NSBFraction$ is 2.7%, the difference in loan growth would be 10.81%. This estimate of the

²⁸Please see Section 7.4 for more details.

²⁹Denote the loans from nonstate banks before 2018 as n and the loans from state banks before 2018 as s ; Δ here

difference in loan growth is close to that at the bank level (9.1%).

6.4 Between-firm borrowing analysis

While the loan-level data provide a means to test Implication 3, one caveat is that NEEQ is a relatively small market in the Chinese economy and may thus lack representativeness. To address this limitation, we examine firm-level loans using firms listed in the Chinese A share market. However, individual loan-level data for A share firms are generally not available in commercial databases, so it is not feasible to identify the borrowing from state and nonstate banks within the same firm. Alternatively, we test Implication 4, which states that when the banking system shifts from Scenario 1 to Scenario 2, the change in the loans made to non-SOEs is greater than the change in the loans made to SOEs. The model derives the implication based on the observation that the state bank makes a higher fraction of loans to SOEs than nonstate banks, which is supported by Cong et al. (2019).

To test Implication 4, we adopt a DiD framework to examine changes in the difference between bank loans made to SOEs and non-SOEs before and after 2018, using quarterly data from 2015Q1 to 2019Q4. Since A share firms do not directly disclosure their bank loans, we instead use the log of a firm's borrowings, $\log(Borrowing_{i,t})$, as the dependent variable. A firm's borrowings equal the sum of short-term borrowings, long-term borrowings, and the current portion of non-current liabilities. This borrowing term mainly consists of bank loans and non-bank lending and excludes the majority of bonds.³⁰ We think this variable tracks bank loans most closely among those that can be constructed using public data. In Online Appendix C.2, we provide evidence that non-bank lending accounts for only a small and declining share of this borrowing measure. We estimate the following model:

refers to the change before and after 2018. Then we have

$$\begin{aligned}
 \Delta \log(n) - \Delta \log(s) &\approx \frac{\Delta n}{n} - \frac{\Delta s}{s} \\
 &= \frac{s\Delta n - n\Delta s}{ns} \\
 &= \frac{(n+s)(n+s+\Delta n+\Delta s)}{ns} \cdot \frac{(s\Delta n - n\Delta s)}{(n+s)(n+s+\Delta n+\Delta s)} \\
 &= \frac{(n+s)}{n} \cdot \frac{(n+s+\Delta n+\Delta s)}{s} \cdot \left(\frac{n+\Delta n}{n+s+\Delta n+\Delta s} - \frac{n}{n+s} \right) \\
 &= \frac{(n+s)}{n} \cdot \frac{(n+s+\Delta n+\Delta s)}{s} \cdot \Delta NSBFraction \\
 &\approx \frac{\Delta NSBFraction}{NSBFraction \cdot (1 - NSBFraction)}.
 \end{aligned}$$

³⁰The majority of bonds are recorded as bonds payable or notes payable, but bonds with maturities less than one year are included in the current portion of non-current liabilities.

$$\begin{aligned}\log(\text{Borrowing}_{i,t}) = & \beta_0 \text{After}_t \times \text{NSE}_i + \beta_1 \log(\text{FirmTypeEndowed}_{i,t}) \\ & + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}.\end{aligned}\quad (20)$$

Here After_t is a dummy variable that equals one since the quarter 2018Q1, and NSE_i is a dummy variable that equals one if the firm is not a SOE. The variation in the overall endowed funds of the state and nonstate banks could also impact the loans made to SOEs and non-SOEs. To control for this effect, we include $\log(\text{FirmTypeEndowed}_{i,t})$ in the regression. For SOEs (non-SOEs), this variable corresponds to the log of all state (nonstate) banks' endowed funds.

As this regression is a comparison in borrowing between firms, we include a group of variables to control for firms' demand for financing as follows:

$$Z_{i,t-1} = \{\log(\text{Revenue}_{i,t-1}), \text{BM}_{i,t-1}, \text{PE}_{i,t-1}, \text{ROA}_{i,t-1}, \text{QuickRatio}_{i,t-1}, \text{OperCashRatio}_{i,t-1}\}.$$

Here, $\log(\text{Revenue}_{i,t-1})$ is the log of firm i 's revenue, controlling for the firm's scale and business condition, and $\text{BM}_{i,t-1}$ and $\text{PE}_{i,t-1}$ are firm i 's book-to-market ratio and price-to-earnings ratio respectively, controlling for the firm's growth opportunities. Finally, $\text{ROA}_{i,t-1}$ is firm i 's return on assets, $\text{QuickRatio}_{i,t-1}$ is firm i 's current assets excluding inventory divided by current liabilities, and $\text{OperCashRatio}_{i,t-1}$ is firm i 's cash flow from operating activities divided by current liabilities. They control for the firm's profitability and liquidity. We also control for firm fixed effects and quarter fixed effects and cluster standard errors at the firm level.

To test the parallel trend assumption, similar to the specification in model (19), we replace After_t in model (20) with a year dummy for quarters in 2015 and quarter dummies for later quarters, and use the quarter 2017Q4 as the omitted category. Panel A of Figure 11 reports the estimates of β_0 when $\log(\text{FirmTypeEndowed}_{i,t})$ is not included as a regressor, and Panel B reports those when it is included. The results exhibit a similar pattern to those on the sample of NEEQ firms. Controlling for endowed funds, we see that in the three years leading up to 2018, the difference in borrowings between non-SOEs and SOEs is stable and close to zero but it has increased significantly since 2018. Panel A of Table 11 shows the estimation results of model (20). Across various specifications, the estimates of β_0 remain significantly positive at the 1% level. The point estimate in Column (4) with full controls suggests that non-SOE's borrowings have increased 16.3% more than SOE's since 2018.

We next examine whether the additional credit flowing to non-SOEs was deployed into real investment. Specifically, we replace the dependent variable in model (20) with the log of fixed

assets as follows:

$$\log(FixedAssets_{i,t}) = \beta_0 After_t \times NSE_i + \beta_1 \log(FirmTypeEndowed_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}.$$

while maintaining the same control variables, firm and quarter fixed effects, and clustering as in the between-firm borrowing analysis above. Since fixed assets adjust to credit with a delay, we set the baseline treatment onset to 2018Q4 (three quarters after the reallocation shift), using 2018Q3 and 2019Q1 for robustness.

Figure 12 plots the estimates of the $NSE_i \times$ quarter interactions, with 2017Q4 as the reference period. The coefficients remain statistically insignificant through 2018Q2, confirming the absence of a pre-trend in non-SOEs' fixed assets prior to the reallocation. From 2018Q4 onward, the coefficients turn significantly positive and rise to approximately 0.12 by 2019Q4. Table 12 reports the estimation results. Panel A presents the baseline estimates (2018Q4). The coefficient on the interaction term is positive and significant, both with and without controls for endowed funds and lagged firm characteristics. Under the full specification in column (4), non-SOEs increase fixed assets by about 6% relative to SOEs following the reallocation. Panels B and C shift the onset to 2018Q3 and 2019Q1 and show similar results to Panel A. These results indicate that the credit reallocation toward non-SOEs documented above was not merely absorbed as financial slack but was channeled into higher real investment.

7 Alternative explanations

The analyses in the previous sections suggest that high interest rates and the boom in non-bank investment vehicles from 2017m1 to 2018m7 shifted state banks' positions on the NCD market from being on the lending side to being on the borrowing side of the market. Consequently, the gap between banks' shadow costs of funds closed, and nonstate banks financed more investments relative to state banks. Several regulatory changes and forces also occurred during this period, which could potentially affect our results. This section examines potential alternative explanations for our empirical findings. Our discussion suggests that they are unlikely to be the main driving force.

7.1 The real estate sector

In the past two decades, real estate has been the most important sector in the Chinese economy. A large fraction of bank loans are related to the real estate sector, so shocks to the credit demand of the sector and regulation related to the credit supply to the sector may cause substantial changes in banks' entire loan portfolios. Moreover, if state and nonstate banks have different exposure to the

real estate sector or undergo the regulatory pressures to a different extent, then their lending to the sector may diverge and drive the difference in bank loans we observe.

To alleviate the concern about the particular impact of the real estate sector, we re-conduct our lending analysis after excluding this sector from our analysis. We collect the breakdown of bank loans from banks' annual reports and redo our bank-level lending analysis on loans to firms outside the real-estate sector. Similar to Figure 9, Figure 13 plots the time trend of loans to non-real-estate sectors. Controlling for endowed funds, we see a statistically significant increase in loans since 2018. Panel B of Table 9 reports the estimation results of equation (16) with the dependent variable being the log of loans to firms outside the non-real-estate sector. All the estimates, whether for all banks or for a subset of banks, are close to those in Panel A.

The within-firm borrowing results are unlikely driven by the real estate sector because only one firm in the NEEQ sample belongs to the sector. For the between-firm borrowing analysis, we redo the analysis, excluding firms operating in real estate. Panel A of Figure 14 and Panel B of Table 11 report the results and align closely with those from the full sample in Section 6.4. Overall, these results suggest that the change in bank loans since 2018 is not solely driven by the real estate sector.

7.2 Shifts in firms' borrowings

In general, firms borrow externally through three channels: bank loans, non-bank lending, and credit markets. Therefore, a natural concern is that changes in the supply of other borrowings may affect firms' demand for bank loans. In 2018, an important regulatory change in the financial sector, new regulation on asset management (NRAM), might have caused such changes at large scales. The purpose of the new regulation is to control the rapidly growing systemic risk in the financial system induced by shadow banking. The regulation explicitly bans any form of guarantees embedded in the wealth management products and imposes clear restriction on the investment of wealth management companies in nonstandardized debt assets. Consequently, the aggressive growth of shadow banking products was reversed, and the investment in nonstandardized debt assets declined. Potentially, firms that previously relied on funds channeled through shadow banking were substantially affected by the regulatory change and might have sought traditional loans from commercial banks as substitutes. If these firms obtained more loans from nonstate banks than state banks, this shift from shadow banking to commercial banks could generate the lending results we document in the previous subsections.

To investigate this alternative explanation, we first examine the industry distribution of trust products, which are the main vehicle for facilitating shadow banking (Chen et al. 2018). The real estate sector receives the largest amount of investment from trust products. Allen et al. (2023) find

that 24–45% of trust products are invested in the real estate sector and point out that an important factor contributing to the dominance of this sector in shadow banking investment is the regulatory restriction on banks’ lending to the sector since 2010. Furthermore, more restrictive regulatory measures were issued in 2020 to control the real estate sector’s leverage ratio. Firms that fail to meet regulatory requirements are not allowed to increase debts. Due to the lending restriction, it is unlikely that firms in the real estate sector could easily replace their shadow banking loans with traditional bank loans. Also, the bank-level lending analysis in the previous section implies that the real estate sector does not drive our results.

Second, we confirm that the firm-level results are not driven by shifts in firms’ borrowings. Regarding the within-firm borrowing analysis, the credit market channel is unlikely to drive the results since only fewer than 7% of the firms in the NEEQ sample issued bonds. Indeed, if we exclude firms issuing bonds, the estimates presented in Panel B of Table 10 are almost the same as those in Panel A. In terms of the nonbank lending channel, if the shift from nonbank borrowing to bank loans drove the results, then the increase of nonstate banks’ lending relative to state banks’ would mainly exist for the firms relying on nonbank borrowing before 2018. By contrast, our story predicts that the increase exists for all firms. We can rely on this difference to distinguish between the two explanations. To this end, we split NEEQ firms into two subsamples: one consisting of firms with nonbank borrowing before 2018, which accounts for 49% of the sample, and another consisting of firms without. With endowed funds being controlled, Figure 15 shows that firms with nonbank borrowing and those without exhibit similar trends to Panel B of Figure 10, suggesting that the same pattern exists in both subsamples. To test the difference between the two subsamples statistically, we add $After_t \times Shadow_i$ to equation (19), where $Shadow_i$ equals one if the firm had nonbank borrowing before 2018. The estimated coefficient of the interaction term in Columns (3), (4), (7), and (8) are almost zero and not significant, consistent with the mechanisms we lay out rather than shifts in firms’ borrowings.

Regarding the between-firm borrowing analysis, note that we examine firms’ borrowings that include both bank loans and nonbank borrowing, so the results cannot be driven by the shift from nonbank borrowing to bank loans. The shift from credit markets to bank loans could be a serious concern, as the sample of A share firms contains large firms that frequently tap credit markets. Moreover, Geng and Pan (2024) demonstrate that NRAM indeed worsened non-SOEs’ credit conditions relative to SOEs in bond markets since 2018Q2. Due to the heightened borrowing costs in bond markets, non-SOEs have more incentive to shift to bank loans than SOEs. This mechanism could also drive the between-firm borrowing results. To exclude the impact of credit market dynamics, we restrict to the sample of A share firms without bond issuance and redo the analysis in Section 6.4. Panel B of Figure 14 and Panel C of Table 11 report the results and align closely with those from the full sample in Section 6.4. Thus, the between-firm borrowing results are not driven

by the shift from credits markets to bank loans.

Finally, while Figure 8, Panel A shows an absolute increase in state banks' loans after 2018, Panel B of the same figure shows that, once we control for pre-trends and endowed funds, it is nonstate banks' loans that rise persistently above trend relative to state banks'. The modest upward drift in state-bank loans is consistent with the overall expansionary monetary stance during this period, but the differential pattern, that nonstate banks grow faster despite no relative increase in their endowed funds, is what our reallocation mechanism explains and what NRAM cannot easily account for.

7.3 The regulation on NCDs

Since 2018Q1, regulators tightened regulations on NCDs. The new regulation requires that NCDs be counted as interbank liabilities, and the latter cannot exceed one-third of total liabilities. Since some nonstate banks have more interbank borrowing and are significantly closer to the threshold than state banks, an alternative story could be that 1) the regulation reduced nonstate banks' demand for NCDs, then due to the reduced demand, 2) the NCD issuing yield decreased, and finally 3) state banks started to issue NCDs to take advantage of the lower issuing cost.

This story suggests that the lower demand for NCDs induced by the regulation pushed down the NCD issuing yield. As shown in Panel A of Figure 4, the interest rates remained high from 2017m1 to 2018m7. The termination of high interest rates was largely due to a wave of RRR cuts starting in 2018m4. RRR cuts were considered to be the most powerful monetary policy instruments to inject liquidity to the financial system. This wave of cuts reduced the RRR by 3.5%. Actually, this was not the first time that RRR cuts were used in such a fashion. Figure 4 shows that a similar wave of RRR cuts in 2015 also dramatically reduced the interest rates. Therefore, the tightened regulation on NCD is unlikely to have led to the dramatic changes in the interest rates.

We do observe 1), 2), and 3) around 2018, but the timeline is not consistent with what this alternative story suggests. Panel A of Figure 4 shows that the decrease in the NCD issuing yield occurred in mid-2018. Panel C of Figure 3 instead shows that decrease in nonstate banks' demand for NCDs occurred in 2019,³¹ and state banks started to systematically issue NCDs at the beginning of 2018. That means, the actual timeline is that 1) was later than 2), and 2) was later than 3). Hence, it is hard to think that 1) caused 2) and 2) caused 3).

Finally, if the regulation induced state banks to issue NCDs by restraining nonstate banks from doing so, we should expect that since 2018, nonstate banks would have faced deteriorated funding conditions and lent less compared to state banks. This is at odds with our findings on banks' lending. Therefore, the regulation on NCDs was unlikely the main driving force behind our

³¹Nonstate banks' NCD issuance in 2018 was as high as that in 2017.

findings.

7.4 Lending to SMB

In our sample period, the Chinese government had various policies to facilitate lending to SMBs. For example, on June 1, 2018, the PBC expanded the scope of the collateral for MLF to include bonds of SMB with credit ratings not lower than AA and high-quality SMB bank loans.³² In the empirical analysis, we try to avoid the influence of this type of policies on our lending results. For the within-firm lending analysis, we remove SMBs that may be subject to these policies by focusing on firms with large borrowings from banks. For the between-firm lending analysis, the sample of A share firms naturally excludes SMBs. Hence, the firm-level results are not substantially affected by policies targeting SMBs. Still, we find that nonstate banks' lending grew faster than state banks'.

For the bank-level lending analysis, the data do not allow us to remove SMB loans or control for policy influence. However, SMB loans accounted for only a small fraction of bank loans. According to the *Chinese Small and Micro Business Financial Service Report* published by the PBC, as of the end of 2018, the balance of inclusive small and micro business loans was 8 trillion RMB; among them, the balance of the credit accounts with a credit limit of less than 5 million RMB was 1.8 trillion RMB; and the balance of all loans was 136 trillion RMB. Although SMB loans grew fast, it is unlikely that they drove the lending results.

8 Conclusion

We document several novel stylized facts about the conduct of monetary policy in China and the Chinese banking system. Informed by these facts, we build a simple model characterizing the reallocation mechanism following quantity-based monetary policy instruments through the Chinese banking system. State banks are conservative in lending to other banks, which impedes the reallocation of funds when state banks have ample resources and lend on wholesale funding markets.

Empirically, we test the model's implications using data for the period from 2013 to 2020. We show that while state and nonstate banks have a similar degree of substitution of central bank borrowing for wholesale borrowing, nonstate banks have a stronger substitution of central bank borrowing for wholesale lending. The substitution effect operates mainly through the NCDs instead of traditional wholesale transactions. Employing a shift in state banks' positions on the NCD market, we find that the reallocation mechanism has considerable effects on banks' asset allocations and lending to the real economy. Our analysis demonstrates that the endowment of financial

³²A bank's loan to a firm qualifies as a SMB loan if the firm has a total credit line smaller than 10 million RMB with the bank.

resources affects the distribution of resources across banks and ultimately in the real economy. To improve the final allocation of financial resources, monetary policy instruments should be designed to avoid and eliminate funding imbalances within the banking system. Facilitating financial liberalization, e.g. through MMFs, could be one step towards that goal.

References

- Acharya, V. V., R. S. Chauhan, R. Rajan, and S. Steffen (2023). Liquidity dependence and the waxing and waning of central bank balance sheets. Technical report. [1](#)
- Adrian, T., H. Hoyle, and F. Natalucci (2019). Patterns of trading in china’s bond market. In *The Future of China’s Bond Market*. International Monetary Fund. [2.3](#)
- Allen, F., X. Gu, C. W. Li, J. â. Qian, and Y. Qian (2023). Implicit guarantees and the rise of shadow banking: The case of trust products. *Journal of Financial Economics* 149(2), 115–141. [7.2](#)
- Allen, F., Y. Qian, G. Tu, and F. Yu (2019). Entrusted loans: A close look at China’s shadow banking system. *Journal of Financial Economics* 133(1), 18–41. [C.2](#)
- Altavilla, C., M. Boucinha, L. Burlon, M. Giannetti, and J. Schumacher (2022a). Central bank liquidity reallocation and bank lending: Evidence from the tiering system. [1](#)
- Altavilla, C., M. Boucinha, L. Burlon, M. Giannetti, and J. Schumacher (2022b). Money Markets and Bank Lending: Evidence from the Tiering Adoption. *SSRN Electronic Journal*. [1](#)
- Amstad, M. and Z. He (2020). CHINESE BOND MARKETS AND INTERBANK MARKET. In *The Handbook of China’s Financial System*. [2](#), [2.3](#)
- Autor, D. H., D. Dorn, and G. H. Hanson (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review* 103(6), 2121–2168. [1](#), [5.1](#)
- Bank for International Settlements (2019). Annual Economic Report. Technical report, Bank for International Settlements, Basel, Switzerland. [1](#)
- Bernanke, B. (1983). Nonmonetary effects of the financial crisis in the great depression. *American Economic Review* 73(3), 257–276. [1](#)
- Bernanke, B. and A. S. Blinder (1988). Credit , Money , and Aggregate Demand. *American Economic Review Papers and Proceedings* 78(2), 435–439. [1](#)

- Bernanke, B. and M. Gertler (1989). Agency Costs, Net Worth, and Business Fluctuations. *American Economic Review* 79(1), 14–31. [1](#)
- Bernanke, B. S. and A. S. Blinder (1992). The federal funds rate and the channels of monetary transmission. *American Economic Review* 82(4), 901–921. [1](#)
- Bianchi, J. and S. Bigio (2022). Banks, Liquidity Management, and Monetary Policy. *Econometrica* 90(1), 391–454. [1](#)
- Borusyak, K., P. Hull, and X. Jaravel (2022). Quasi-Experimental Shift-Share Research Designs. *Review of Economic Studies* 89(1), 181–213. [1](#), [5.1](#)
- Breuer, M. (2021). Bartik Instruments: An Applied Introduction. *SSRN Electronic Journal*. [1](#), [5.1](#)
- Brunnermeier, M. K. and Y. Sannikov (2014). A macroeconomic model with a financial sector. *American Economic Review* 104(2), 379–421. [1](#)
- Buchak, G., J. Hu, and S.-J. Wei (2021). Fintech as a Financial Liberator. *SSRN Electronic Journal*. [2.3](#), [23](#)
- Chen, K., H. Gao, P. Higgins, D. F. Waggoner, and T. Zha (2023). Monetary stimulus amidst the infrastructure investment spree: Evidence from china’s loan-level data. *The Journal of Finance* 78(2), 1147–1204. [4](#)
- Chen, K., J. Ren, and T. Zha (2018). The nexus of monetary policy and shadow banking in China. *American Economic Review* 108(12), 3891–3936. [4](#), [2.1](#), [2.2](#), [10](#), [2.4](#), [7.2](#), [C.1](#), [13](#)
- Chen, K., Y. Xiao, and T. Zha (2025). A Trade-off Between Monetary Policy Transmission and Systemic Risk in China. NBER Working Paper 34056, National Bureau of Economic Research. [1](#)
- Coimbra, N. and H. Rey (2024). Financial cycles with heterogeneous intermediaries. *The Review of Economic Studies* 91(2), 817–857. [1](#)
- Cong, L. W., H. Gao, J. Ponticelli, and X. Yang (2019). Credit Allocation under Economic Stimulus: Evidence from China. *Review of Financial Studies* 32(9), 3412–3460. [1](#), [1](#), [3.1](#), [5.1](#), [6.4](#)
- Copeland, A., D. Duffie, and Y. Yang (2021). Reserves were not so ample after all. Technical report. [1](#)
- D’Avernas, A. and Q. Vandeweyer (2023). Treasury bill shortages and the pricing of short-term assets. *The Journal of Finance*. [1](#)

- Deng, Y., R. Morck, J. Wu, and B. Yeung (2011). Monetary and Fiscal Stimuli, Ownership Structure, and China's Housing Market. NBER Working Paper 16871, National Bureau of Economic Research. 2.4
- Drechsler, I., A. Savov, and P. Schnabl (2017). The deposits channel of monetary policy. *Quarterly Journal of Economics* 132(4), 1819–1876. 3, 1, 2.3, 2.4, 3.1
- Du, W., K. Forbes, and M. N. Luzzetti (2024). Quantitative tightening around the globe: What have we learned? Technical report. 1
- Eisenschmidt, J., Y. Ma, and A. L. Zhang (2022). Monetary Policy Transmission in Segmented Markets. *SSRN Electronic Journal*. 1
- Fang, H., Y. Wang, and X. Wu (2020). The Collateral Channel of Monetary Policy: Evidence from China. *SSRN Electronic Journal*. 1
- Gao, B., X. Chen, and C. Li (2020). Banking property rights heterogeneity, shadow banking and the effectiveness of monetary policy. *Economic Research Journal* 55(4), 53–69. 2.4
- Geng, Z. and J. Pan (2024). The soe premium and government support in china's credit market. *The Journal of Finance* 79(5), 3041–3103. 1, 3.1, 7.2
- Gertler, M. and N. Kiyotaki (2010). Financial intermediation and credit policy in business cycle analysis. In *Handbook of Monetary Economics*, Volume 3, pp. 547–599. 1
- Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020). Bartik instruments: What, when, why, and how. *American Economic Review* 110(8), 2586–2624. 1, 5.1
- Hachem, K. and Z. Song (2021). Liquidity rules and credit booms. *Journal of Political Economy* 129(10), 2721–2765. 1, 2.2, 2.4
- He, Z. and A. Krishnamurthy (2013). Intermediary asset pricing. *American Economic Review* 103(2), 732–770. 1
- Huang, Y., T. Ge, and C. Wang (2020). MONETARY POLICY FRAMEWORK AND TRANSMISSION MECHANISMS. In *The Handbook of China's Financial System*. 2, 2.4
- Huang, Y., M. Pagano, and U. Panizza (2020). Local crowding-out in china. *The Journal of Finance* 75(6), 2855–2898. 1
- Kashyap, A. and J. Stein (1994). Monetary Policy and Bank Lending. In *Monetary Policy*, pp. 221–261. 1

- Kiyotaki, N. and J. Moore (1997). Credit Cycles. *Journal of Political Economy* 105(2), 211–248. [1](#)
- Li, X., Z. Liu, Y. Peng, and Z. Xu (2022). Bank Risk-Taking and Monetary Policy Transmission: Evidence from China. *Federal Reserve Bank of San Francisco, Working Paper Series*. [4](#)
- Ma, J. and X. He (2020). CHINA’S INTEREST RATE LIBERALIZATION. In *The Handbook of China’s Financial System*. [2.4](#), [4.2](#)
- Miranda-Agrippino, S., T. Nenova, and H. Rey (2026, jan). The Ins & Outs of Chinese Monetary Policy Transmission. NBER Working Paper 34626, National Bureau of Economic Research. [2.1](#), [10](#), [C.1](#)
- Scharfstein, D. and A. Sunderam (2016). Market Power in Mortgage Lending and the Transmission of Monetary Policy. *Working Paper*. [1](#)
- Sun, G. (2020). BANKING INSTITUTIONS AND BANKING REGULATIONS. In *The Handbook of China’s Financial System*. [2](#)
- Sun, G., J. Zhang, and X. Zhu (2021). Monetary Policy Transmission with Heterogeneous Banks and Firms : The Case of China. (72003102), 1–25. [4](#), [2.2](#)
- Vari, M. (2020, mar). Monetary Policy Transmission with Interbank Market Fragmentation. *Journal of Money, Credit and Banking* 52(2-3), 409–440. [1](#)
- Wang, T. (2020). MONETARY POLICY INSTRUMENTS. In *The Handbook of China’s Financial System*. [2](#)
- Wang, X. and Z. Jin (2021). Interbank Market Frictions, Bank Heterogeneity and the Transmission of Monetary Policy. *Economic Research Journal* 56(10), 56–71. [4](#), [2.2](#)
- Wang, Y., T. M. Whited, Y. Wu, and K. Xiao (2022). Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation. *Journal of Finance* 77(4), 2093–2141. [1](#)
- Xiao, K. (2020). Monetary transmission through shadow banks. *Review of Financial Studies* 33, 2379–2420. [1](#)

Appendix

A Tables and Figures

Table 2: PBC monetary policy instruments

This table lists the quantity based instruments used extensively by the PBC from 2013 to 2019. The purpose, liquidity receivers, terms, types of collaterals, and balance in September 2022 (in trillion RMB) of the instruments are also presented.

		Purpose	Liquidity receivers	Collateral	Balance as of 2022.9
MLF	Medium-term lending Facility	Supply base money over the medium term	banks that meet the requirements of macro-prudential management	High-quality bonds	4.55 tn
PSL	Pledged supplementary lending	Provide fund for rural and urban development	Policy banks	High-quality bonds and loans	2.65 tn
Re-lending		Support agriculture, small enterprises in the private sector, and loans related to green causes, technology innovation, pension and logistics	All banks		2.35 tn
Re-discount		Support commercial activities in important industries, especially agriculture, small and medium enterprises, and the private sector	All banks	Commercial drafts	0.54 tn
RRP	Open market operation through reverse repo	Provide short term liquidity for financial institutions	PDs	Treasury bonds	0.97 tn
CB FX holding	Central bank foreign exchange holding	Sustain FX reserve and exchange rate, provide RMB liquidity	PDs		21.31 tn
RRR	Required Reserve ratios in both RMB and FX	Supply base money through RRR cut	All banks		12.34 tn
SLF	Standing lending facility	Discount window liquidity upon request	All banks	High-quality bonds and loans	0 tn
TMLF	Targeted medium term lending facility	Provide medium -term liquidity to support private sector and SMEs	PDs		0 tn

Table 3: Descriptive statistics of banks

This table shows the summary statistics of banks' quarterly financial information from 2013Q4 to 2019Q4. Banks are categorized into three groups as state banks, Nonstate PD banks and non-PD banks. Assets are banks' total assets at the ends of quarters, and other balance sheet variables are scaled by assets. Banks' active liabilities include NCD outstanding, debt securities excluding NCD outstanding, three traditional wholesale borrowings (deposits from financial institutions, placement from financial institutions, repo), and central bank borrowing. On the asset side, excess reserves are calculated as total reserves minus the product of deposits and required reserve ratios. Three traditional wholesale lending items are deposits to financial institutions, placement to financial institutions, and reverse repo. Financial investment includes bonds, mutual funds, and account receivable investments. Real-estate loans' information is acquired from banks' annual reports, from 2015 to 2019.

	Panel A: State Banks						Panel B: Nonstate PD						Panel C: Non-PD					
	N	mean	sd	p25	Median	p75	N	mean	sd	p25	Median	p75	N	mean	sd	p25	Median	p75
Assets (trillion RMB)	139	17.35	6.45	9.93	18.34	22.21	458	2.51	2.13	.63	1.79	4.14	2365	.12	.19	.02	.06	.14
ROA (%)	139	.68	.32	.37	.64	.92	457	.57	.27	.34	.56	.77	2328	.58	.35	.32	.54	.78
NCD outstanding/assets (%)	139	.38	.82	.00	.05	.31	458	8.54	4.95	4.84	9.03	11.74	2365	5.31	6.68	.00	2.01	9.24
Debt securities issued excl. NCD	139	1.87	.74	1.43	2.01	2.42	442	2.58	1.42	1.80	2.50	3.30	1934	.08	4.80	.41	1.31	2.00
Deposits from financial institutions/assets (%)	139	7.47	4.01	5.11	6.28	9.66	444	13.43	7.36	7.61	13.40	18.66	2136	5.86	7.13	.70	3.29	8.56
Placement from financial institutions/assets (%)	139	2.01	1.20	1.37	1.77	2.20	444	2.24	1.42	1.32	2.11	2.87	2131	.93	2.06	.00	.18	1.18
Repo/assets (%)	139	1.04	.86	.35	.84	1.46	444	2.77	2.09	1.26	2.27	3.81	2134	3.67	3.91	.65	2.77	5.30
CB borrowing/assets (%)	139	1.77	1.89	.00	1.47	2.53	444	2.23	2.06	.27	1.68	3.87	2130	1.04	1.76	.00	.46	1.37
Deposits/assets (%)	139	75.65	8.29	72.02	75.90	79.24	445	61.93	7.51	56.85	61.54	66.53	2147	74.95	11.80	67.20	75.61	83.51
Excess reserve/assets (%)	139	.94	.93	.20	.98	1.53	444	.70	1.14	-.03	.43	1.12	2125	.94	2.49	-.42	.64	1.76
Deposits to financial institutions/assets (%)	139	2.15	1.08	1.38	1.93	2.84	444	2.81	2.74	1.10	1.81	3.60	2130	5.40	5.55	1.74	3.40	7.09
Placement to financial institutions/assets (%)	139	2.80	1.29	2.00	2.43	3.07	444	2.09	2.29	.48	1.54	4.09	2130	.96	2.20	.00	.19	1.07
Reverse repo/assets (%)	139	2.13	1.23	1.17	1.94	2.17	444	3.55	4.38	.79	2.08	4.36	2130	2.10	3.41	.00	.83	2.88
Financial investments/assets (%)	139	25.55	5.17	22.34	24.10	28.67	439	37.08	11.21	27.66	36.48	46.30	2102	31.58	11.17	24.45	32.84	39.07
Total loans/assets (%)	139	52.36	4.67	49.81	53.69	55.50	445	44.65	8.28	38.08	45.03	50.62	2139	48.56	14.87	42.43	48.47	54.77
Real-estate loans/assets (%)	28	2.61	1.06	2.15	2.49	2.91	103	3.98	1.74	2.63	4.08	5.06	350	3.12	2.67	1.02	2.49	4.71
Non-real-estate loans/assets (%)	28	49.51	3.62	46.94	49.92	51.79	103	41.53	7.71	35.97	40.56	47.57	347	45.59	10.04	39.46	45.41	52.00

Table 4: Descriptive statistics of firms

This table shows the summary statistics of firms' quarterly financial information from 2015Q1 to 2019Q4. Panel A shows the summary statistics of NEEQ firms. Loans here refers to the average daily balance of firms' loans. We further decompose firms' loans into those from state banks and nonstate banks, and NSB fraction is computed as loans from nonstate banks divided by total loans. Panel B shows the summary statistics of A share firms in two groups: SOEs and Non-SOEs. Borrowing here refers to the closing balance of firms' borrowing. The quick ratio is calculated as current assets excluding inventory divided by current liabilities. The operating cashflow ratio is calculated as cash flows from operating activities divided by current liabilities.

Panel A: NEEQ firms						
	N	mean	sd	p25	Median	p75
Total loans (billion RMB)	11,626	.500	2.227	.062	.128	.300
Loans from state banks (billion RMB)	11,626	.228	.801	.018	.046	.127
Loans from nonstate banks (billion RMB)	11,626	.272	1.934	.023	.057	.141
N fraction (%)	11,626	53.182	32.636	26.275	54.519	81.220
Panel B: A share firms						
	SOEs					
	N	mean	sd	p25	Median	p75
Borrowing (billion RMB)	15,428	9.884	30.957	.363	1.437	5.922
Revenue (billion RMB)	15,428	16.688	91.726	.947	2.598	8.068
ROA (%)	15,428	2.354	2.754	.572	1.537	3.254
Quick ratio (%)	15,428	117.964	120.788	58.867	89.234	136.127
Operating cashflow ratio (%)	15,428	8.351	24.945	-2.831	4.432	15.816
Book-to-market ratio (%)	15,428	70.252	26.013	50.311	72.487	91.732
Price-to-earnings ratio	14,350	93.718	298.015	17.472	34.096	73.606
	Non-SOEs					
	N	mean	sd	p25	Median	p75
Borrowing (billion RMB)	30,090	1.949	6.340	.121	.432	1.328
Revenue (billion RMB)	30,090	3.254	10.994	.422	.976	2.476
ROA (%)	30,090	3.201	3.312	.958	2.270	4.489
Quick ratio (%)	30,090	177.885	193.688	86.385	129.599	202.370
Operating cashflow ratio (%)	30,090	7.087	31.171	-5.266	3.939	16.067
Book-to-market ratio (%)	30,090	58.047	22.296	41.472	57.341	73.904
Price-to-earnings ratio	28,454	157.525	4206.610	27.084	46.071	84.847

Table 5: The substitution of central bank borrowing for wholesale funding
This table shows the substitution of central bank borrowing for wholesale funding estimated by the model at the bank level.

$$\Delta Y_{i,t} = \beta_1 \Delta CBBorrow_{i,t} + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \alpha_i + \lambda_q + u_{i,t}.$$

The dependent variable $\Delta Y_{i,t}$ is the quarterly change in a wholesale funding term at time t , and $\Delta CBBorrow_{i,t}$ is the quarterly change in central bank borrowing. In 2SLS estimation, $\Delta CBBorrow_{i,t}$ is instrumented by $\Delta MLF_t \times PD_{i,t}$ and $\Delta MLF_t \times SB_i$. We include the lagged central borrowing $CBBorrow_{i,t-1}$, bank controls $D_{i,t} = \{Deposit_{i,t-1}, Deposit_{i,t}, \Delta Deposit_{i,t}, \Delta RR_{i,t}, X Reserve_{i,t-1}\}$, fundamental controls $X_{i,t} = \{PD_{i,t}, \Delta r_t \times SB_i, \Delta r_t \times PD_{i,t}\}$, bank fixed effects, and quarter fixed effects. For each bank, we include the quarters since the first time that its NCD outstanding was greater than both 1 billion RMB and 0.1% of its total assets. Standard errors are clustered by banks.

	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
	Panel A: Wholesale borrowing											
	NCD outstanding			Debt securities issued excl. NCD outstanding			Deposits from financial institutions			Placement from financial institutions		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\Delta CBBorrow_{it}$	-0.082 (0.099)	-1.339** (0.575)	-0.059 (0.100)	-0.582 (0.614)	-0.232 (0.158)	0.801 (0.888)	-0.053 (0.046)	-0.028 (0.219)	-0.308* (0.159)	-0.385 (0.485)		
$CBBorrow_{it-1}$	0.070 (0.070)	-0.371 (0.239)	0.030 (0.062)	-0.154 (0.225)	-0.085 (0.088)	0.277 (0.337)	-0.036 (0.026)	-0.028 (0.088)	0.057 (0.076)	0.030 (0.198)		
Observations	1,658	1,658	1,631	1,631	1,658	1,658	1,658	1,658	1,658	1,658		
R-squared	0.122		0.064		0.058		0.029		0.098			
	Panel B: Wholesale lending											
	Financial investments			Deposits to financial institutions			Placement to financial institutions			Reverse repo		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\Delta CBBorrow_{it}$	-0.260 (0.253)	-1.393 (1.006)	-0.062 (0.175)	0.422 (0.439)	-0.062 (0.175)	0.422 (0.439)	0.002 (0.032)	0.136 (0.160)	0.293 (0.180)	-0.259 (0.443)		
$CBBorrow_{it-1}$	0.149 (0.152)	-0.247 (0.372)	0.067 (0.060)	0.237 (0.194)	0.067 (0.060)	0.237 (0.194)	0.016 (0.020)	0.063 (0.063)	-0.012 (0.064)	-0.206 (0.208)		
Observations	1,625	1,625	1,658	1,658	1,658	1,658	1,658	1,658	1,658	1,658		
R-squared	0.138		0.076				0.043		0.112			
Bank controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Fundamental Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Instruments' F-statistics		20.82		20.82		20.82		20.82		20.82		

Table 6: The substitution of CB borrowing for NCD outstanding with bank-characteristic controls
This table augments Table 5 with bank characteristics as controls. It estimates

$$\Delta Y_{i,t} = \beta_1 \Delta CBBorrow_{i,t} + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \theta' B_{i,t-1} + \alpha_i + \lambda_t + u_{i,t},$$

where $\Delta Y_{i,t}$ is the quarterly change in NCD outstanding, $D_{i,t}$ and $X_{i,t}$ are the bank and fundamental controls in model (13), and $B_{i,t-1}$ denotes the four lagged bank characteristics listed below. Columns (1)–(2) reproduce the baseline specification on the full estimation sample, and columns (3)–(4) add the lagged leverage, capital adequacy ratio, loans-to-deposits ratio, and log assets as controls. $\Delta CBBorrow_{i,t}$ is instrumented by $\Delta MLF_t \times PD_{i,t}$ and $\Delta MLF_t \times SB_i$. Standard errors are clustered by banks. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Baseline		Characteristic controls	
	OLS (1)	IV (2SLS) (2)	OLS (3)	IV (2SLS) (4)
$\Delta CBBorrow_{i,t}$	-0.082 (0.099)	-1.339** (0.575)	-0.133 (0.116)	-1.375** (0.639)
$Leverage_{i,t-1}$			-0.033 (0.231)	-0.089 (0.230)
$CAR_{i,t-1}$			0.270** (0.116)	0.263** (0.114)
$LDR_{i,t-1}$			-0.012 (0.020)	0.007 (0.022)
$\log(Assets_{i,t-1})$			-2.712** (1.140)	-3.192*** (1.229)
Observations	1,658	1,658	1,499	1,499
Banks	179	179	175	175
Bank fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
First-stage F	–	20.82	–	17.07

Table 7: The difference in the substitution for NCD outstanding

This table reports the difference in the substitution for NCD outstanding with and without bank characteristics as controls. It estimates

$$\Delta Y_{i,t} = \beta_0(\Delta CBBorrow_{i,t} \times SB_i) + \beta_1 \Delta CBBorrow_{i,t} + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \theta' B_{i,t-1} + \alpha_i + \lambda_q + u_{i,t},$$

where $\Delta Y_{i,t}$ is the quarterly change in NCD outstanding, $D_{i,t}$ and $X_{i,t}$ are the bank and fundamental controls in model (14), and $B_{i,t-1}$ denotes the four lagged bank characteristics listed below. Panel A reports the all-banks specification and Panel B reports the PD-bank specification. In both panels, columns (1)-(2) reproduce the baseline specification on the full estimation sample, while columns (3)-(4) add lagged leverage, capital adequacy ratio, loans-to-deposits ratio, and log assets. In Panel A, $\Delta CBBorrow_{i,t}$ and $\Delta CBBorrow_{i,t} \times SB_i$ are jointly instrumented by $\Delta MLF_t \times PD_{i,t}$ and $\Delta MLF_t \times SB_i$; in Panel B, the state-bank interaction is instrumented by $\Delta MLF_t \times SB_i$. Standard errors are clustered by banks. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Panel A: All banks				Panel B: PD banks			
	Baseline		Characteristic controls		Baseline		Characteristic controls	
	OLS (1)	IV (2SLS) (2)	OLS (3)	IV (2SLS) (4)	OLS (1)	IV (2SLS) (2)	OLS (3)	IV (2SLS) (4)
$\Delta CBBorrow_{i,t}$	-0.080 (0.099)	-1.338** (0.659)	-0.131 (0.117)	-1.411** (0.708)	-0.216 (0.181)	-0.202 (0.179)	-0.179 (0.132)	-0.169 (0.130)
$\Delta CBBorrow_{i,t} \times SB_i$	-0.265 (0.222)	-0.001 (0.410)	-0.206 (0.224)	0.144 (0.423)	-0.116 (0.372)	-0.709 (0.433)	-0.058 (0.336)	-0.466 (0.498)
$Leverage_{i,t-1}$			-0.034 (0.232)	-0.090 (0.230)			-0.133 (0.269)	-0.150 (0.275)
$CAR_{i,t-1}$			0.269** (0.116)	0.263** (0.114)			-0.040 (0.110)	-0.041 (0.110)
$LDR_{i,t-1}$			-0.012 (0.020)	0.008 (0.022)			-0.005 (0.023)	-0.004 (0.023)
$\log(Assets_{i,t-1})$			-2.707** (1.140)	-3.209*** (1.239)			-4.423*** (1.438)	-4.401*** (1.446)
Observations	1,658	1,658	1,499	1,499	456	456	423	423
Banks	179	179	175	175	38	38	38	38
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F: $\Delta CBBorrow_{i,t}$	-	20.82	-	17.07	-	-	-	-
First-stage F: SB_i interaction	-	99.25	-	100.30	-	245.22	-	235.96

Table 8: The difference in the substitution for NCD holdings and NCD outstanding

This table reports the difference in the substitution for NCD holdings and NCD outstanding with and without bank-type characteristics as controls. All panels estimate the monthly bank-type analogue of model (14):

$$\Delta Y_{i,t} = \beta_0(\Delta CBBorrow_{i,t} \times SB_i) + \beta_1 \Delta CBBorrow_{i,t} + \beta_2 CBBorrow_{i,t-1} + \kappa D_{i,t} + \gamma X_{i,t} + \theta' B_{i,t-1} + \alpha_i + \lambda_t + u_{i,t},$$

where $\Delta Y_{i,t}$ is the monthly change in the indicated NCD outcome, $D_{i,t}$ and $X_{i,t}$ collect the bank and fundamental controls, and $B_{i,t-1}$ denotes the three lagged bank-type characteristics listed below. Panel A reports the Big 6's NCD holdings, Panel B the Big 6's NCD outstanding, and Panel C the Big 4's NCD outstanding. In every panel, columns (1)-(2) reproduce the specification without the characteristic controls, while columns (3)-(4) add lagged bank-type leverage, log aggregate assets, and loans-to-deposits ratio. $\Delta CBBorrow_{i,t} \times SB_i$ is instrumented by $\Delta MFL_i \times SB_i$. We use Huber-White robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Panel A: Big 6's NCD holdings				Panel B: Big 6's NCD outstanding				Panel C: Big 4's NCD outstanding			
	Baseline		Characteristic controls		Baseline		Characteristic controls		Baseline		Characteristic controls	
	OLS	IV (2SLS)	OLS	IV (2SLS)	OLS	IV (2SLS)	OLS	IV (2SLS)	OLS	IV (2SLS)	OLS	IV (2SLS)
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
$\Delta CBBorrow_{i,t} \times SB_i$	-0.389* (0.203)	-0.382*** (0.141)	-0.324* (0.191)	-0.299** (0.132)	-0.212 (0.227)	-0.202 (0.153)	-0.036 (0.217)	-0.012 (0.143)	-0.174 (0.234)	-0.164 (0.158)	0.004 (0.224)	0.026 (0.147)
$\Delta CBBorrow_{i,t}$	0.309 (0.257)	0.303* (0.170)	0.307 (0.248)	0.287* (0.163)	-0.164 (0.309)	-0.172 (0.213)	-0.084 (0.291)	-0.104 (0.192)	-0.200 (0.314)	-0.208 (0.217)	-0.127 (0.297)	-0.145 (0.197)
$Leverage_{i,t-1}$			0.0015 (0.0011)	0.0015** (0.0007)			0.0011 (0.0022)	0.0011 (0.0014)			0.0012 (0.0022)	0.0012 (0.0013)
$\log(Assets_{i,t-1})$			-0.0007 (0.0012)	-0.0007 (0.0007)			-0.0007 (0.0025)	-0.0007 (0.0016)			-0.0001 (0.0024)	-0.0001 (0.0015)
$LDK_{i,t-1}$			0.0000 (0.0001)	0.0000 (0.0000)			-0.0003** (0.0001)	-0.0003*** (0.0001)			-0.0003** (0.0001)	-0.0003*** (0.0001)
Observations	118	118	118	118	118	118	118	118	118	118	118	118
Bank-type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F	-	259.07	-	226.74	-	259.07	-	226.74	-	259.07	-	226.74

Table 9: Bank-level lending

This table examines the change in bank lending by estimating the following model using a yearly sample from 2015 to 2019

$$\log(Lending_{i,t}) = \beta_0 After_t \times NSB_i + \beta_1 \log(Endowed_{i,t}) + \alpha_i + \lambda_t + u_{i,t}.$$

For Panel A, the dependent variable $\log(Lending_{i,t})$ is the log of bank i 's loans in year t and for Panel B it is the log of bank i 's loans to non-real-estate sectors in year t . The dummy variable $After_t$ equals one since the year 2018, and $\log(Endowed_{i,t})$ is the log of bank i 's endowed funds in year t . We also control for bank fixed effects and year fixed effects. Standard errors are clustered by banks. We do the estimation in the full sample and four subsamples that include state banks and different types of nonstate banks.

Panel A: All bank loans							
	(1)	All banks (2)	State banks & joint-stock banks (3)	State banks & joint-stock banks (4)	State banks & urban commercial banks (5)	State banks & rural commercial banks (6)	State banks & rural commercial banks (7)
$After_t \times NSB_i$	0.137*** (0.034)	0.091*** (0.034)	0.129*** (0.052)	0.123*** (0.039)	0.232*** (0.037)	0.151*** (0.035)	0.058* (0.032)
$\log(Endowed_{i,t})$		0.347*** (0.128)		0.865*** (0.163)		0.616*** (0.061)	0.458*** (0.183)
Bank FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Observations	2,793	2,772	89	89	664	664	1,851
R-squared	0.984	0.993	0.995	0.998	0.993	0.995	0.996
Panel B: Bank loans to non-real-estate sectors							
	(9)	All banks (10)	State banks & joint-stock banks (11)	State banks & joint-stock banks (12)	State banks & urban commercial banks (13)	State banks & urban commercial banks (14)	State banks & rural commercial banks (15)
$After_t \times NSB_i$	0.160*** (0.036)	0.101*** (0.033)	0.122*** (0.052)	0.116*** (0.038)	0.242*** (0.040)	0.173*** (0.040)	0.065* (0.034)
$\log(Endowed_{i,t})$		0.692*** (0.062)		0.854*** (0.150)		0.581*** (0.114)	0.668*** (0.077)
Bank FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Observations	1,014	1,007	89	89	453	453	443
R-squared	0.995	0.997	0.995	0.998	0.994	0.995	0.999

Table 10: Within-firm lending analysis

This table examines firms' loan compositions by estimating the following model using a yearly sample from 2015 to 2019:

$$NSBFraction_{i,t} = \beta_0 After_t + \beta_1 NSBEndowedFraction_t + \alpha_i + u_{i,t}.$$

Here $NSBFraction_{i,t}$ is the fraction of the average daily balance of loans from nonstate banks to that from nonstate and state banks, as defined by Equation (17), and $NSBEndowedFraction_t$ is the fraction of the quarterly average endowed funds of all nonstate banks to those of all nonstate banks and the Big 4 banks, calculated using banks' monthly *Sources and Uses of Credit Funds* published by the PBC. The dummy variable $After_t$ equals one since the quarter 2018Q1. We control for firm fixed effects. Standard errors are clustered by firms. For Columns (3) to (6), the model is augmented with $After_t \times SOE_i$ or $After_t \times Shadow_i$ to test the difference between different groups of firms. The term SOE_i equals one if the firm is a state-owned enterprise, and $Shadow_i$ equals one if the firm had nonbank borrowing before 2018. We do the estimation on all firms for panel A and on firms excluding real-estate and bond-issuing firms for panel B. Panel B excludes 1 real-estate firm and 38 bond-issuing firms.

Panel A: All firms				
	(1)	(2)	(3)	(4)
$After_t$	0.050*** (0.010)	0.027*** (0.009)	0.053*** (0.014)	0.029** (0.014)
$NSBEndowedFraction_t$		0.738** (0.304)		0.739** (0.303)
$After_t \times Shadow_i$			-0.004 (0.020)	-0.005 (0.020)
Firm FE	Y	Y	Y	Y
Observations	11,626	11,626	11,626	11,626
R-squared	0.616	0.617	0.616	0.617
Panel B: Excluding real-estate and bond-issuing firms				
	(5)	(6)	(7)	(8)
$After_t$	0.055*** (0.010)	0.029*** (0.009)	0.056*** (0.014)	0.031** (0.014)
$NSBEndowedFraction_t$		0.800** (0.314)		0.801** (0.314)
$After_t \times Shadow_i$			-0.003 (0.021)	-0.004 (0.021)
Firm FE	Y	Y	Y	Y
Observations	10,915	10,915	10,915	10,915
R-squared	0.613	0.615	0.613	0.615

Table 11: Between-firm lending analysis

This table examines the difference before and after 2018Q1 in non-financial firms' loans between state-owned firms and nonstate firms by estimating the following model using a quarterly sample from 2015 to 2019:

$$\log(\text{Borrowing}_{i,t}) = \beta_0 \text{After}_t \times \text{NSE}_i + \beta_1 \log(\text{FirmTypeEndowed}_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}.$$

The dependent variable $\log(\text{Borrowing}_{i,t})$ is the log of firm i 's borrowing in quarter t . The dummy variable After_t equals one since the quarter 2018Q1. NSE_i is a dummy variable that equals 1 if the firm is not a state-owned enterprise. Furthermore, $\log(\text{FirmTypeEndowed}_{i,t})$ is the log of firm i 's endowed funds at quarter t , where we use state banks' endowed funds as state-owned enterprises' endowed funds and nonstate banks' endowed funds as those of nonstate firms. We also include lagged variables to control for firm's demand for loans, $Z_{i,t-1} = \{\log(\text{Revenue}_{i,t-1}), \text{BM}_{i,t-1}, \text{PE}_{i,t-1}, \text{ROA}_{i,t-1}, \text{QuickRatio}_{i,t-1}, \text{OperCashRatio}_{i,t-1}\}$. We control for firm fixed effects and quarter fixed effects. Standard errors are clustered by firms. We do the estimation on all firms for panel A, on firms excluding real-estate firms for panel B, and on firms excluding bond-issuing firms for panel C.

Panel A: All firms				
	(1)	(2)	(3)	(4)
$\text{After}_t \times \text{NSE}_i$	0.471*** (0.037)	0.223*** (0.035)	0.304*** (0.036)	0.163*** (0.035)
$\log(\text{FirmTypeEndowed}_{i,t})$		1.949*** (0.258)		1.134*** (0.248)
Controls	N	N	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Observations	45,401	45,401	39,113	39,113
R-squared	0.876	0.876	0.898	0.898
Panel B: Excluding real-estate firms				
	(5)	(6)	(7)	(8)
$\text{After}_t \times \text{NSE}_i$	0.511*** (0.039)	0.237*** (0.036)	0.329*** (0.038)	0.176*** (0.037)
$\log(\text{FirmTypeEndowed}_{i,t})$		2.152*** (0.266)		1.236*** (0.256)
Controls	N	N	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Observations	42,445	42,445	36,477	36,477
R-squared	0.866	0.867	0.890	0.890
Panel C: Excluding bond-issuing firms				
	(9)	(10)	(11)	(12)
$\text{After}_t \times \text{NSE}_i$	0.569*** (0.056)	0.289*** (0.051)	0.363*** (0.055)	0.217*** (0.053)
$\log(\text{FirmTypeEndowed}_{i,t})$		2.224*** (0.373)		1.198*** (0.369)
Controls	N	N	Y	Y
Firm FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y
Observations	29,403	29,403	25,172	25,172
R-squared	0.813	0.814	0.845	0.845

Table 12: The real effects on fixed assets

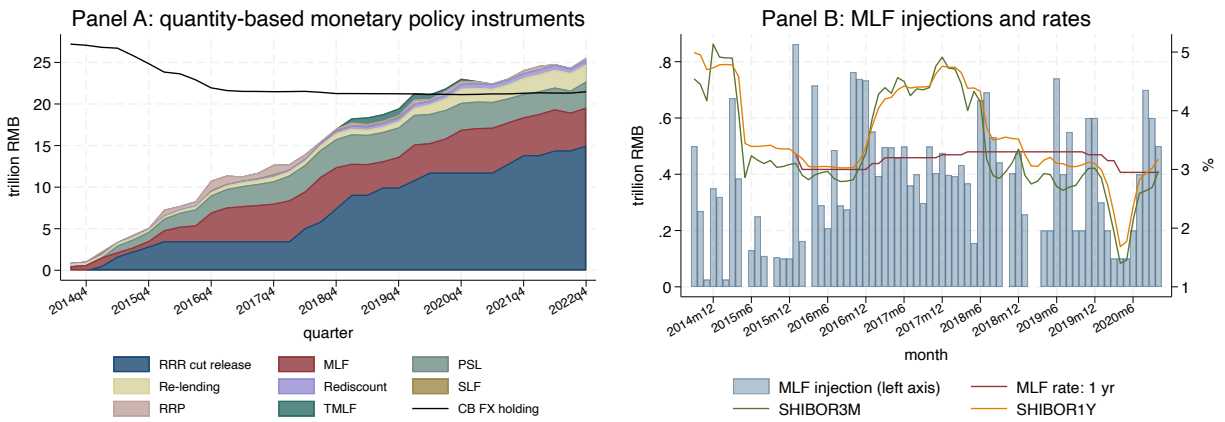
This table estimates

$$\log(FixedAssets_{i,t}) = \beta_0 After_t \times NSE_i + \beta_1 \log(FirmTypeEndowed_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}$$

on a quarterly sample of A-share nonfinancial listed firms from 2015 to 2019. The dependent variable is the log of firm fixed assets. Panel A defines $After_t$ as one from 2018Q4 and is the baseline; Panels B and C instead start treatment in 2018Q3 and 2019Q1. NSE_i equals one for nonstate enterprises, and $\log(FirmTypeEndowed_{i,t})$ is the log of endowed funds. The controls are lagged log revenue, book-to-market, price-to-earnings ratio, return on assets, quick ratio, and operating cash ratio. All panels use identical estimation samples, firm fixed effects, and quarter fixed effects. Standard errors are clustered by firms. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

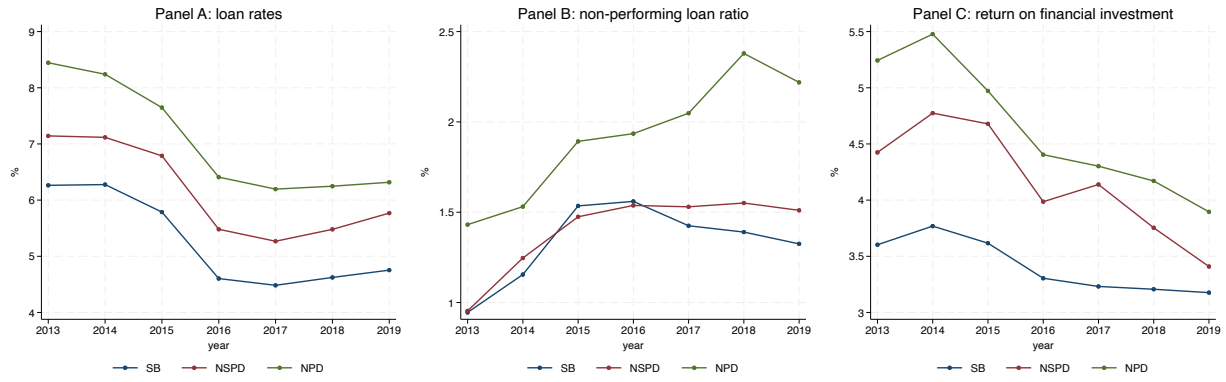
	Panel A: Treatment from 2018Q4 (baseline)				Panel B: Treatment from 2018Q3				Panel C: Treatment from 2019Q1			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
$After_t \times NSE_i$	0.158*** (0.020)	0.062*** (0.017)	0.087*** (0.019)	0.061*** (0.015)	0.155*** (0.020)	0.062*** (0.017)	0.075*** (0.019)	0.043*** (0.015)	0.158*** (0.019)	0.059*** (0.016)	0.092*** (0.018)	0.064*** (0.015)
$\log(FirmTypeEndowed_{i,t})$		0.777*** (0.155)		0.226 (0.145)		0.754*** (0.153)		0.273* (0.143)		0.817*** (0.155)		0.246* (0.145)
Controls	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	51,590	51,590	44,518	44,518	51,590	51,590	44,518	44,518	51,590	51,590	44,518	44,518
R^2	0.946	0.947	0.962	0.962	0.946	0.947	0.962	0.962	0.946	0.947	0.962	0.962

Figure 1: The volumes of quantity-based monetary policy instruments



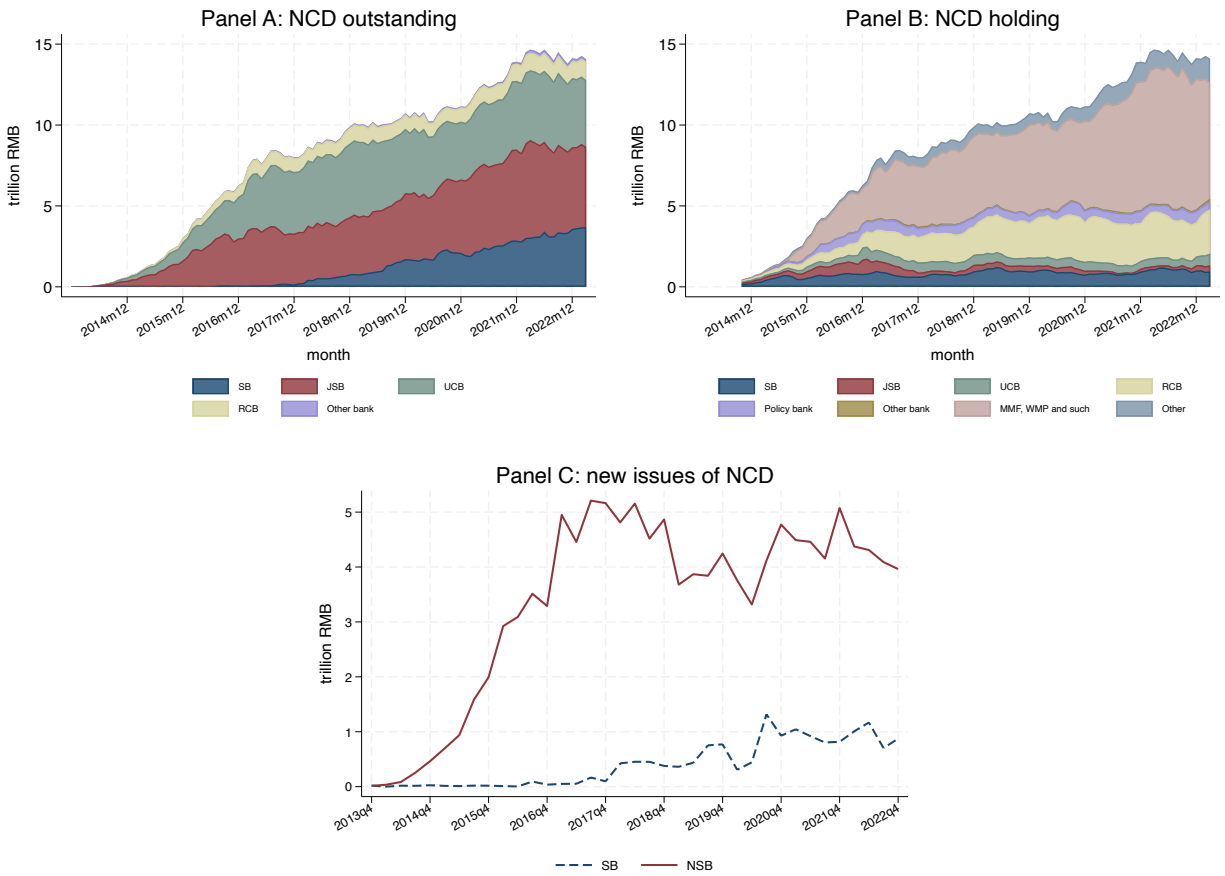
Panel A shows the quarterly amount of base money supplied through monetary policy instruments from 2014Q3 to 2021Q2. Liquidity injected by RRR cuts is estimated by using aggregate deposits times the percentage of RRR cut, and we set 2014Q3 as the base period (when RRR cut injection balance equals zero). RRP and MLF quarterly balances are backed out using PBC open market operation announcements. Data come from the PBC website. Panel B shows the time series of MLF injection, the one-year MLF rate, the three-month and the one-year Shibo.

Figure 2: Banks' investment styles



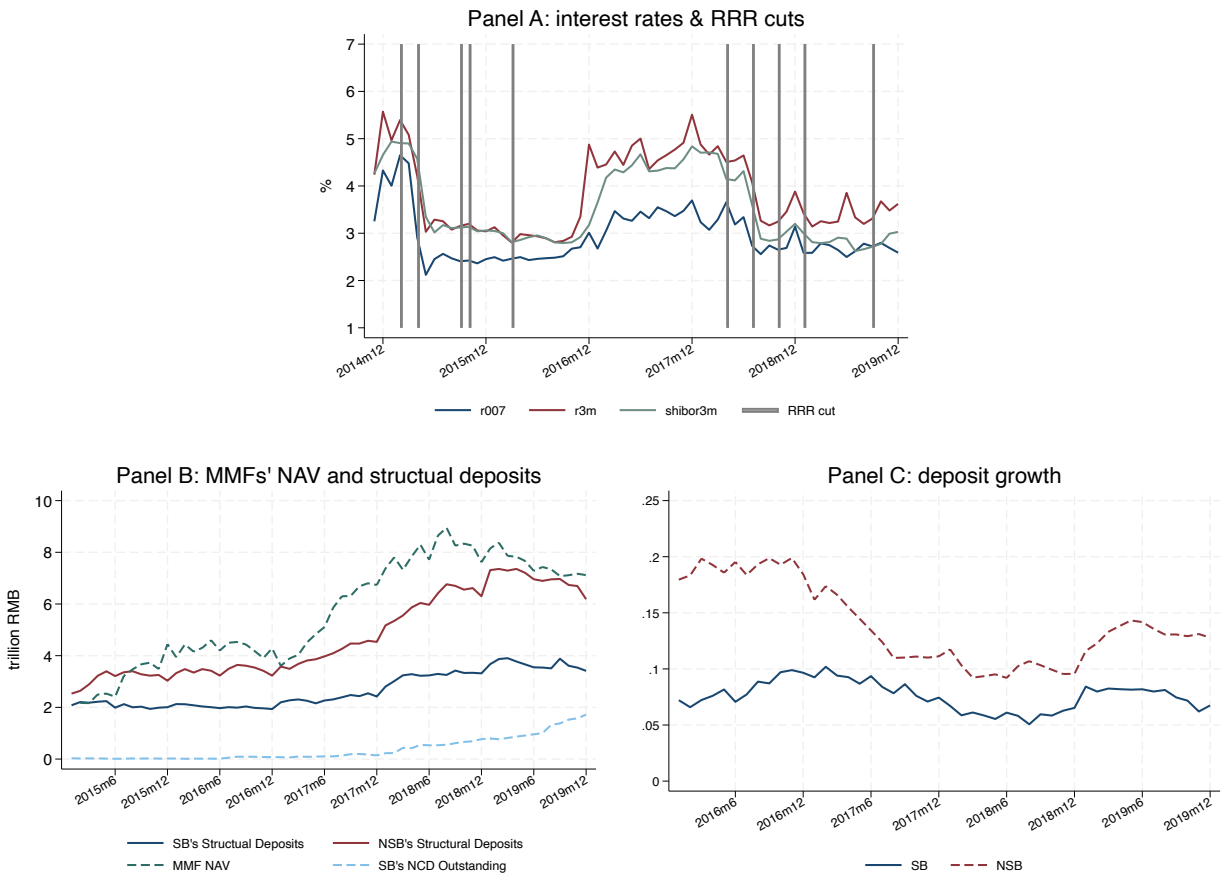
This figure plots the yearly average of loan rate, nonperforming loan ratio, and return on financial investments of different bank groups from 2013--2019. Banks are grouped into state banks (SB), nonstate PD banks (NSPD) and non-PD banks (NPD). Loan rates (return of financial investment) are calculated using annual interest income from loans (financial investments) scaled by the average of loan (financial investments) balance at year start and year end. The nonperforming loan ratio is directly collected from bank annual reports.

Figure 3: NCD holding & issuance



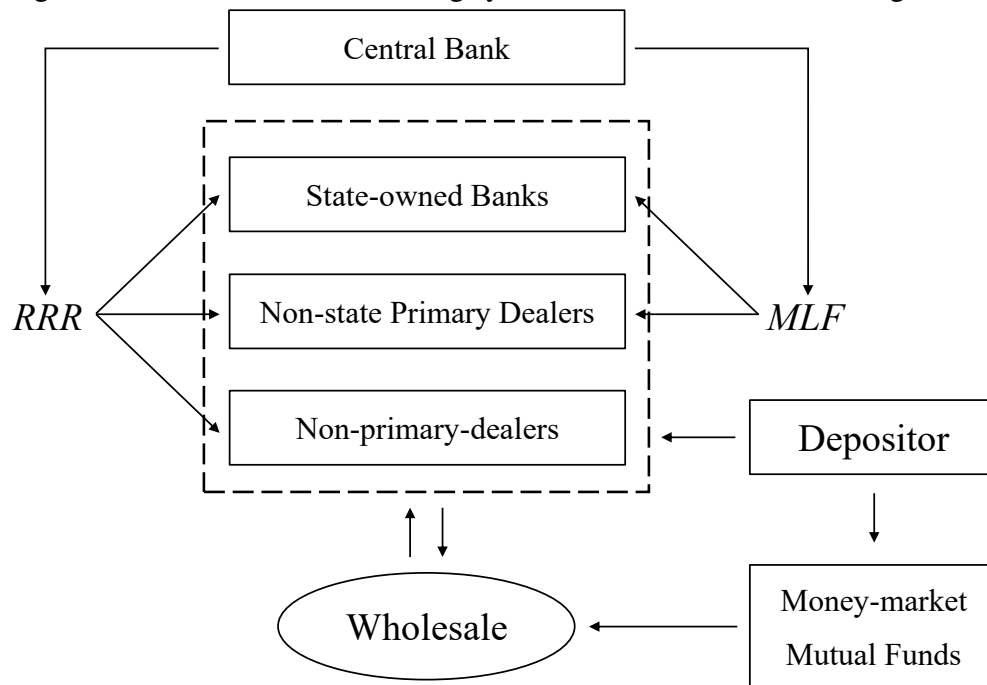
Panel A shows the monthly volume of NCD outstanding by different types of issuing banks from December 2013 to July 2021. Banks are grouped into state banks (SB), joint-stock banks (JSB), urban commercial banks (UCB), and rural commercial banks (RCB), policy banks, and other banks. Panel B shows the monthly volume of NCD holding by different types of investors from October 2014 to July 2022. Panel C shows the quarterly volume of new issues of NCD by state and nonstate banks.

Figure 4: Important variables around 2018



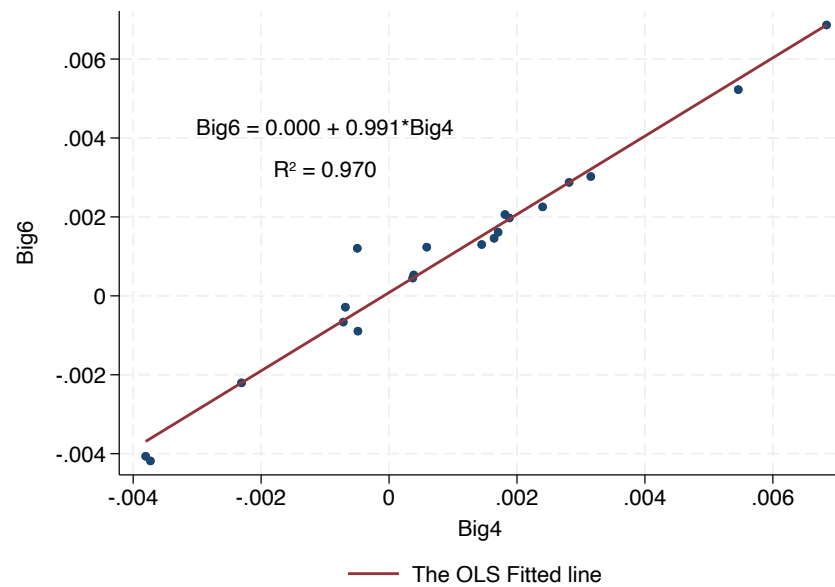
Panel A shows three interbank interest rates, r007, r3m, Shibor3m, and marks the months with RRR cuts. Panel B shows the volumes of money market funds' NAV, the structural deposits of state banks (SBs) and nonstate banks (NSBs), and state banks' NCD outstanding. Panel C plots the year-on-year growth rates of bank deposits. Here, bank deposits include only domestic individual deposits and domestic corporate deposits, but not deposits of financial institutions.

Figure 5: Fund flows in the banking system and the wholesale funding market



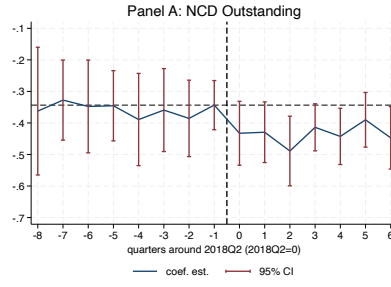
The PBC injects funds to banks mainly through quantity-based instruments such as RRR cuts and MLF, which may leave a substantial funding imbalance within the banking system. Therefore, the reallocation of funds on the wholesale funding market is crucial to monetary policy transmission and resource allocation. Traditionally, depositors mainly deposit money in banks. Recently, attracted by higher returns, depositors alternatively deposited money at money-market mutual funds, which further allocate funds to banks. Directing funds on the wholesale funding market, money-market mutual funds have a profound impact on the resource allocation across banks.

Figure 6: Comparing the Big 6's and the Big 4's changes in central bank borrowing



This figure shows the scatter plot of the Big 6 banks' quarterly changes in central bank borrowing scaled by their lagged total assets against those of the Big 4 banks, for the sample period from 2014Q4 to 2019Q4. We regress the former on the latter. The figure also shows the regression results of them and the fitted line.

Figure 7: The second wave of RRR cuts

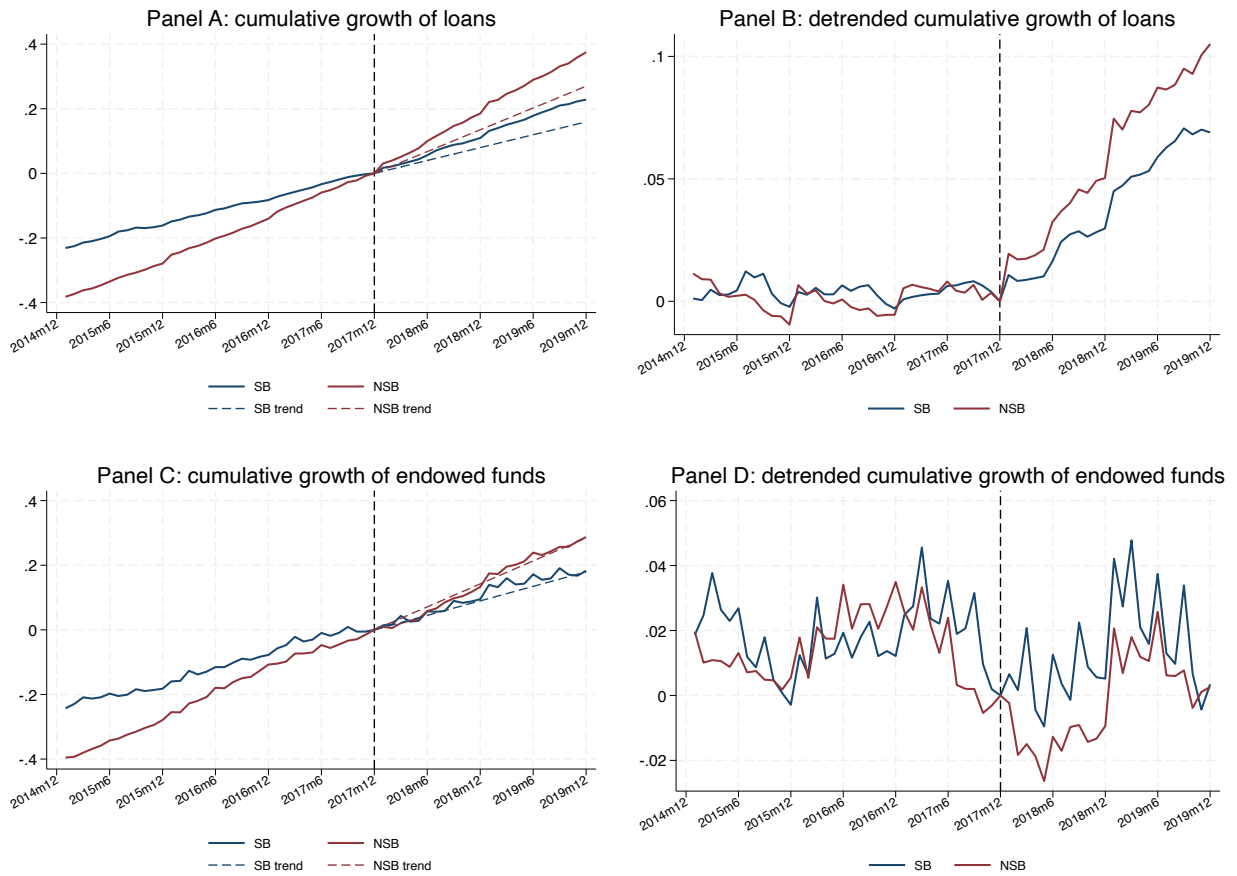


This figure plots the estimates of β_1 and associated 95% confidence intervals from the regression

$$Y_{i,t} = \beta_1 \text{Deposit}_{i,2018Q1} + \beta_2 (\text{Deposit}_{i,t} - \text{Deposit}_{i,2018Q1}) + \beta_3 \text{CBBorrow}_{i,t} + \lambda_t + \gamma X_{i,t} + u_{i,t}$$

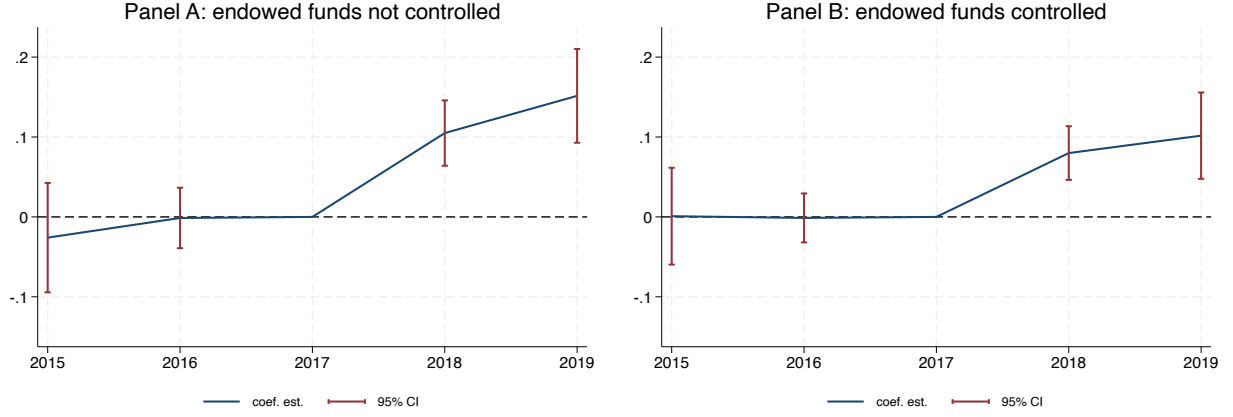
with NCD outstanding being the dependent variable around the wave of RRR cuts starting in 2018Q2. $\text{Deposit}_{i,2018Q1}$ is bank i 's deposits in 2018Q1, the quarter right before the start of the wave, and $\text{Deposit}_{i,t}$ is bank i 's deposits at time t . We include $\text{CBBorrow}_{i,t}$ to control for the impacts of central bank borrowing. Finally, $X_{i,t} = \{\text{NSB}_i, \text{PD}_{i,t}, \text{XReserve}_{i,t-1}\}$ are fundamental controls. 2018Q2 is labeled as Quarter 0. The 15 quarters from the eighth quarter before the start of the wave to the end of 2019 are plotted.

Figure 8: Cumulative and detrended growth of bank loans and endowed funds



Panel A plots the arithmetic cumulative growth of loans using the level of loans in 2017m12 as the benchmark. The time series of the level of loans are scaled by the level in 2017m12, and normalized to 0 in 2017m12. Dash lines are the predicted trends since 2018, which are estimated using OLS models and the time series before 2018. Panel B plot the detrended time series. Panel C and Panel D plot the same time series for endowed funds.

Figure 9: Bank lending before and since the shift

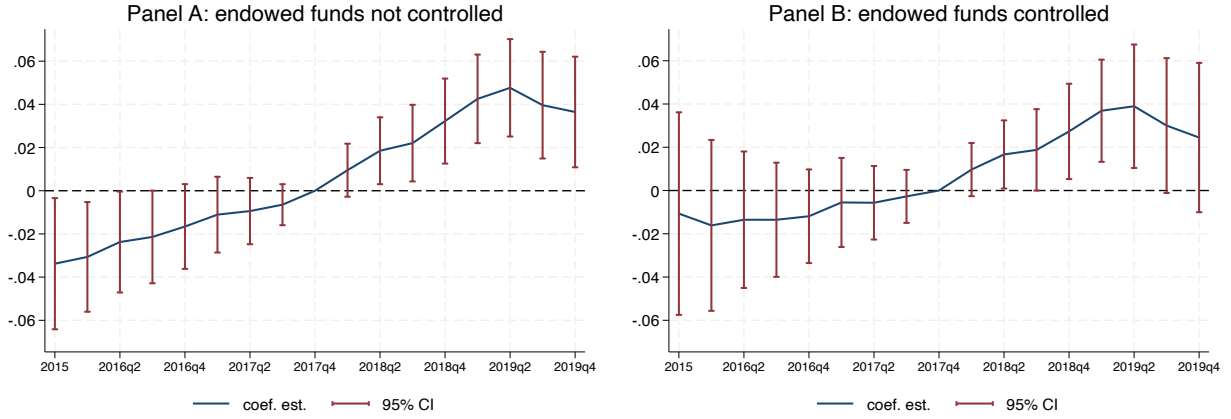


Panel B plots the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from the regression

$$\log(Lending_{i,t}) = \sum_{\tau=2015}^{2016} \beta_{0,\tau} Y_{\tau,t} \times NSB_i + \sum_{\tau=2018}^{2019} \beta_{0,\tau} Y_{\tau,t} \times NSB_i + \beta_1 \log(Endowed_{i,t}) + \alpha_i + \lambda_t + u_{i,t}.$$

Here, $\log(Lending_{i,t})$ is the log of bank i 's loans in year t ; $\log(Endowed_{i,t})$ is the log of bank i 's endowed funds in year t ; and $Y_{\tau,t}$ is the year dummy variable that equals one if $t = \tau$. We use the year 2017 as the benchmark and control for bank fixed effects and year fixed effects. Standard errors are clustered by banks. Panel A plots the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from a regression that does not include $\log(Endowed_{i,t})$ as a regressor.

Figure 10: Firms' loan compositions

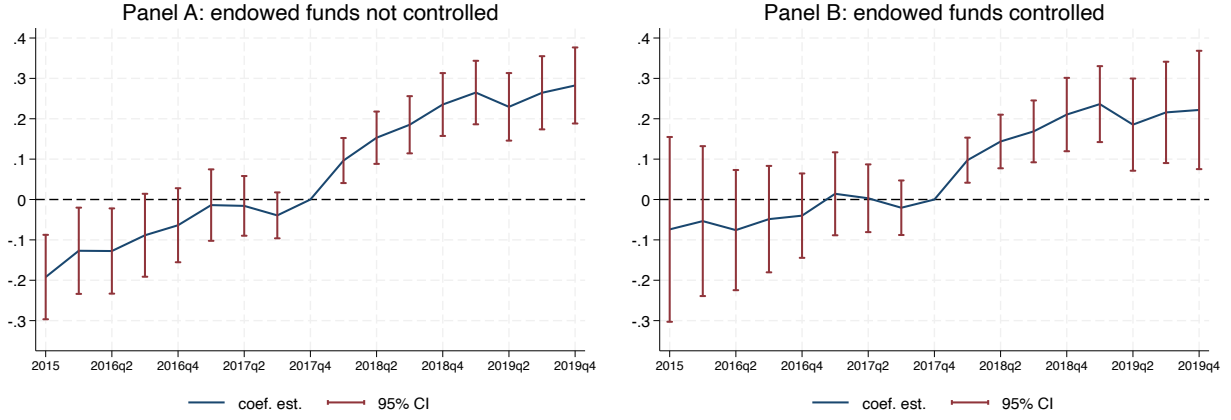


Panel B plots the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from the regression

$$NSBFraction_{i,t} = \beta_{0,2015}Y_{2015,t} + \sum_{\tau=2016Q1}^{2017Q3} \beta_{0,\tau}Y_{\tau,t} + \sum_{\tau=2018Q1}^{2019Q4} \beta_{0,\tau}Y_{\tau,t} + \beta_1 NSBEndowedFraction_t + \alpha_i + u_{i,t}.$$

Here $NSBFraction_{i,t}$ is the fraction of the quarterly average daily balance of loans from nonstate banks to that from nonstate and state banks, as defined by Equation (17); $NSBEndowedFraction_t$ is the fraction of the quarterly average endowed funds of all nonstate banks to those of all banks, calculated using banks' monthly *Sources and Uses of Credit Funds* published by the PBC. $Y_{2015,t}$ is the quarter dummy variable that equals one if the quarter is in 2015; and $Y_{\tau,t}$ is the quarter dummy variable that equals one if $t = \tau$. We use the quarter 2017Q4 as the benchmark and control for firm fixed effects. Standard errors are clustered by firms. Panel A plots the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from a regression that does not include $NSBEndowedFraction_t$ as a regressor.

Figure 11: A share-listed nonfinancial firms' loans

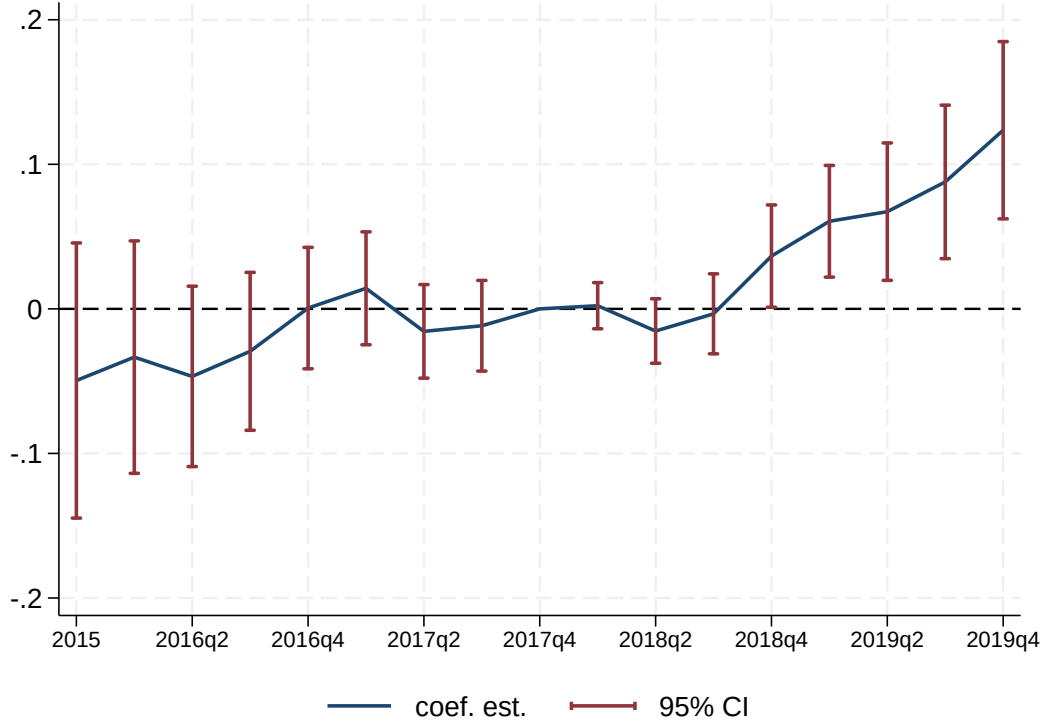


Panel B plots the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from the regression

$$\begin{aligned} \log(Loan_{i,t}) = & \beta_{0,2015}Y_{2015,t} \times NSE_i + \sum_{\tau=2016Q1}^{2017Q3} \beta_{0,\tau}Y_{\tau,t} \times NSE_i + \sum_{\tau=2018Q1}^{2019Q4} \beta_{0,\tau}Y_{\tau,t} \times NSE_i \\ & + \beta_1 \log(FirmTypeEndowed_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}. \end{aligned}$$

The dependent variable $\log(Loan_{i,t})$ is the log of firm i 's loans at quarter t ; $Y_{2015,t}$ is the quarter dummy variable that equals 1 if the quarter is in 2015; $Y_{\tau,t}$ is the quarter dummy variable that equals one if $t = \tau$; NSE_i is a dummy variable that equals one if the firm is not a state-owned enterprise; and $\log(FirmTypeEndowed_{i,t})$ is the log of firm i 's endowed funds at quarter t , where we use state banks' endowed funds as state-owned enterprises' endowed funds and nonstate banks' endowed funds as those of nonstate firms. We also include lagged variables to control for firm's demands for loans, $Z_{i,t-1} = \{\log(Revenue_{i,t-1}), BM_{i,t-1}, PE_{i,t-1}, ROA_{i,t-1}, QuickRatio_{i,t-1}, OperCashRatio_{i,t-1}\}$. We use the quarter 2017Q4 as the benchmark and control for firm fixed effects and quarter fixed effects. Standard errors are clustered by firms. Panel A plots the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from the regression that does not include $\log(FirmTypeEndowed_{i,t})$ as a regressor. The estimation is done on all A share-listed nonfinancial firms.

Figure 12: A-share listed non-financial firms' fixed assets (endowed funds controlled)

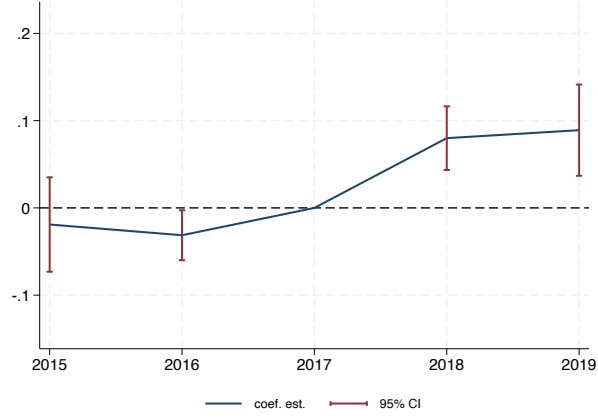


This figure plots the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from the regression

$$\begin{aligned} \log(FixedAssets_{i,t}) = & \beta_{0,2015}Y_{2015,t} \times NSE_i + \sum_{\tau=2016Q1}^{2017Q3} \beta_{0,\tau}Y_{\tau,t} \times NSE_i + \sum_{\tau=2018Q1}^{2019Q4} \beta_{0,\tau}Y_{\tau,t} \times NSE_i \\ & + \beta_1 \log(FirmTypeEndowed_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}. \end{aligned}$$

The dependent variable $\log(FixedAssets_{i,t})$ is the log of firm i 's fixed assets at quarter t ; $Y_{2015,t}$ is the quarter dummy variable that equals 1 if the quarter is in 2015; $Y_{\tau,t}$ is the quarter dummy variable that equals one if $t = \tau$; NSE_i is a dummy variable that equals one if the firm is not a state-owned enterprise; and $\log(FirmTypeEndowed_{i,t})$ is the log of firm i 's endowed funds at quarter t , where we use state banks' endowed funds as state-owned enterprises' endowed funds and nonstate banks' endowed funds as those of nonstate firms. We also include lagged variables to control for firm's demands for loans, $Z_{i,t-1} = \{\log(Revenue_{i,t-1}), BM_{i,t-1}, PE_{i,t-1}, ROA_{i,t-1}, QuickRatio_{i,t-1}, OperCashRatio_{i,t-1}\}$. We use the quarter 2017Q4 as the benchmark and control for firm fixed effects and quarter fixed effects. Standard errors are clustered by firms. The estimation is done on all A-share listed nonfinancial firms.

Figure 13: Bank lending to sectors other than real estate

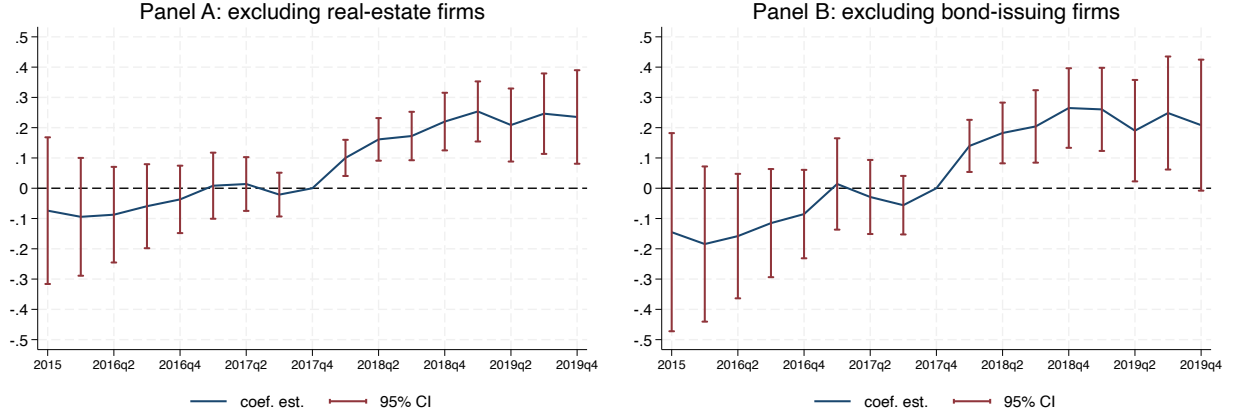


This figure plots the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from the regression

$$\log(Lending_{i,t}) = \sum_{\tau=2015}^{2016} \beta_{0,\tau} Y_{\tau,t} \times NSB_i + \sum_{\tau=2018}^{2019} \beta_{0,\tau} Y_{\tau,t} \times NSB_i + \beta_1 \log(Endowed_{i,t}) + \alpha_i + \lambda_t + u_{i,t}.$$

Here $\log(Lending_{i,t})$ is the log of bank i 's loans to non-real-estate in year t ; $\log(Endowed_{i,t})$ is the log of bank i 's endowed funds in year t ; and $Y_{\tau,t}$ is the year dummy variable that equals one if $t = \tau$. We use the year 2017 as the benchmark and control for bank fixed effects and year fixed effects. Standard errors are clustered by banks.

Figure 14: A-share listed non-financial firms' borrowing excluding real-estate and bond-issuing firms

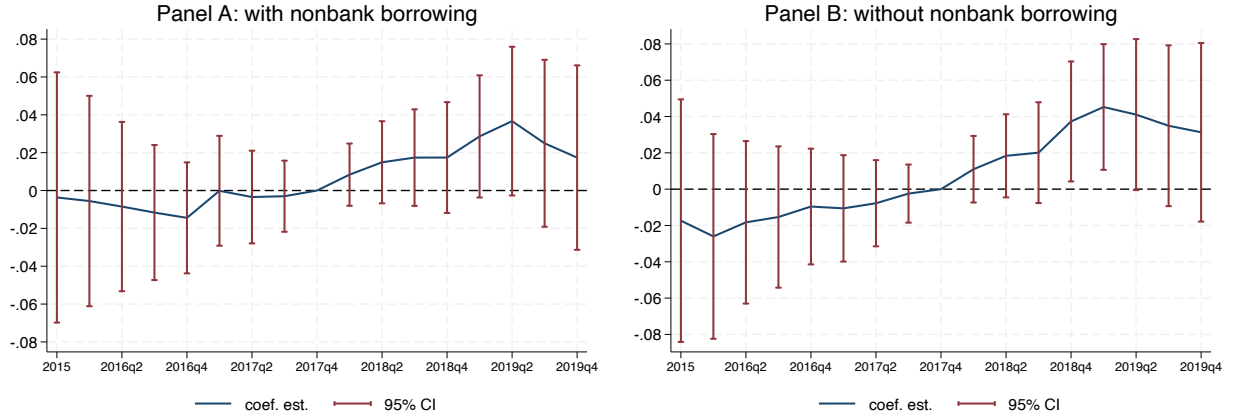


Both panels plot the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from the regression

$$\begin{aligned} \log(\text{Borrowing}_{i,t}) = & \beta_{0,2015} Y_{2015,t} \times NSE_i + \sum_{\tau=2016Q1}^{2017Q3} \beta_{0,\tau} Y_{\tau,t} \times NSE_i + \sum_{\tau=2018Q1}^{2019Q4} \beta_{0,\tau} Y_{\tau,t} \times NSE_i \\ & + \beta_1 \log(\text{FirmTypeEndowed}_{i,t}) + \delta Z_{i,t-1} + \alpha_i + \lambda_t + u_{i,t}. \end{aligned}$$

The dependent variable $\log(\text{Borrowing}_{i,t})$ is the log of firm i 's loans at quarter t ; $Y_{2015,t}$ is the quarter dummy variable that equals 1 if the quarter is in 2015; $Y_{\tau,t}$ is the quarter dummy variable that equals one if $t = \tau$; NSE_i is a dummy variable that equals one if the firm is not a state-owned enterprise; and $\log(\text{FirmTypeEndowed}_{i,t})$ is the log of firm i 's endowed funds at quarter t , where we use state banks' endowed funds as state-owned enterprises' endowed funds and nonstate banks' endowed funds as those of nonstate firms. We also include lagged variables to control for firm's demand for loans, $Z_{i,t-1} = \{\log(\text{Revenue}_{i,t-1}), \text{BM}_{i,t-1}, \text{PE}_{i,t-1}, \text{ROA}_{i,t-1}, \text{QuickRatio}_{i,t-1}, \text{OperCashRatio}_{i,t-1}\}$. We use the quarter 2017Q4 as the benchmark and control for firm fixed effects and quarter fixed effects. Standard errors are clustered by firms. Panel A plots the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from the regression that does not include $\log(\text{FirmTypeEndowed}_{i,t})$ as a regressor. We do the estimation on A share--listed nonfinancial firms excluding real-estate firms for Panel A, and on A share--listed nonfinancial firms excluding bond-issuing firms for Panel B.

Figure 15: The loan composition of firms with nonbank borrowing and those without.



Both panels plot the estimates of $\beta_{1,\tau}$ and associated 95% confidence intervals from the regression

$$NSBFraction_{i,t} = \beta_{0,2015}Y_{2015,t} + \sum_{\tau=2016Q1}^{2017Q3} \beta_{0,\tau}Y_{\tau,t} + \sum_{\tau=2018Q1}^{2019Q4} \beta_{0,\tau}Y_{\tau,t} + \beta_1 NSBEndowedFraction_t + \alpha_i + u_{i,t}.$$

Here, $NSBFraction_{i,t}$ is the fraction of the quarterly average daily balance of loans from nonstate banks to that from nonstate and state banks, as defined by Equation (17), and $NSBEndowedFraction_t$ is the fraction of the quarterly average endowed funds of all nonstate banks to those of all banks, calculated using banks' monthly *Sources and Uses of Credit Funds* published by the PBC. The quarter dummy variable $Y_{2015,t}$ equals one if the quarter is in 2015, and $Y_{\tau,t}$ is the quarter dummy variable that equals one if $t = \tau$. We use the quarter 2017Q4 as the benchmark and control firm fixed effects. Standard errors are clustered by firms. Panel A plots the estimates of $\beta_{0,\tau}$ and associated 95% confidence intervals from a regression that does not include $NSBEndowedFraction_t$ as a regressor. We do the estimation on firms with nonbank borrowing for Panel A and on firms without nonbank borrowing for Panel B.

B Proofs

Proof of Proposition 1

In equilibrium, the nonstate bank i must choose a loan portfolio such that the marginal returns of loans to the two types are equal:

$$\begin{aligned} R_K - 2\gamma k_{SOE} &= R_K - 2\gamma k_{NSOE}, \\ k_{SOE} + k_{NSOE} &= k_i. \end{aligned}$$

It implies that the loans to either type must take up half of its loan portfolio:

$$k_{SOE} = k_{NSOE} = \frac{1}{2}k_i.$$

Therefore, a nonstate bank i 's utility from loans can be expressed as a function of the amount of all its loans as follows:

$$\Pi_i = R_K k_i - \frac{1}{2}\gamma k_i^2. \quad (21)$$

Due to the conservatism, the loans to risky firms need to be justified by a higher marginal return:

$$\begin{aligned} R_K - 2\gamma k_{SOE} &= R_K - 2\gamma k_{NSOE} - \eta k_{NSOE}, \\ k_{SOE} + k_{NSOE} &= k_s. \end{aligned}$$

It implies that the loans to SOEs must take up a greater fraction of its loan portfolio than those to non-SOEs:

$$k_{SOE} = \frac{2\gamma + \eta}{4\gamma + \eta} k_s > k_{NSOE} = \frac{2\gamma}{4\gamma + \eta} k_s.$$

Therefore, the state bank's utility from loans can be expressed as a function of the amount of all its loans as follows:

$$\Pi_s - \frac{1}{2}\eta k_{NSOE}^2 = R_K k_s - \frac{2\gamma + \eta}{4\gamma + \eta} \gamma k_s^2. \quad (22)$$

Notice that the state bank makes more than half of its loans to SOEs, while nonstate banks make exactly half. The state bank makes a higher fraction of loans to SOEs than nonstate banks.

Plugging equations (3),(4),(21), and (22) into expressions (1) and (2), we can rewrite a nonstate bank's utility as expression (5) and the state bank's utility as expression (6).

Proof of Proposition 3

It is easy to see $R_{WF}^1 > R_{WF}^2$.

For the state bank, the marginal cost of borrowing on the wholesale funding market is R_{WF} ,

and the marginal benefit of lending on the wholesale funding market is $R_{WF} - \delta l_s \cdot \mathbf{1}\{l_s > 0\}$. Due to the conservatism δ , the state bank does not simultaneously borrow and lend.

If the state bank borrows on the wholesale funding market, i.e., $b_s \geq 0 = l_s$,

$$k_s = \frac{R_K - R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}.$$

Equation (10) implies

$$\frac{R_K - R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} + \frac{R_K - R_{WF}}{\gamma} = e_s + e_{ns} + W.$$

Hence, the interest rate on the market must be equal to R_{WF}^2 . This is indeed the equilibrium if the state bank does borrow under R_{WF}^2 , i.e.,

$$k_s - h_s = \frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF} - R_D}{\kappa} \geq e_s.$$

If this condition does not, then the state bank lends on the wholesale funding market, i.e., $l_s \geq 0 = b_s$. Then

$$\begin{aligned} h_s &= \frac{R_{WF} - R_D - \delta l_s}{\kappa}, \\ k_s &= \frac{R_K - R_{WF} + \delta l_s}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}, \\ l_s &= e_s + h_s - k_s. \end{aligned}$$

We obtain

$$\begin{aligned} l_s &= \frac{e_s + \frac{R_{WF} - R_D}{\kappa} - \frac{R_K - R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}}{1 + \frac{\delta}{\kappa} + \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}}, \\ k_s &= \left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta}\right) \frac{R_K - R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} + \delta \frac{e_s + \frac{R_{WF} - R_D}{\kappa}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{\delta}{\kappa} \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \delta}, \\ h_s &= \left(1 - \frac{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta}\right) \frac{R_{WF} - R_D}{\kappa} - \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma \frac{\delta}{\kappa} \frac{e_s - \frac{R_K - R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta}. \end{aligned}$$

Plugging it with $k_i = \frac{R_K - R_{WF}}{\gamma}$ into equation (10), we obtain

$$\begin{aligned} & \left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \right) \frac{R_K - R_{WF}^1}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} + \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \frac{R_{WF}^1 - R_D}{\kappa} + \frac{R_K - R_{WF}^1}{\gamma} \\ &= \left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \right) e_s + e_{ns} + W, \end{aligned}$$

Hence, the interest rate on the market must be equal to R_{WF}^1 . This is indeed the equilibrium, since the state bank does lend under R_{WF}^1 , i.e.,

$$\begin{aligned} k_s - h_s &= \left(1 - \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \right) \frac{R_K - R_{WF}^1}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} + \delta \frac{e_s + \frac{R_{WF}^1 - R_D}{\kappa}}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{\delta}{\kappa} \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \delta} \\ &\quad - \frac{R_{WF}^1 - R_D - \delta \frac{e_s + \frac{R_{WF}^1 - R_D}{\kappa} - \frac{R_K - R_{WF}^1}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}}{1 + \frac{\delta}{\kappa} + \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}}}{\kappa} \\ &= \frac{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \left(\frac{R_K - R_{WF}^1}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF}^1 - R_D}{\kappa} \right) + \frac{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} e_s \\ &< \frac{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} \left(\frac{R_K - R_{WF}^2}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma} - \frac{R_{WF}^2 - R_D}{\kappa} \right) + \frac{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta}{\frac{4\gamma+2\eta}{4\gamma+\eta}\gamma + \frac{4\gamma+2\eta}{4\gamma+\eta}\gamma\frac{\delta}{\kappa} + \delta} e_s \\ &< e_s. \end{aligned}$$

Proof of Implication 1

For nonstate banks, the change in wholesale funding is given by

$$\Delta(b_i - l_i) = \Delta k_i - \Delta h_i - \Delta e_i = \frac{\Delta R_K - \Delta R_{WF}}{\gamma} - \frac{\Delta R_{WF} - \Delta R_D}{\kappa} - \Delta e_i.$$

When they borrow on the interbank market,

$$\frac{\partial \Delta b_i}{\partial \Delta e_i} = -1.$$

When they lend on the interbank market,

$$\frac{\partial \Delta l_i}{\partial \Delta e_i} = 1.$$

For the state bank,

$$\begin{aligned}\Delta(b_s - l_s) &= \Delta k_s - \Delta h_s - \Delta e_s \\ &= \frac{\Delta R_K - \Delta R_{WF} + \delta \Delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma} - \frac{\Delta R_{WF} - \Delta R_D - \delta l_s \cdot \mathbf{1}\{l_s > 0\}}{\kappa} - \Delta e_s.\end{aligned}$$

When it borrows on the wholesale funding market, $b_s \geq 0 = l_s$, and

$$\Delta b_s = \frac{\Delta R_K - \Delta R_{WF}}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma} - \frac{\Delta R_{WF} - \Delta R_D}{\kappa} - \Delta e_s,$$

so

$$\frac{\partial \Delta b_s}{\partial \Delta e_s} = -1.$$

When it lends on the wholesale funding market, $l_s \geq 0 = b_s$, and

$$-\Delta l_s = \frac{\Delta R_K - \Delta R_{WF} + \delta \Delta l_s}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma} - \frac{\Delta R_{WF} - \Delta R_D - \delta l_s}{\kappa} - \Delta e_s,$$

so

$$\frac{\partial \Delta l_s}{\partial \Delta e_s} = \frac{1}{1 + \frac{\delta}{\kappa} + \frac{\delta}{\frac{4\gamma+2\eta}{4\gamma+\eta} \gamma}} < 1.$$

Proof of Implication 2

Suppose that there is no other monetary policy intervention, i.e., m_j stays unchanged. Consider two banks j_1 and j_2 that borrow on the market, and assume $d_{j_1} < d_{j_2}$.

$$\frac{b_{j_1} - b_{j_2}}{d_{j_1} - d_{j_2}} = -(1 - \rho) + \frac{(k_{j_1} - h_{j_1} - m_{j_1}) - (k_{j_2} - h_{j_2} - m_{j_2})}{d_{j_1} - d_{j_2}}.$$

Then

$$\partial \frac{b_{j_1} - b_{j_2}}{d_{j_1} - d_{j_2}} / \partial \rho > 0$$

Similarly, consider two banks j_1 and j_2 that lend on the market, and assume $d_{j_1} < d_{j_2}$.

$$\frac{l_{j_1} - l_{j_2}}{d_{j_1} - d_{j_2}} = (1 - \rho) + \frac{(m_{j_1} - k_{j_1}) - (m_{j_2} - k_{j_2})}{d_{j_1} - d_{j_2}}.$$

Then

$$\partial \frac{l_{j_1} - l_{j_2}}{d_{j_1} - d_{j_2}} / \partial \rho < 0$$

Proof of Implication 3

We use the superscript 1 (2) to refer to terms in Scenario 1 (2). We prove

$$R_{WF}^1 - \delta l_s^1 < R_{WF}^2.$$

Suppose the opposite. Then

$$R_{WF}^1 > R_{WF}^1 - \delta l_s^1 \geq R_{WF}^2,$$

so $k_s^1 \leq k_s^2$, and $k_i^1 < k_i^2$. This implies that the sum of the total loans in Scenario 1 is smaller than that in Scenario 2. Since this sum is equal to the total available funds, this implication contradicts that the total available funds are the same in the two scenarios. With $R_{WF}^1 - \delta l_s^1 < R_{WF}^2$, we immediately obtain $k_s^1 > k_s^2$. Following a similar logic, we can also show that

$$R_{WF}^1 > R_{WF}^2,$$

which implies $k_i^1 < k_i^2$. Taken together, $k_i^2 - k_i^1 > 0 > k_s^2 - k_s^1$.

Proof of Implication 4

The loans made to non-SOEs are $\frac{1}{2}k_i^1 + \frac{2\gamma}{4\gamma+\eta}k_s^1$ in Scenario 1 and $\frac{1}{2}k_i^2 + \frac{2\gamma}{4\gamma+\eta}k_s^2$ in Scenario 2. Hence, when the banking system shifts from Scenario 1 to Scenario 2, the change in the loans made to non-SOEs is $\frac{1}{2}(k_i^2 - k_i^1) + \frac{2\gamma}{4\gamma+\eta}(k_s^2 - k_s^1)$. Similarly, the change in the loans made to SOEs is $\frac{1}{2}(k_i^2 - k_i^1) + \frac{2\gamma+\eta}{4\gamma+\eta}(k_s^2 - k_s^1)$. Since $k_s^2 < k_s^1$, the change in the loans made to non-SOEs is greater than the change in the loans made to SOEs.

C Online Appendix

C.1 The quantity-based monetary policy rule

To characterize how the quantity-based tools respond to macroeconomic conditions, we begin with the CRZ endogenous-switching policy rule developed by [Chen et al. \(2018\)](#) and compare it with the MNR extension proposed by [Miranda-Agrippino et al. \(2026\)](#):

$$y_t = \gamma_0 + \gamma_m y_{t-1} + \gamma_\pi (\pi_{t-1} - \pi^*) + \gamma_{fx} \tilde{x}_{t-1} + \gamma_y (s_t) gap_{t-1} + \varepsilon_t,$$

where y_t is the quarterly log change in the quantity-based tools and the regime s_t is determined endogenously by the sign of the lagged output gap relative to a threshold $\bar{y}$, so that the response to the output gap is allowed to differ between the above-threshold and below-threshold states. In CRZ, the growth rate of M2 is the policy variable and the output-gap response switches according to whether economic growth is above or below the government's growth objective. MNR generalizes this framework to accommodate the evolution of the PBC's operating framework: they replace M2 with a composite monetary policy indicator that combines price- and quantity-based instruments, add an exchange-rate term constructed from CFETS trade weights and central parity rates, estimate rather than fix the output-gap threshold, use time-varying official inflation targets, and estimate the rule at monthly frequency. For our post-2014 quarterly sample, the MNR-style variant brings the corresponding modeling elements into the quantity-tool rule through a single disturbance variance, an exchange-rate term, a grid-estimated threshold, and a time-varying official inflation target, while retaining our quantity-tool dependent variables and quarterly frequency. Because several modeling choices are debatable in this short sample, we estimate all eight combinations of three binary choices: the CRZ specification versus the MNR-style variant described above, raw versus seasonally adjusted data, and the sum of quantity-based tools excluding PSL versus MLF plus RRR as the dependent variable. Table 13 reports the eight estimates.

The central result is the below-threshold output response γ_{yL} . It is negative and statistically significant in every cell, at least at the 10% level. Because the output gap is negative in this regime, the negative coefficient means that the quantity-based tools expand when economic growth falls short of the target, providing countercyclical support when growth is weaker than expected. Thus, like M2 in the original policy rule, the quantity-based monetary policy tools exhibit countercyclical adjustment to below-target growth. This result is robust to the specification, to seasonal adjustment, and to the definition of the dependent variable across all eight estimates.

Table 13: The endogenous-switching quantity-based policy rule

This table reports Gaussian maximum likelihood estimates of the endogenous-switching quantity rule

$$y_t = \gamma_0 + \gamma_m y_{t-1} + \gamma_\pi (\pi_{t-1} - \pi^*) + \gamma_f \tilde{x}_{t-1} + \gamma_y (s_t) gap_{t-1} + \varepsilon_t,$$

where the regime s_t is determined by the sign of the lagged output gap relative to a threshold $\bar{y}$. The CRZ columns follow the specification of [Chen et al. \(2018\)](#): regime-switching volatility σ_H/σ_L , no exchange-rate term, $\bar{y} = 0$, and a constant inflation target. The MNR columns use a single σ , an exchange-rate term $\tilde{x} \times 100$, a grid-estimated $\bar{y}$, and a time-varying inflation target π^* . In the SA columns every variable is seasonally adjusted with X-13ARIMA-SEATS, including the FX term (additively, since $\tilde{x}$ is signed); the Raw columns keep all variables unadjusted. The dependent variable is the quarterly log difference of either the sum of quantity-based tools excluding PSL or MLF+RRR. The GDP gap is year-over-year and the inflation gap is quarter-over-quarter, following the CRZ horizons. The sample is 2014Q4–2019Q4, and N_a/N_b reports the regime split. Standard errors, in parentheses, are computed from the numerical Hessian. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	CRZ				MNR			
	ToolSum ex PSL		MLF+RRR		ToolSum ex PSL		MLF+RRR	
	Raw	SA	Raw	SA	Raw	SA	Raw	SA
γ_0	0.0598 (0.0586)	-0.0120 (0.0455)	0.0525 (0.0504)	-0.0279 (0.0414)	0.0352 (0.0598)	-0.0311 (0.0652)	0.0072 (0.0630)	-0.0449 (0.0663)
γ_m	0.0156 (0.1487)	0.2076 (0.1275)	0.2610* (0.1547)	0.4095*** (0.1402)	0.1653 (0.2800)	0.1976 (0.2422)	0.4147 (0.2852)	0.4012* (0.2435)
γ_π	-4.0903* (2.4772)	1.8143 (7.5453)	-3.2463 (2.2586)	-2.0146 (6.8455)	-10.6455* (5.5507)	-28.3340*** (9.1106)	-8.1464 (5.5193)	-25.0983** (9.9698)
γ_{fx}	- (17.0999)	- (32.0144**)	- (13.1408)	- (29.0632**)	-2.3372 (4.0713)	0.3437 (3.7872)	-2.5203 (4.6695)	0.3582 (4.3436)
γ_H	-17.0999 (28.1960)	32.0144** (12.7806)	-13.1408 (23.9654)	29.0632** (12.1949)	-11.0679 (27.8441)	4.6760 (24.9354)	4.7932 (29.4910)	10.2134 (26.9157)
γ_L	-55.6718** (27.8722)	-58.0878** (22.5876)	-50.9886* (27.7185)	-53.3958** (21.8732)	-57.4441*** (21.6567)	-48.1421*** (17.4889)	-61.1123*** (22.2633)	-47.3362** (18.6520)
σ_H	0.04786*** (0.01222)	0.04287*** (0.01137)	0.04423*** (0.01138)	0.04311*** (0.01090)	0.14020*** (0.02217)	0.11736*** (0.01856)	0.14450*** (0.02285)	0.12953*** (0.02048)
σ_L	0.19219*** (0.03978)	0.20429*** (0.04808)	0.20074*** (0.04177)	0.19792*** (0.04473)	- (0.0000)	- (0.0010)	- (0.0020)	- (0.0010)
$\bar{y}$	0.0000	0.0000	0.0000	0.0000	-0.0020	0.0010	-0.0020	0.0010
R^2	0.2860	0.1234	0.3600	0.3689	0.3918	0.4923	0.4646	0.5269
N	20	20	20	20	20	20	20	20
N_a/N_b	8/12	9/11	8/12	9/11	15/5	6/14	15/5	6/14

C.2 Non-bank borrowing versus bank loans

We provide both aggregate and micro-level evidence showing that non-bank borrowing, such as entrusted loans and trust loans, constitutes only a small fraction of the borrowing measure, so the documented growth cannot be mainly driven by higher non-bank borrowing demand.

Our firm-level $Borrowing_{i,t}$ is defined as the sum of short-term borrowings, long-term borrowings, and the current portion of non-current liabilities. Mapped onto the components of Aggregate Financing to the Real Economy (AFRE), the loan-type financing captured by this measure consists of

$$\underbrace{\text{RMB loans} + \text{FX loans} + \text{undiscounted acceptances}}_{\text{bank loan types}} + \underbrace{\text{entrusted loans} + \text{trust loans}}_{\text{non-bank loan types}}.$$

Equity financing and the vast majority of bonds are not part of $Borrowing_{i,t}$. Table 14 presents a decomposition of the AFRE data by financing component. The data are from the official AFRE (stock) statistical tables published by the People’s Bank of China. In the calculation of AFRE, data are from the People’s Bank of China, the China Securities Regulatory Commission, the China Insurance Regulatory Commission, the China Central Depository & Clearing Co., Ltd., and the National Association of Financial Market Institutional Investors. Entrusted loans and trust loans together account for a modest share of total loan-type social financing, ranging from 10.7% to 15.2%, peaking in 2017 and declining thereafter. Thus, non-bank lending was not only a minority share but also shrinking during the period when our estimated effects emerge.

To complement the aggregate patterns, we turn to micro-level evidence for further validation. Table 15 disaggregates NEEQ firms’ borrowing into bank and non-bank lending. Entrusted loans and trust loans constitute an even smaller share in this sample, between 3.2% and 4.7%. Because the A-share listed firms in our between-firm analysis have substantially better access to formal bank credit than NEEQ firms, they face even weaker incentives to rely on non-bank channels. Complementing this, Allen et al. (2019) document that 99% of firms borrowing through entrusted loans are non-listed. We can therefore reasonably infer that A-share listed firms in our sample have a similarly small share of non-bank borrowing.

Table 11 shows that non-SOE borrowing increased by 16.3% more than SOE borrowing since 2018. Given that the non-bank share of AFRE was in fact contracting during this period, it is implausible that the documented 16.3% differential growth is driven by non-bank borrowing.

Table 14: Structure of the Aggregate Financing to the Real Economy (AFRE) data, 2015 to 2019
This table shows the structure of yearly AFRE data. Bank loans = RMB loans + FX loans + undiscounted bankers' acceptances; non-bank loans = entrusted loans + trust loans. Corporate bonds, government bonds and equity financing are excluded. Panel A reports the year-end outstanding stock of each borrowing type and Panel B reports the share.

	2015	2016	2017	2018	2019
Panel A: Year-end outstanding stock (trillion RMB)					
Bank loans	101.6	111.7	126.0	140.7	157.0
Entrusted loans	11.0	13.2	14.0	12.4	11.4
Trust loans	5.5	6.3	8.5	7.9	7.5
Total	118.1	131.2	148.5	160.9	175.9
Panel B. Year-end outstanding share (%)					
Bank loans	86.0%	85.1%	84.8%	87.4%	89.3%
Entrusted loans	9.3%	10.1%	9.4%	7.7%	6.5%
Trust loans	4.7%	4.8%	5.7%	4.9%	4.3%
Non-bank loans (entrusted + trust)	14.0%	14.9%	15.2%	12.6%	10.7%

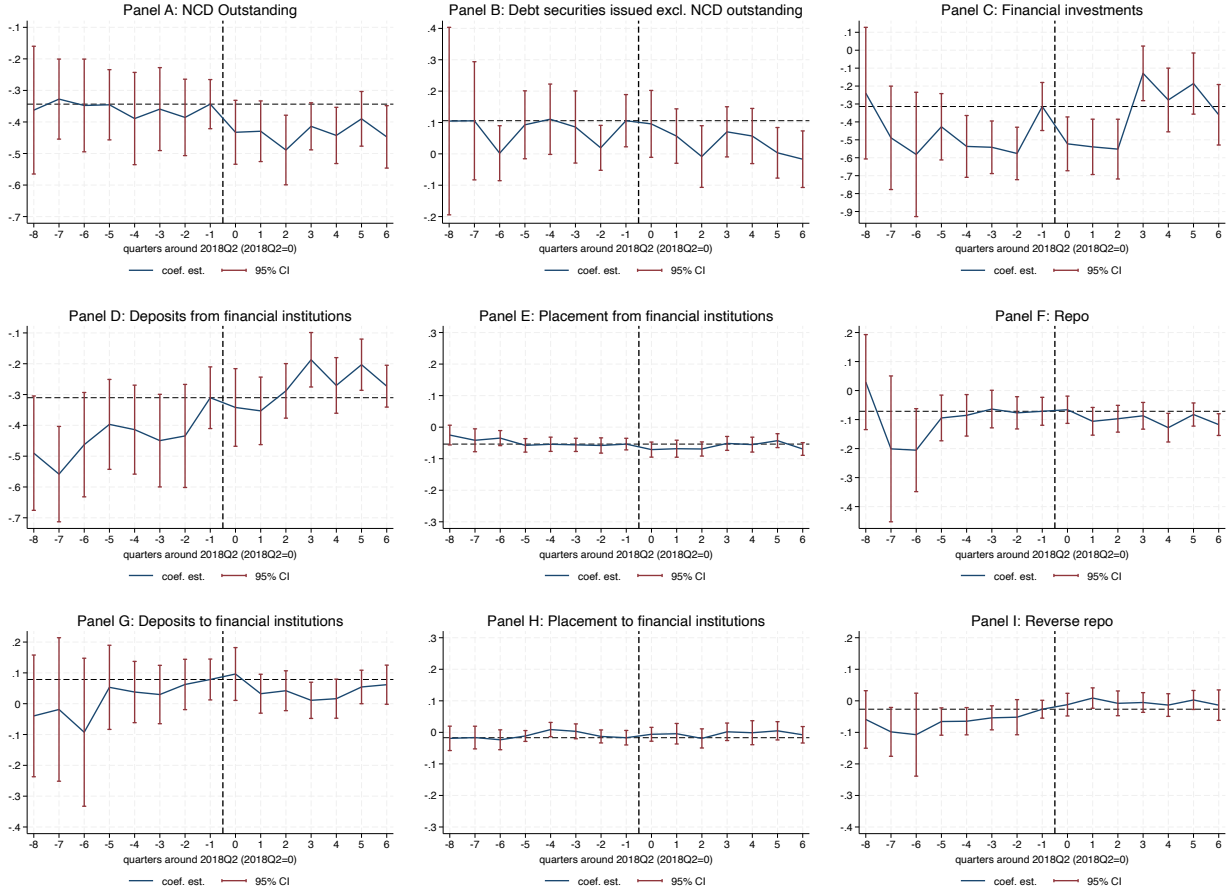
Table 15: Structure of NEEQ firms' borrowing data, 2015 to 2019

This table shows the structure of NEEQ firms' borrowing. Each NEEQ firm's borrowing types are identified from the loan-level disclosures in its annual reports.

	2015	2016	2017	2018	2019
Panel A: Year-end outstanding stock (billion RMB)					
Bank loans	273.8	457.4	496.1	446.8	392.7
Entrusted loans	5.2	9.4	8.0	6.6	5.8
Trust loans	6.1	5.7	12.2	15.4	8.4
Total	285.0	472.4	516.3	468.9	407.0
Panel B. Year-end outstanding share (%)					
Bank loans	96.1%	96.8%	96.1%	95.3%	96.5%
Entrusted loans	1.8%	2.0%	1.5%	1.4%	1.4%
Trust loans	2.1%	1.2%	2.4%	3.3%	2.1%
Non-bank loans (entrusted + trust)	3.9%	3.2%	3.9%	4.7%	3.5%

C.3 Wholesale funding terms around RRR cuts

Figure 16: The second wave of RRR cuts



All panels plot the estimates of β_1 and associated 95% confidence intervals from the regression

$$Y_{i,t} = \beta_1 \text{Deposit}_{i,2018Q1} + \beta_2 (\text{Deposit}_{i,t} - \text{Deposit}_{i,2018Q1}) + \beta_3 \text{CBBorrow}_{i,t} + \lambda_t + \gamma X_{i,t} + u_{i,t}$$

with wholesale funding terms being the dependent variables around the wave of RRR cuts starting in 2018Q2. $\text{Deposit}_{i,2018Q1}$ is bank i 's deposits in 2018Q1, the quarter right before the start of the wave, and $\text{Deposit}_{i,t}$ is bank i 's deposits at time t . We include $\text{CBBorrow}_{i,t}$ to control for the impacts of central bank borrowing. Finally, $X_{i,t} = \{\text{NSB}_i, \text{PD}_{i,t}, \text{XReserve}_{i,t-1}\}$ are fundamental controls. 2018Q2 is labeled as Quarter 0. The 15 quarters from the eighth quarter before the start of the wave to the end of 2019 are plotted.